

(12) **United States Patent**
Vacek

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(45) **Date of Patent:** **Sep. 16, 2025**

(54) **SYSTEMS AND METHODS FOR CONTROLLING AIR PROPERTIES IN STRUCTURES AND INHIBITING MOISTURE ACCUMULATION AND MOLD PROPAGATION IN STRUCTURES**

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(73) Assignee: **VACEK LLC**, Porter, TX (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 159 days.

(21) Appl. No.: **18/379,068**

(22) Filed: **Oct. 11, 2023**

(65) **Prior Publication Data**

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Related U.S. Application Data

(63) Continuation-in-part of application No. 17/556,527, filed on Dec. 20, 2021, now Pat. No. 11,982,466, which is a continuation of application No. 16/273,344, filed on Feb. 12, 2019, now Pat. No. 11,248,814.

(51) **Int. Cl.**

F24F 7/10 (2006.01)
E04B 1/70 (2006.01)
E04B 1/72 (2006.01)
F24F 7/02 (2006.01)
F24F 7/08 (2006.01)
F24F 13/22 (2006.01)
F24F 110/20 (2018.01)
F24F 140/30 (2018.01)

(52) **U.S. Cl.**

CPC **F24F 7/10** (2013.01); **E04B 1/7069** (2013.01); **E04B 1/72** (2013.01); **F24F 7/02** (2013.01); **F24F 7/08** (2013.01); **F24F 2013/221** (2013.01); **F24F 2110/20** (2018.01); **F24F 2140/30** (2018.01)

(58) **Field of Classification Search**

CPC **F24F 7/10**; **F24F 7/02**; **F24F 7/08**; **F24F 2110/20**; **F24F 2140/30**; **F24F 2013/221**; **E04B 1/7069**; **E04B 1/72**

USPC **165/54**
See application file for complete search history.

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Primary Examiner — Claire E Rojohn, III

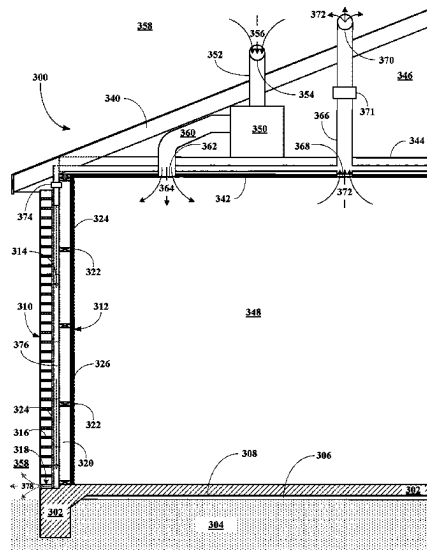
(74) *Attorney, Agent, or Firm* — Robert W. Strozier

(57)

ABSTRACT

A structure comprises at least one outer wall having an internal wall section and an outer wall section with an air flow passage therebetween. A circulation system circulates air through the flow passage to inhibit moisture accumulation and mold growth. A sensing system determines the presence of moisture in the flow passage and generates a signal in response thereto. A controller receives the signal from the sensing system and controls the circulation system to maintain a predetermined temperature and relative humidity in the flow passage.

16 Claims, 55 Drawing Sheets



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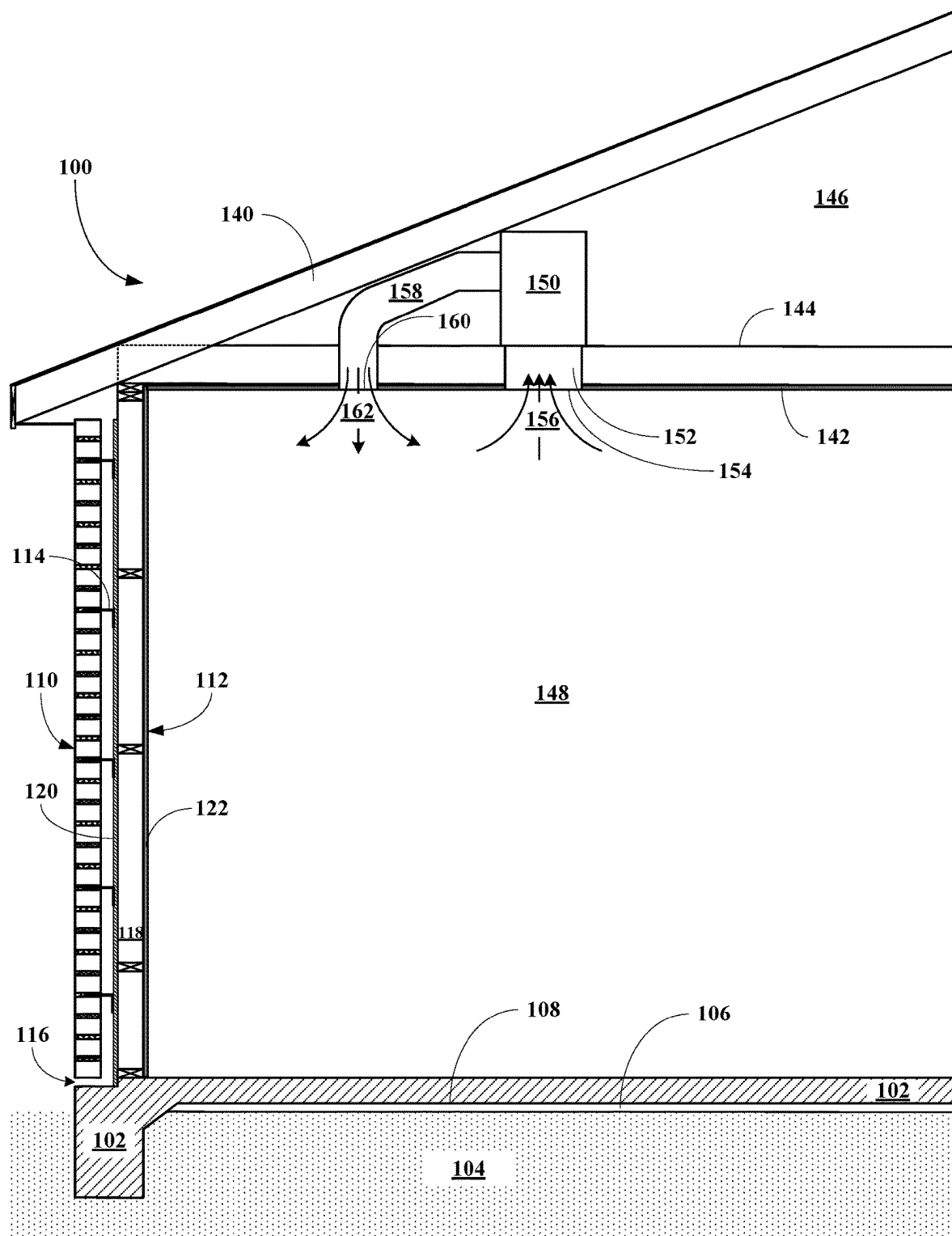


FIG. 1 – Prior Art

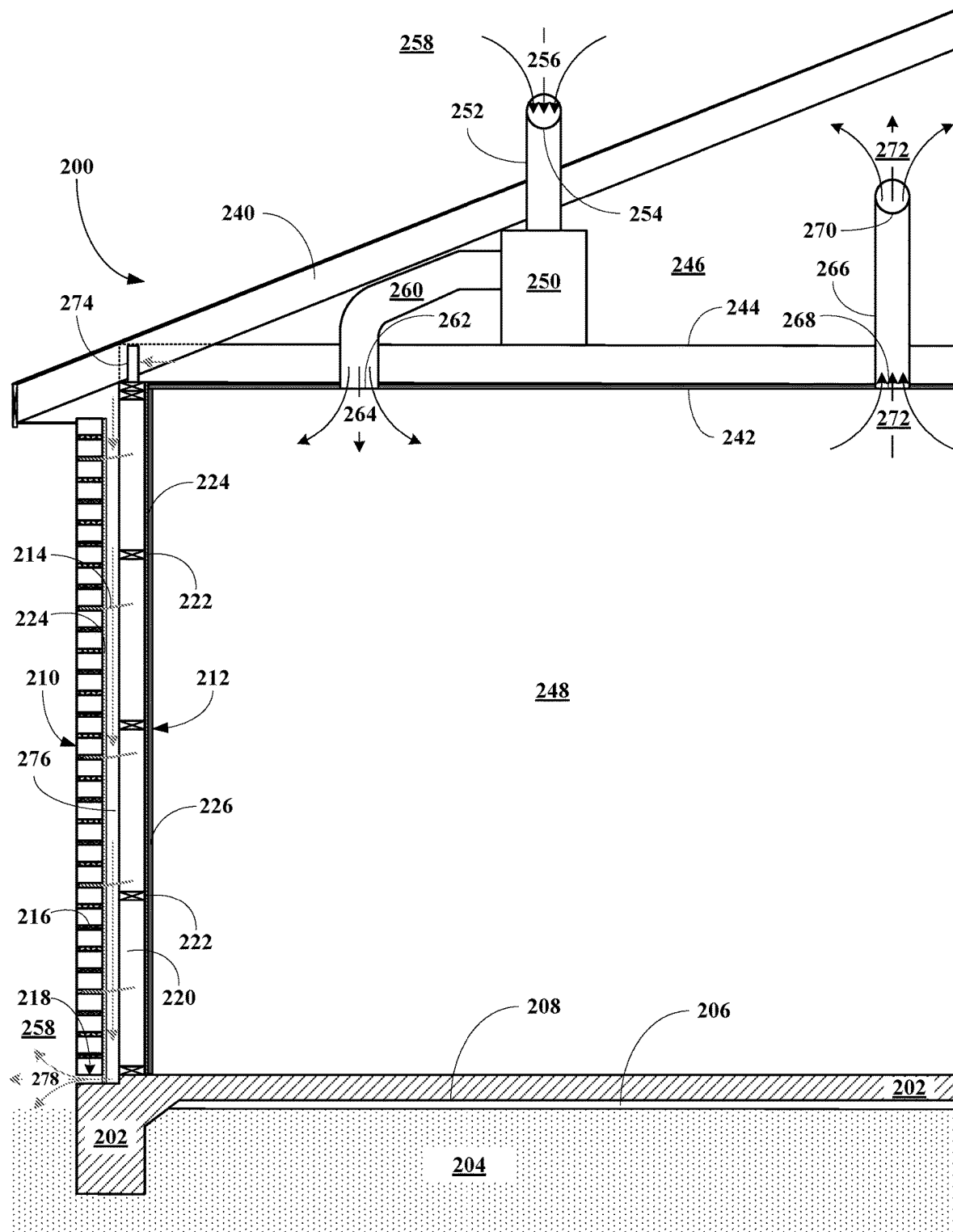


FIG. 2A

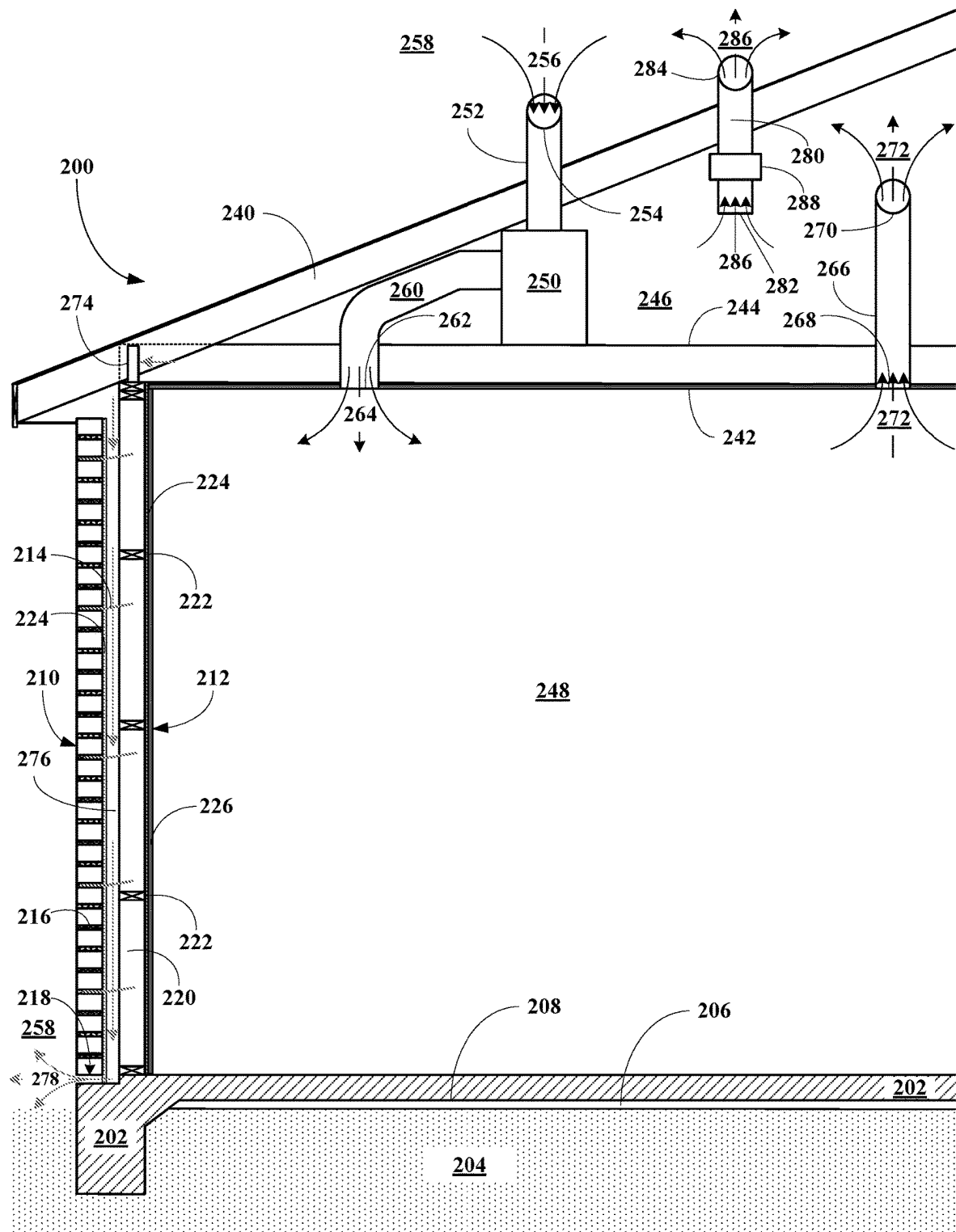


FIG. 2B

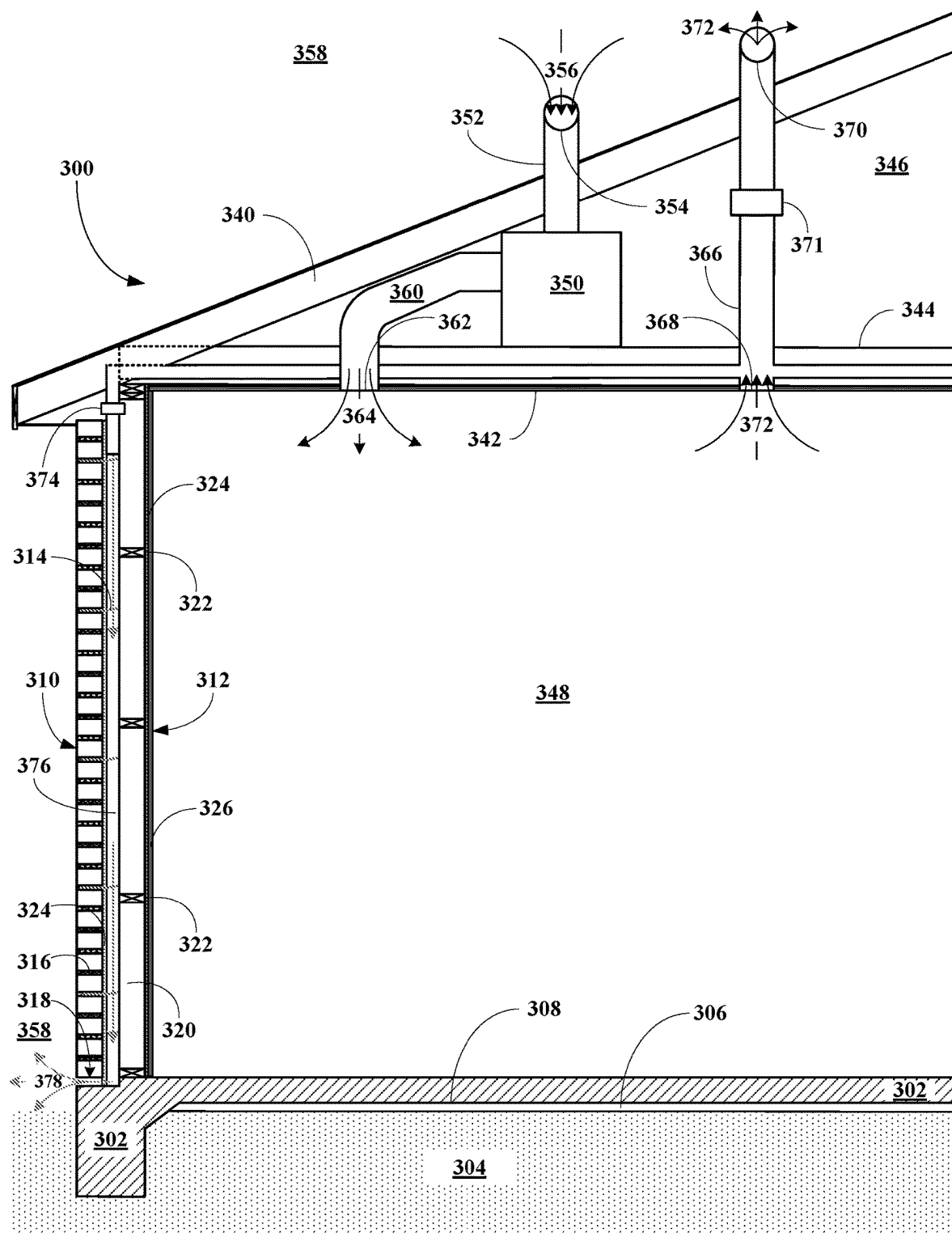


FIG. 3A

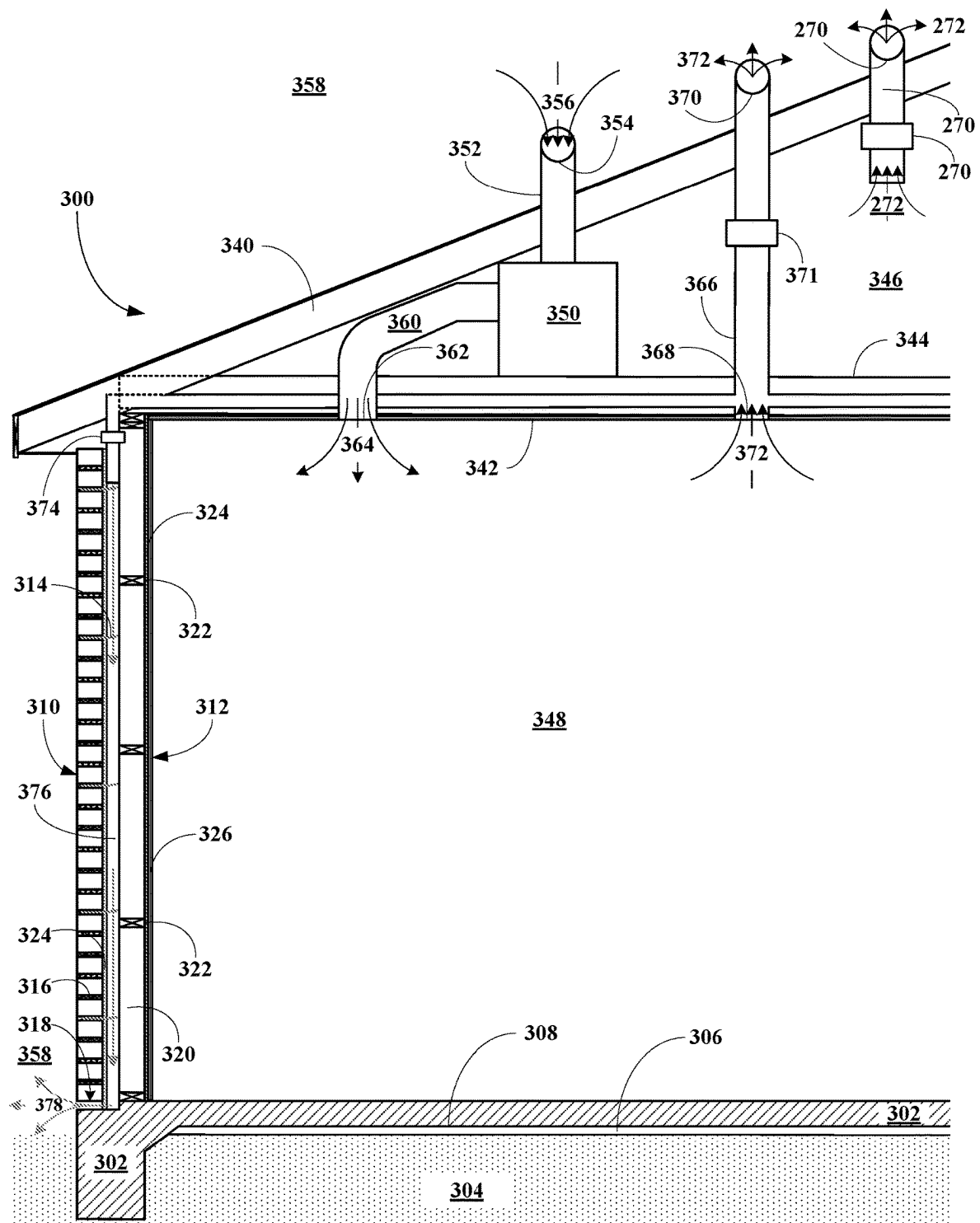


FIG. 3B

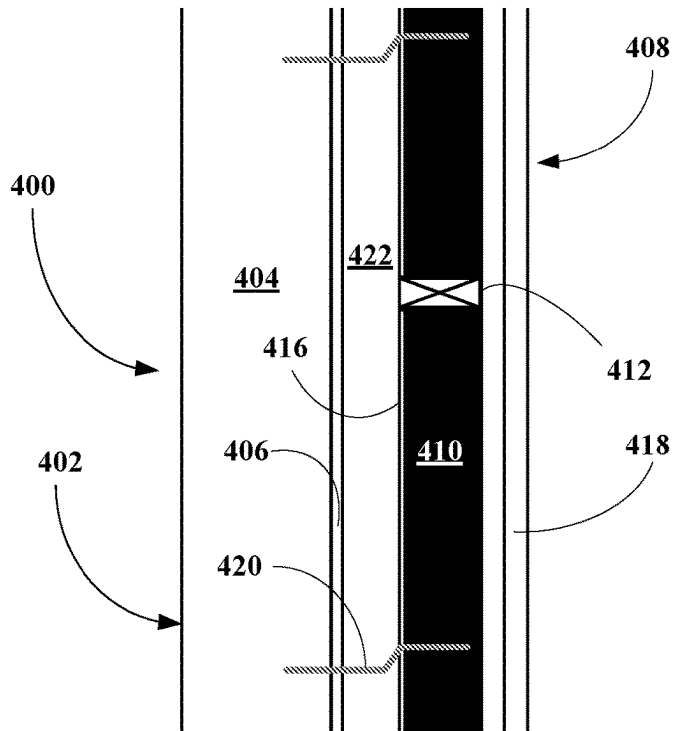


FIG. 4A

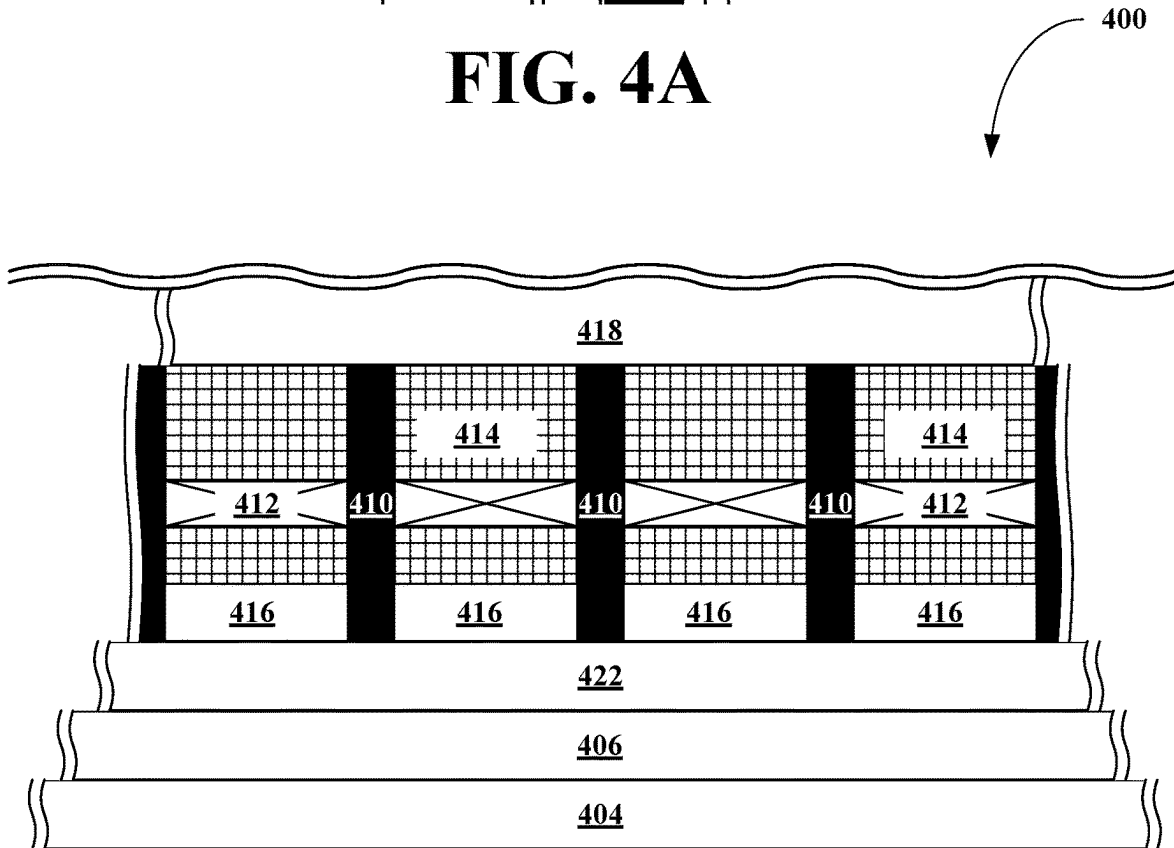


FIG. 4B

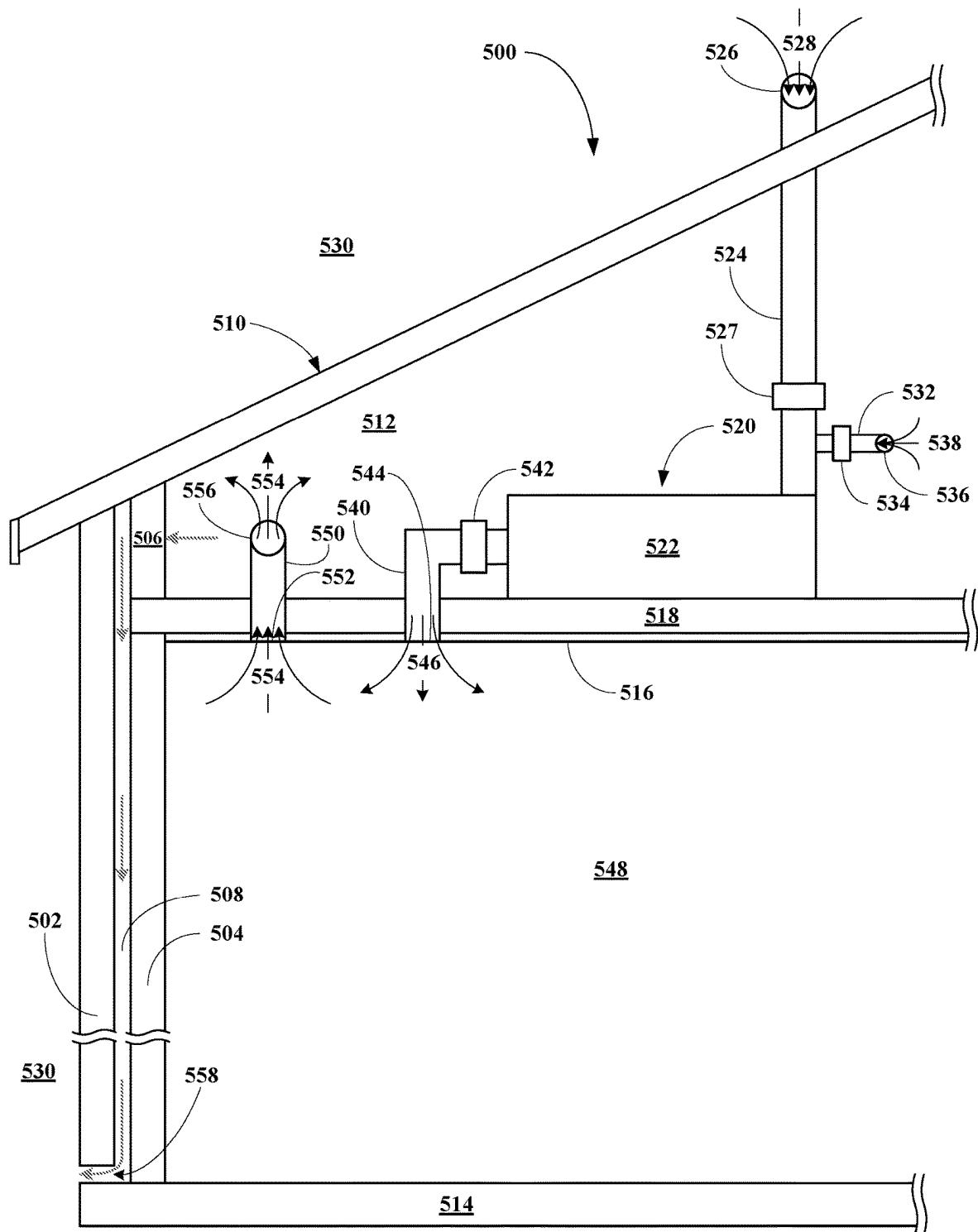


FIG. 5A

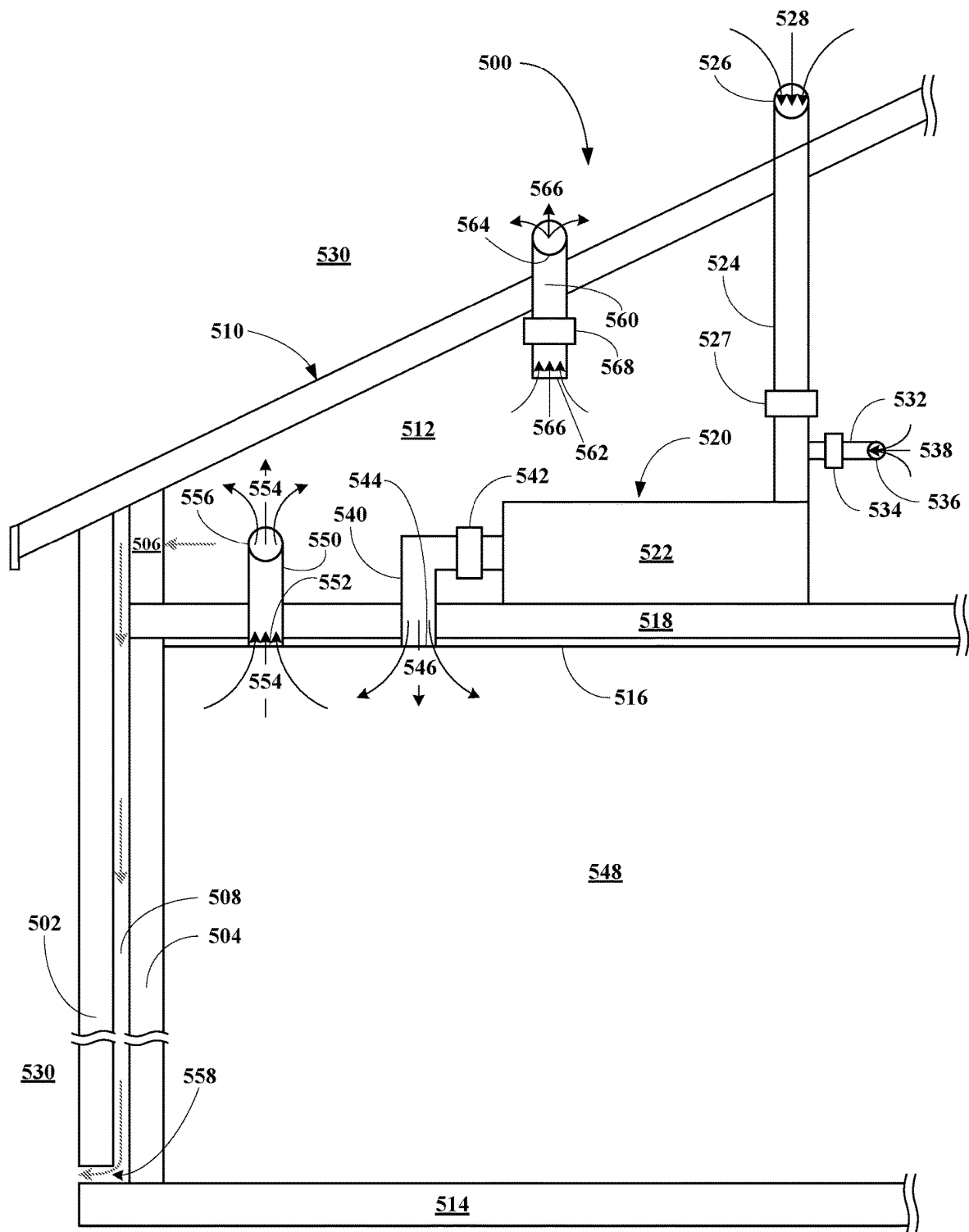


FIG. 5B

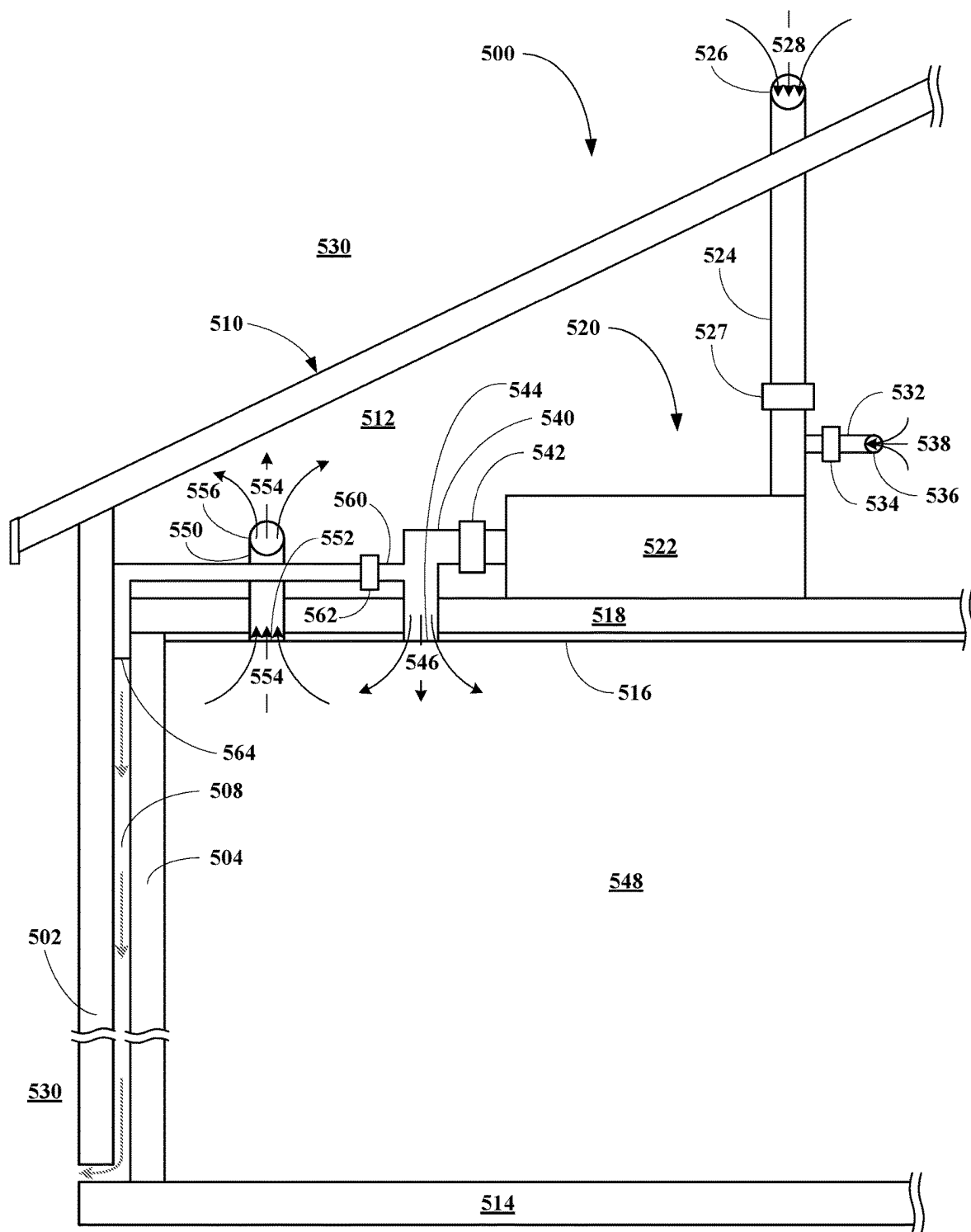


FIG. 5C

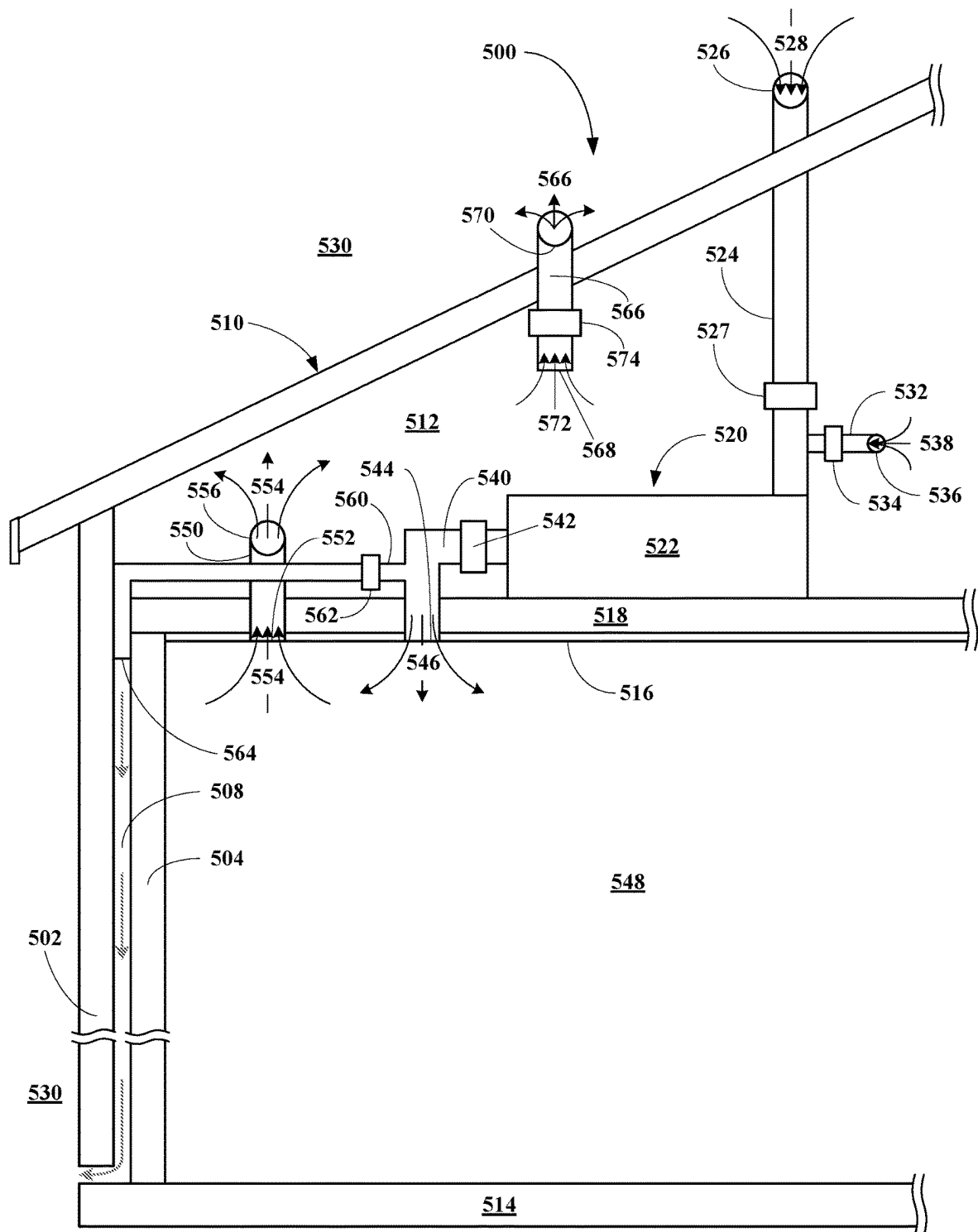
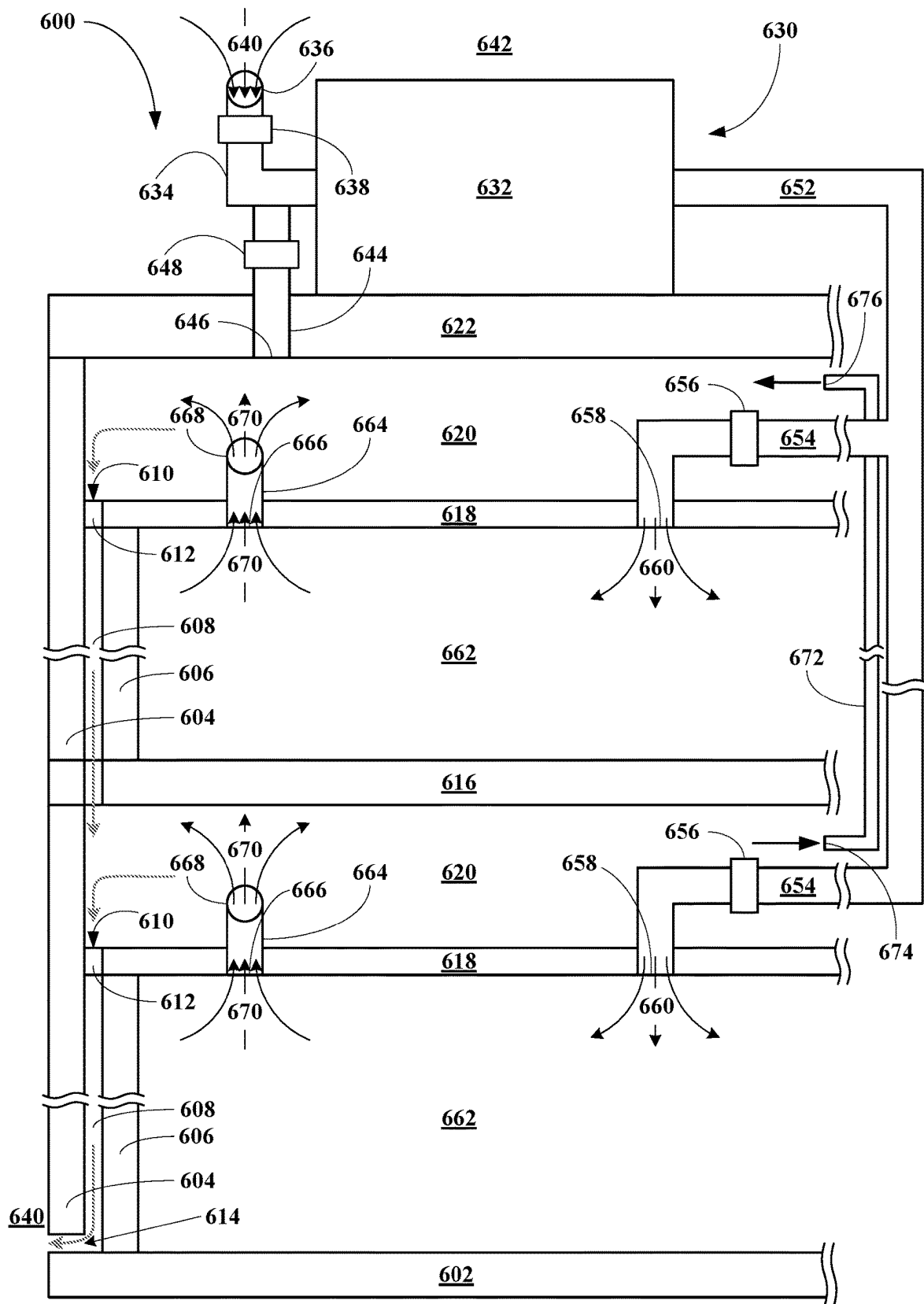


FIG. 5D

**FIG. 6A**

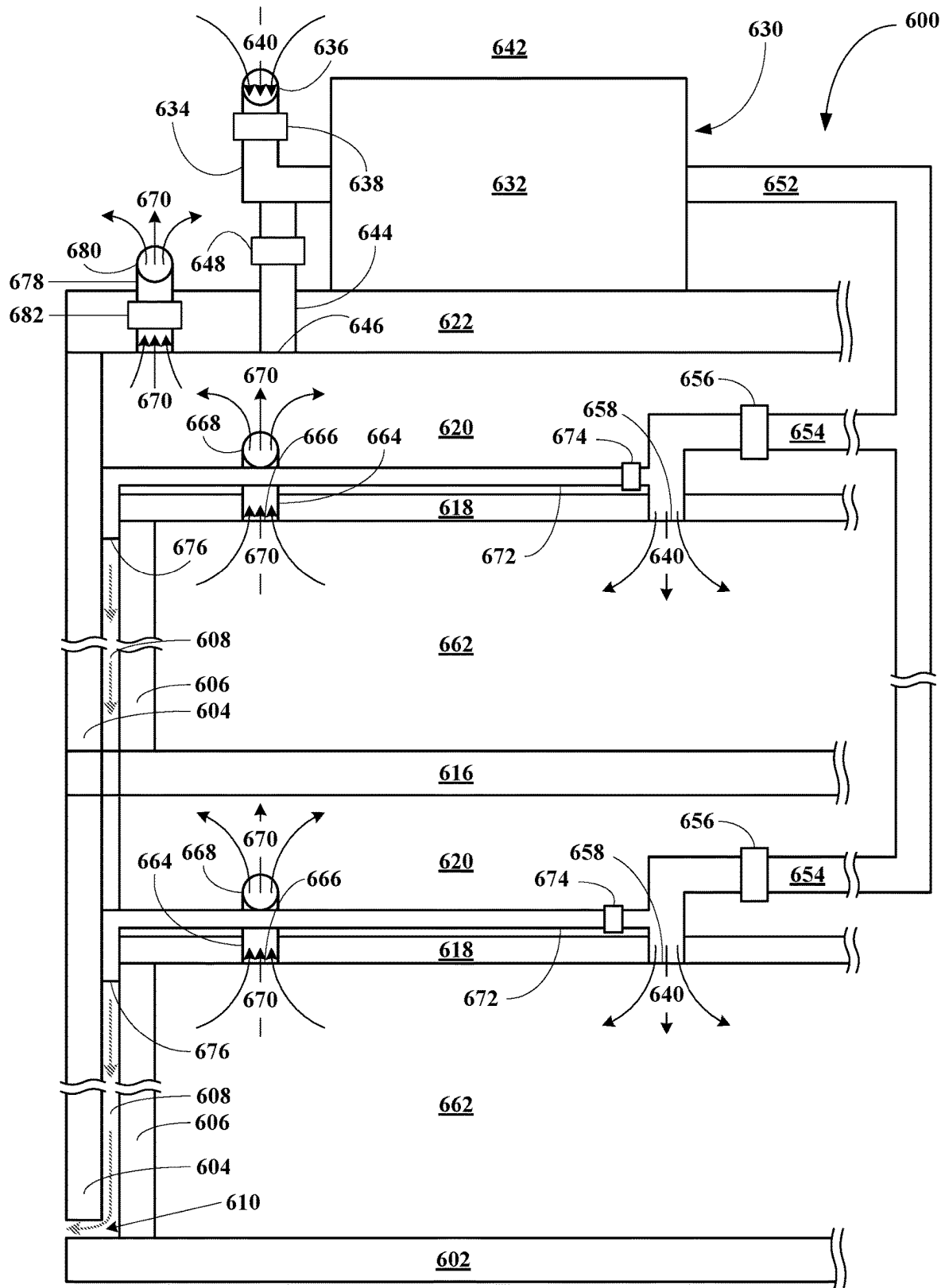


FIG. 6B

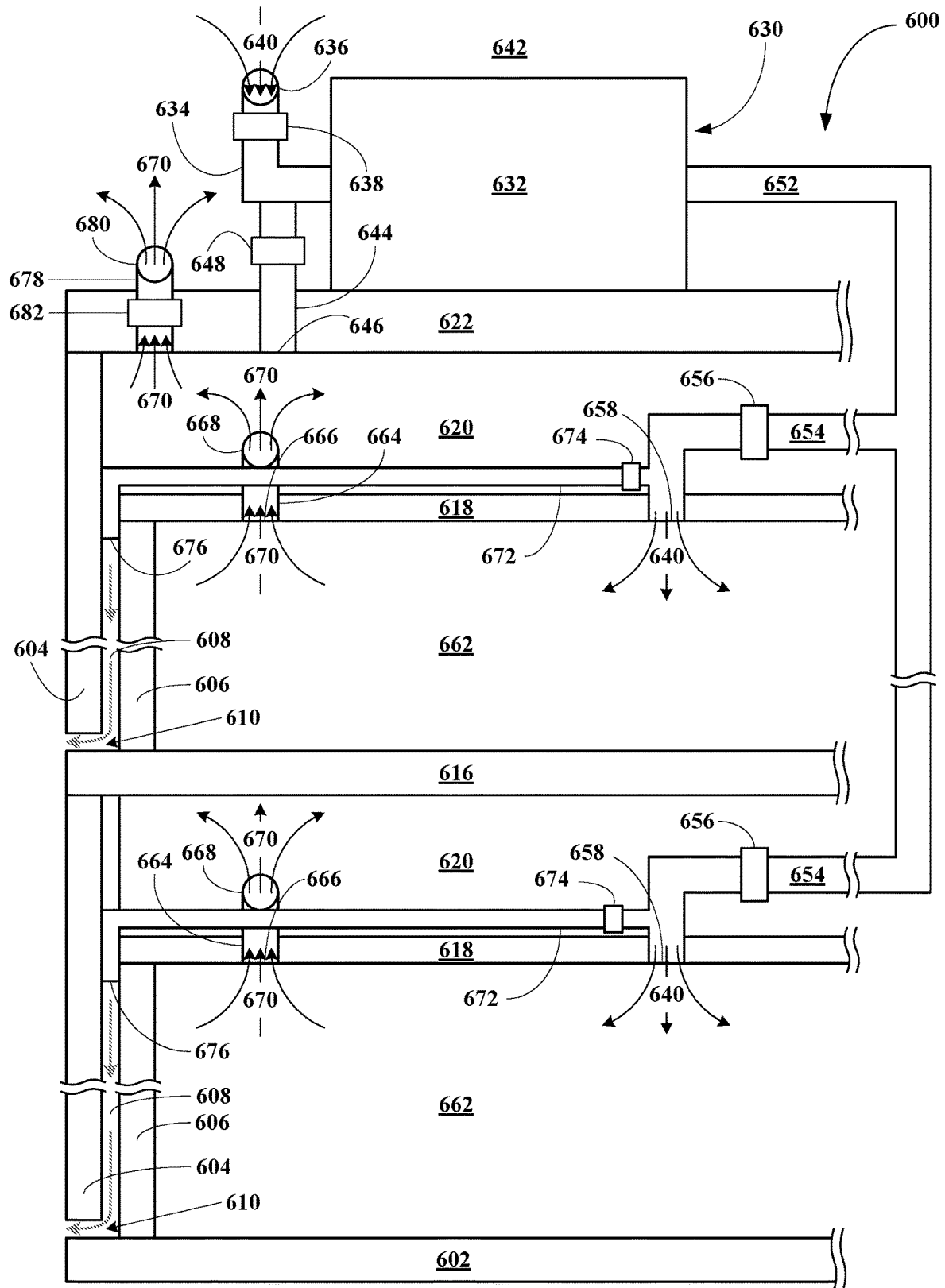


FIG. 6C

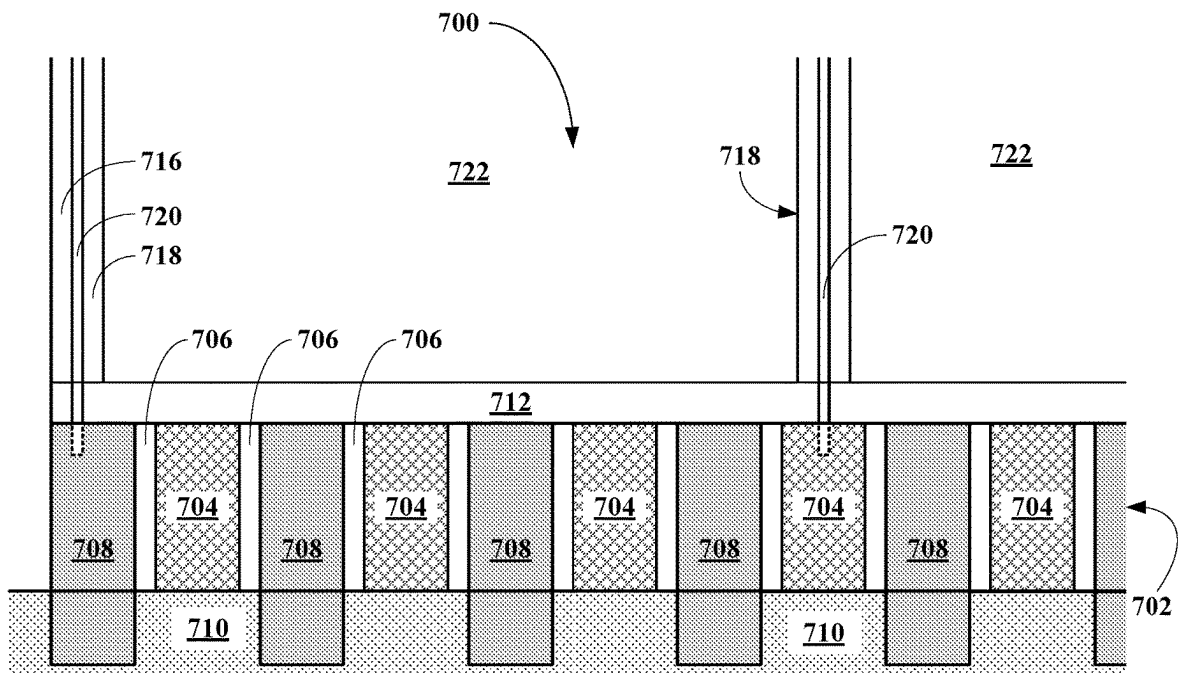


FIG. 7A

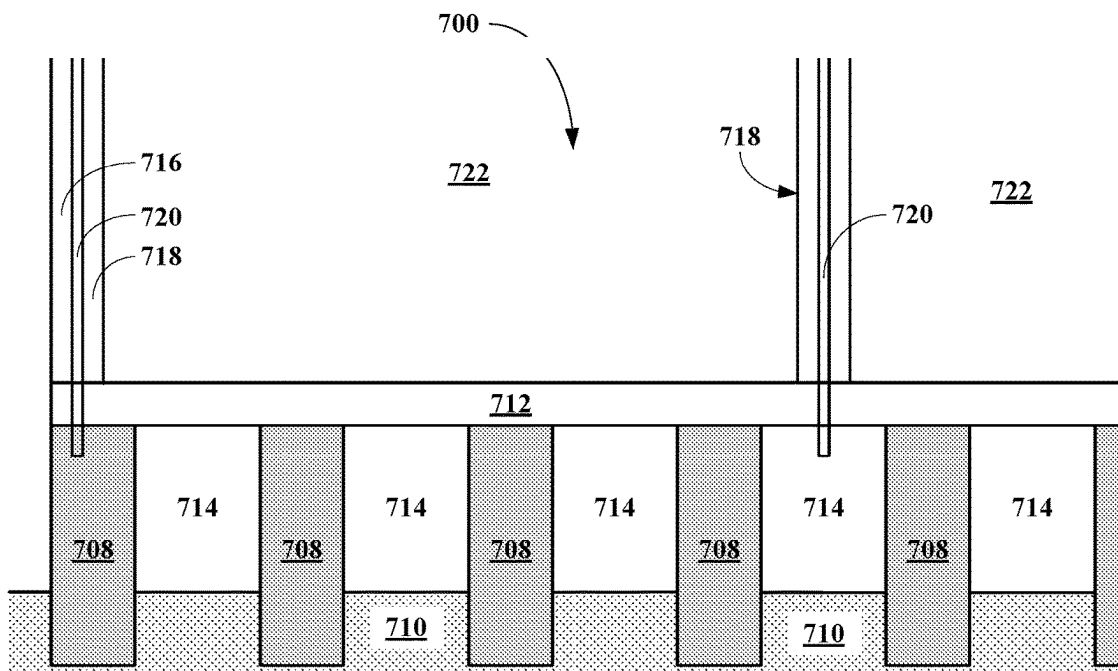


FIG. 7B

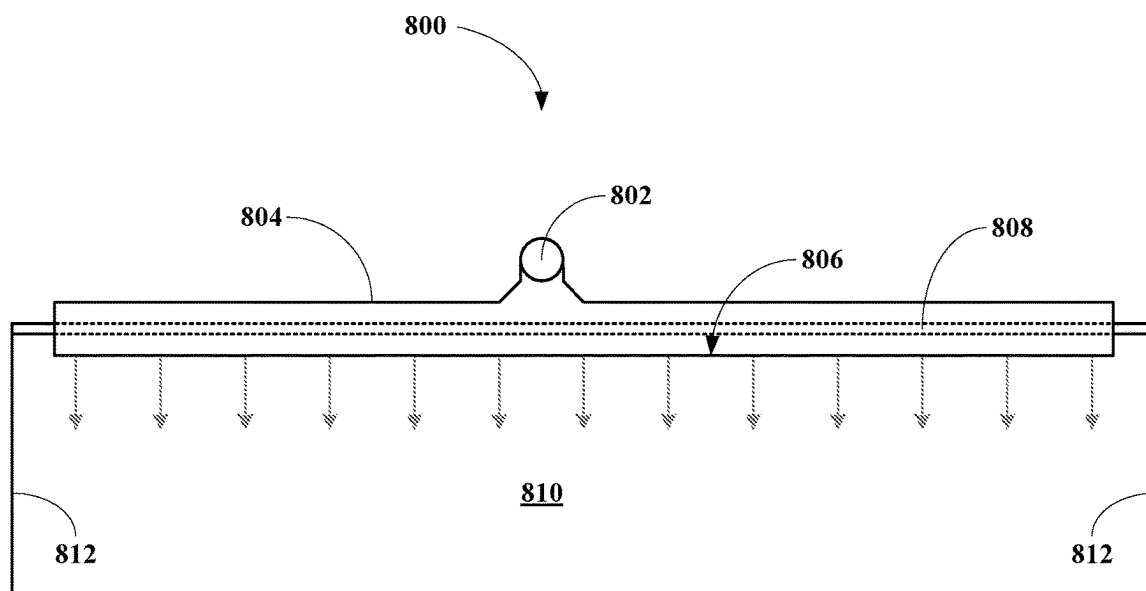


FIG. 8A

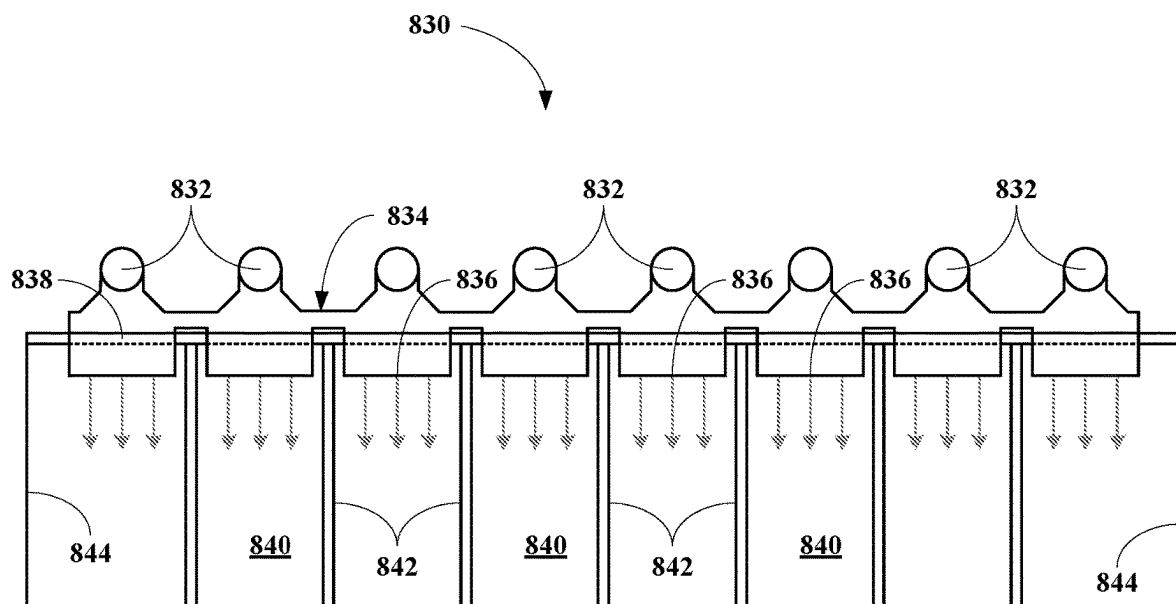


FIG. 8B

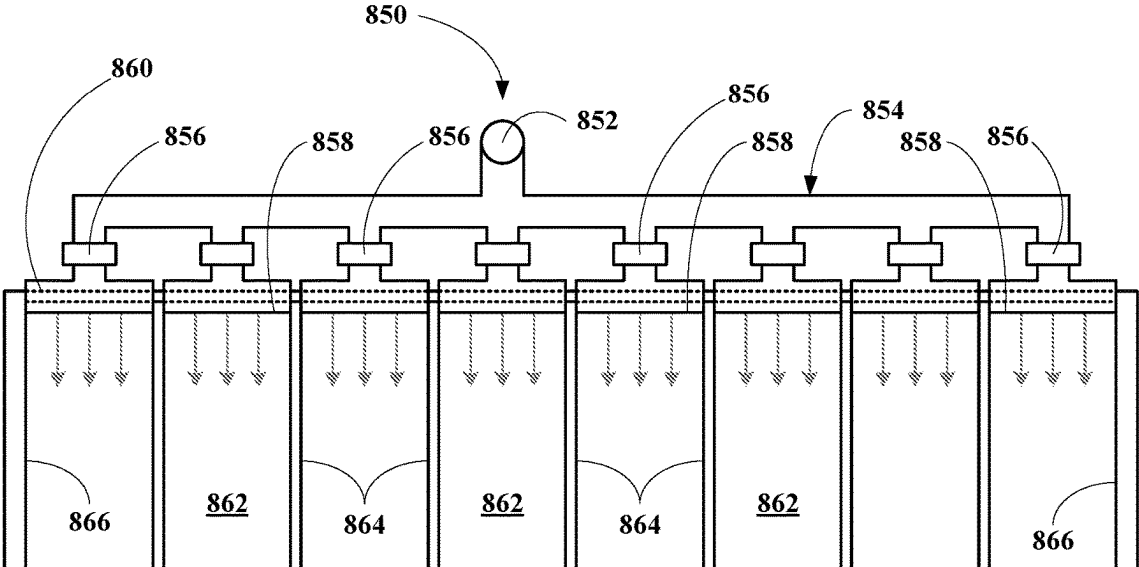


FIG. 8C

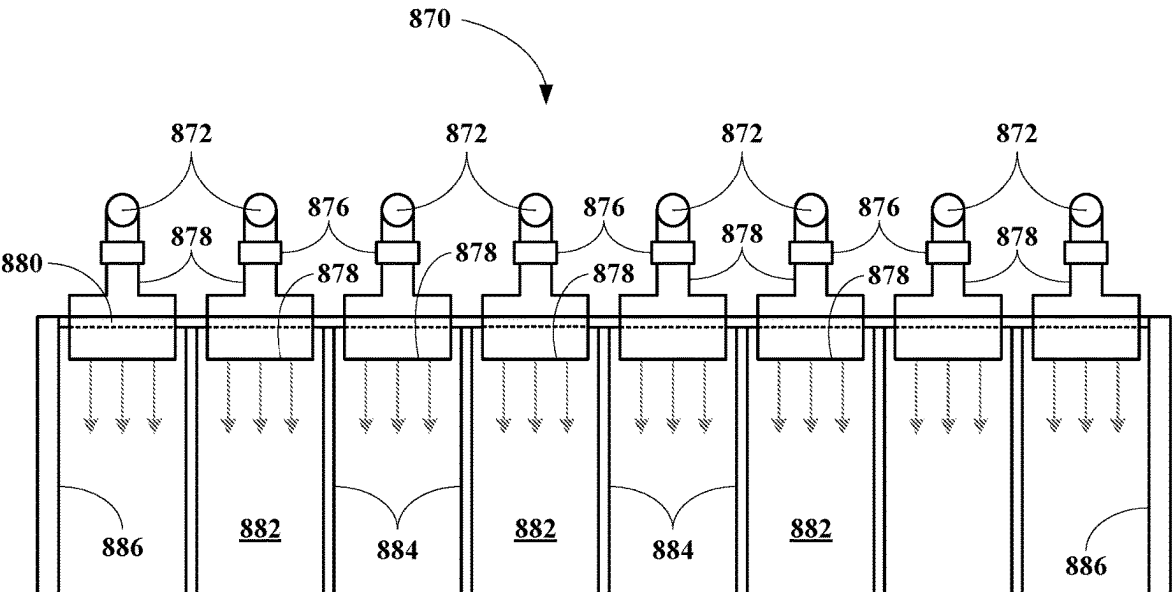


FIG. 8D

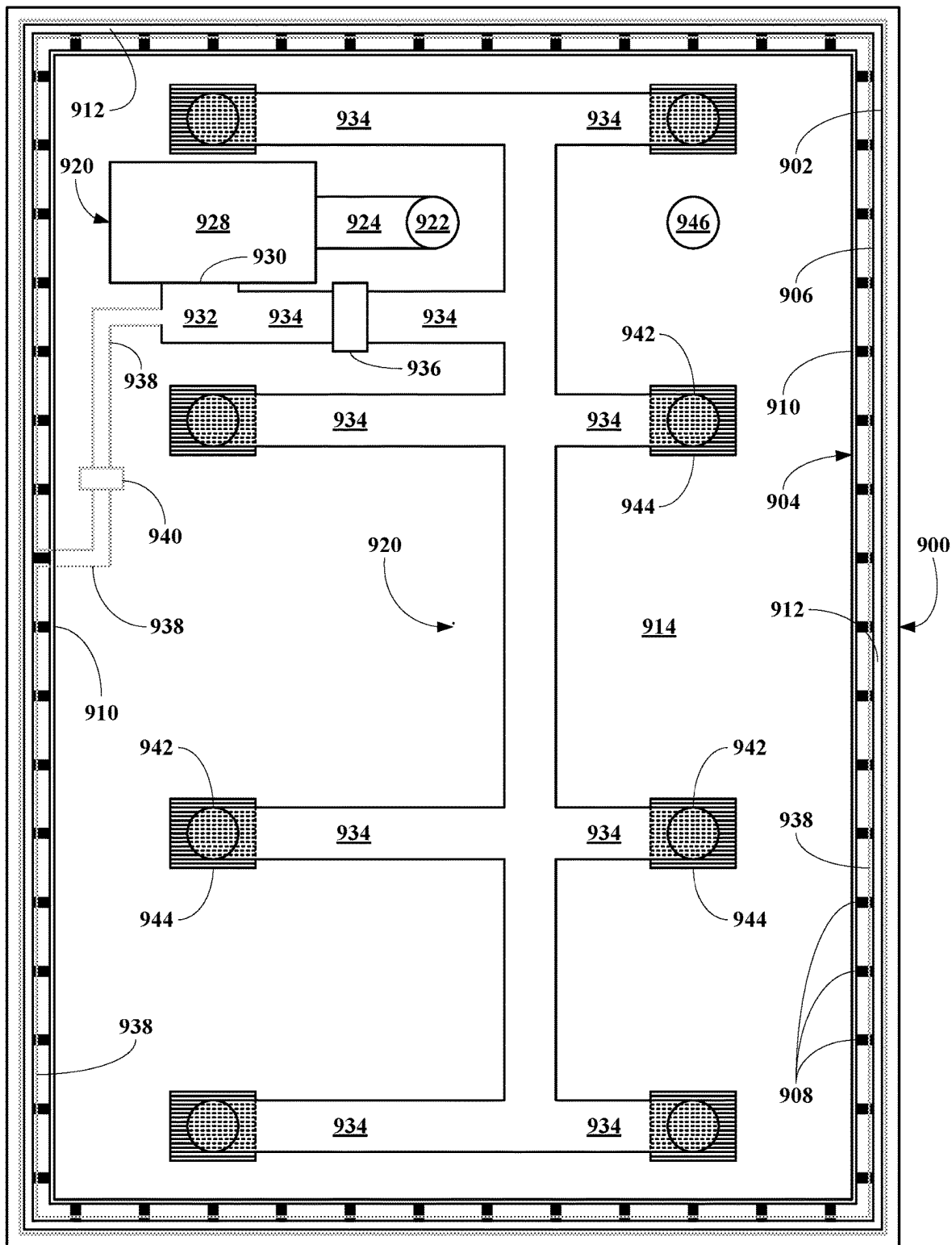
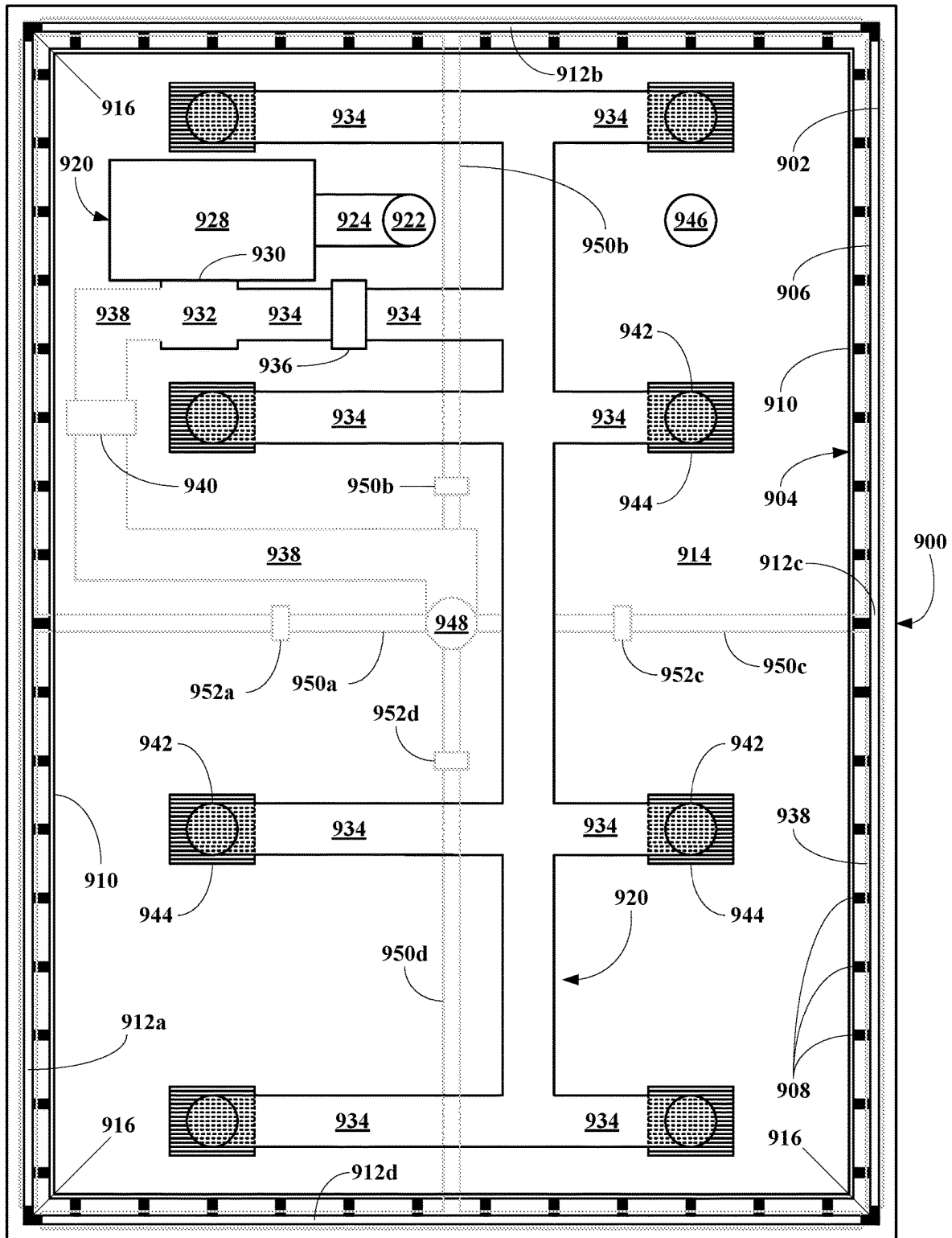
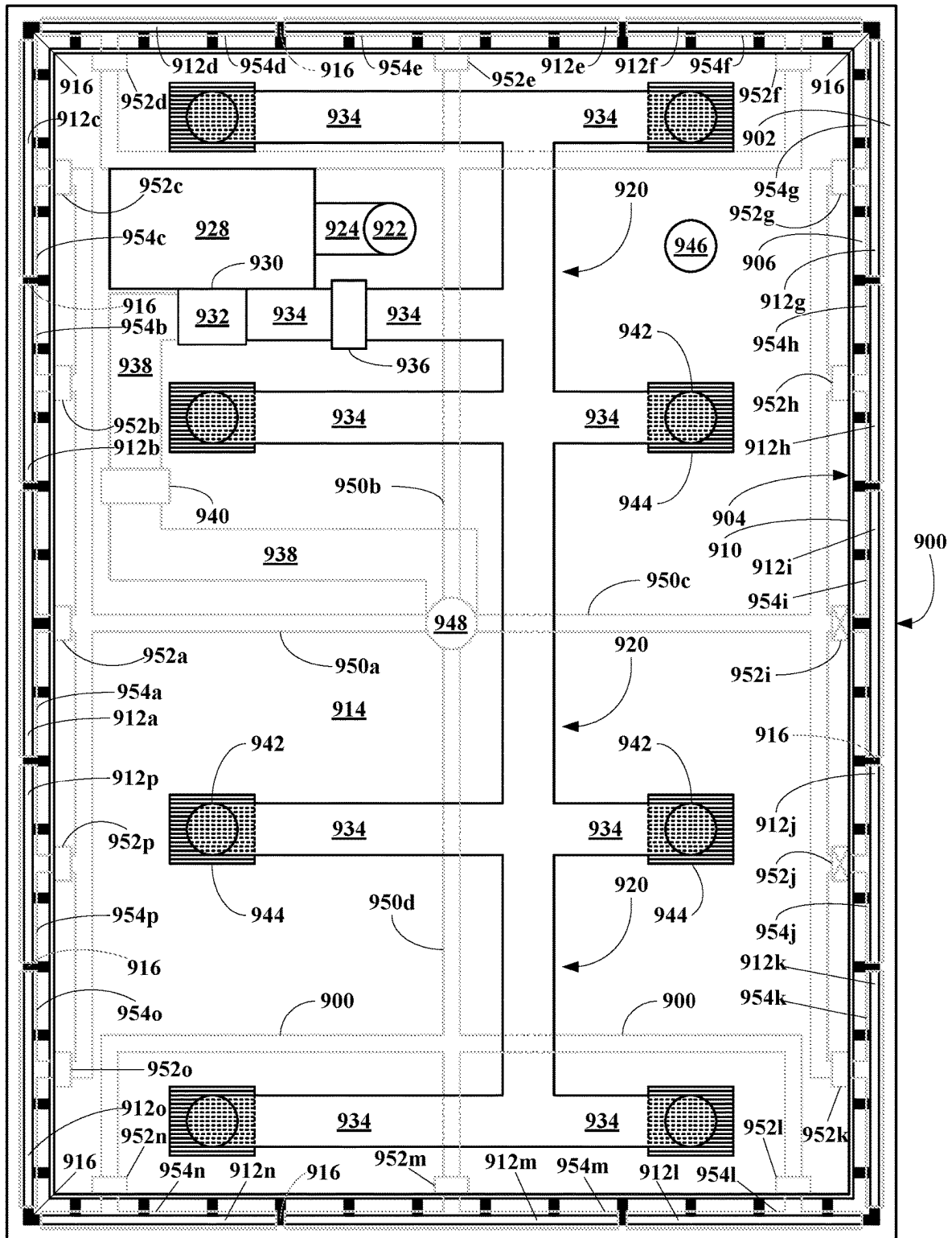


FIG. 9A

**FIG. 9B**

**FIG. 9C**

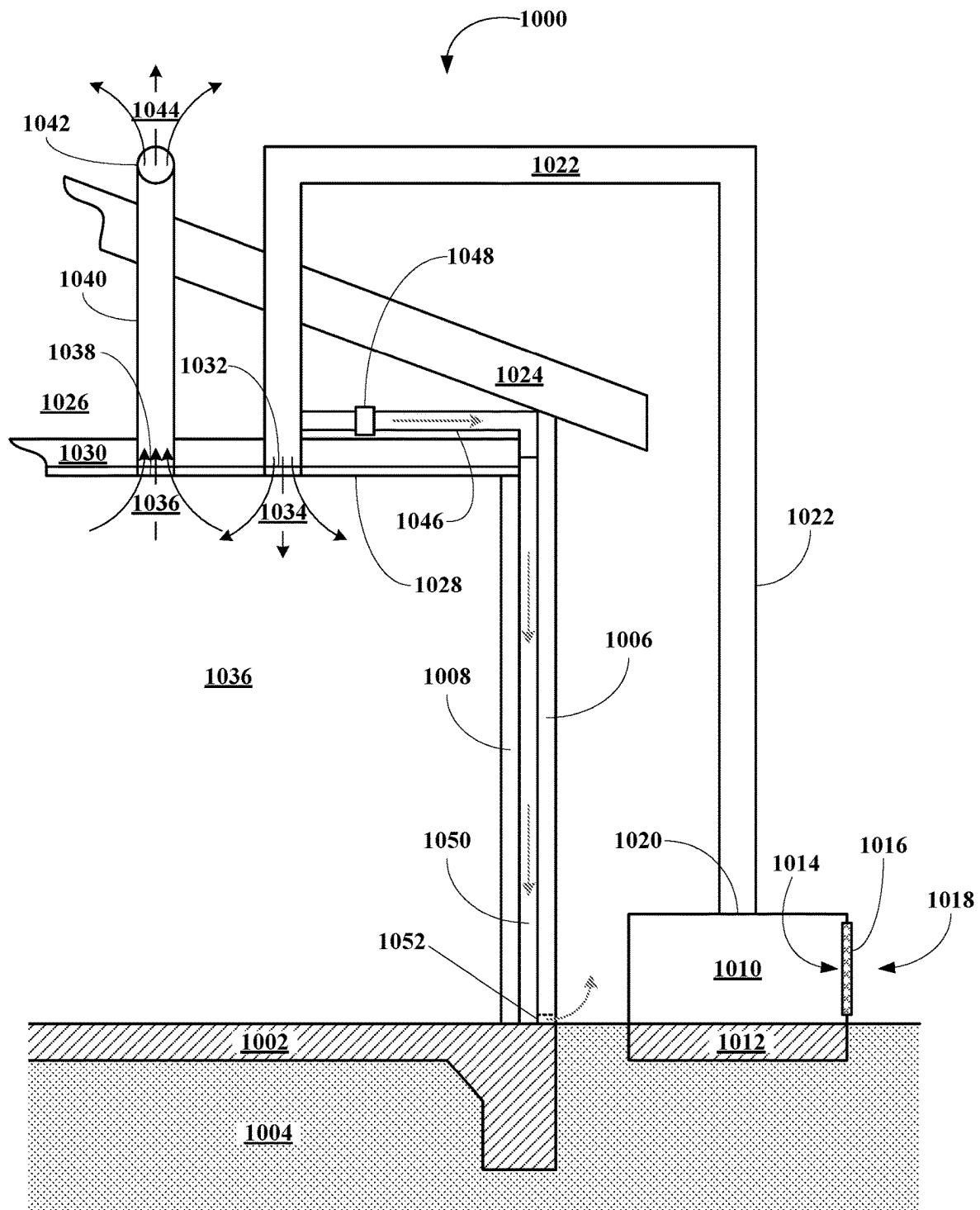


FIG. 10A

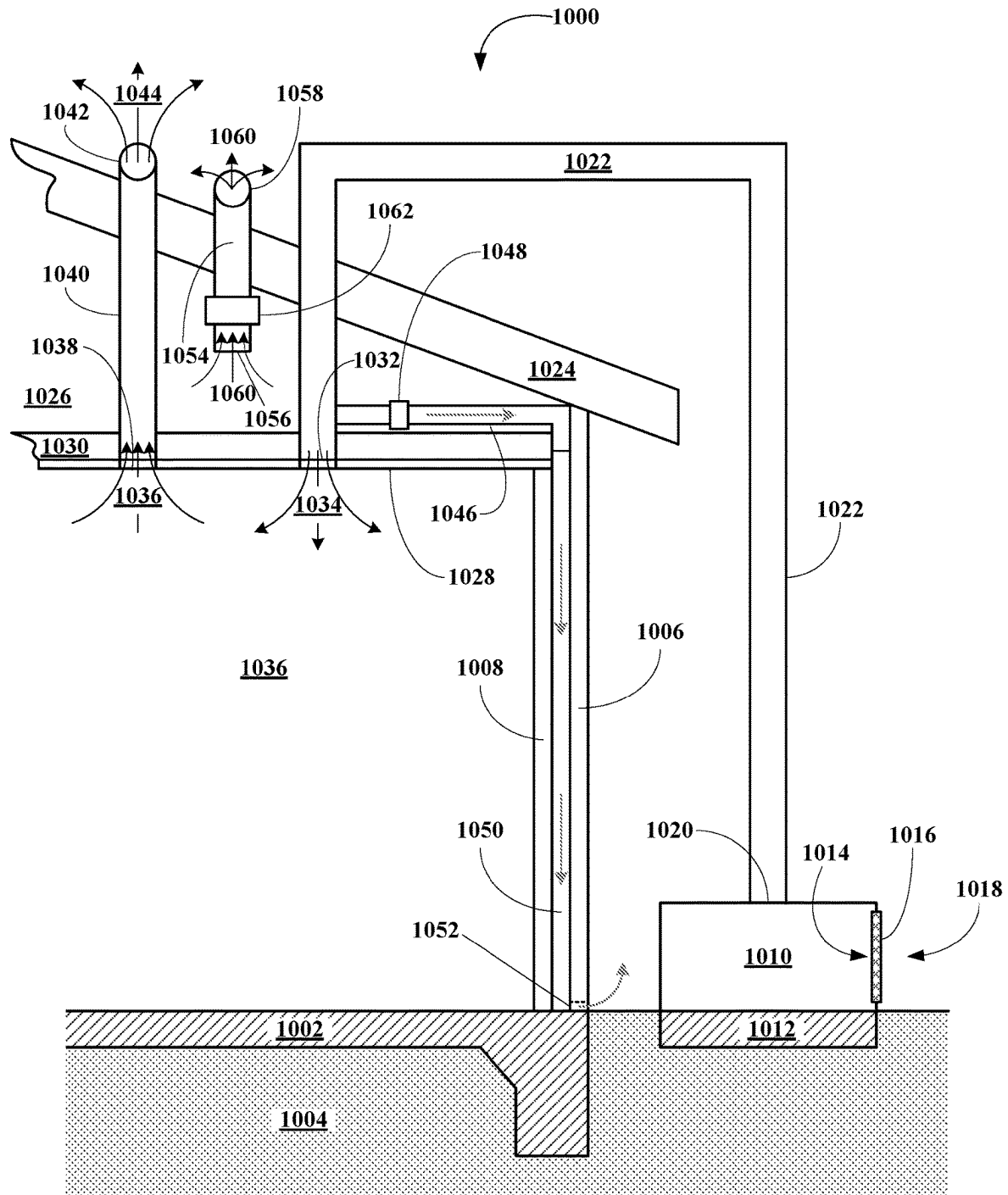


FIG. 10B

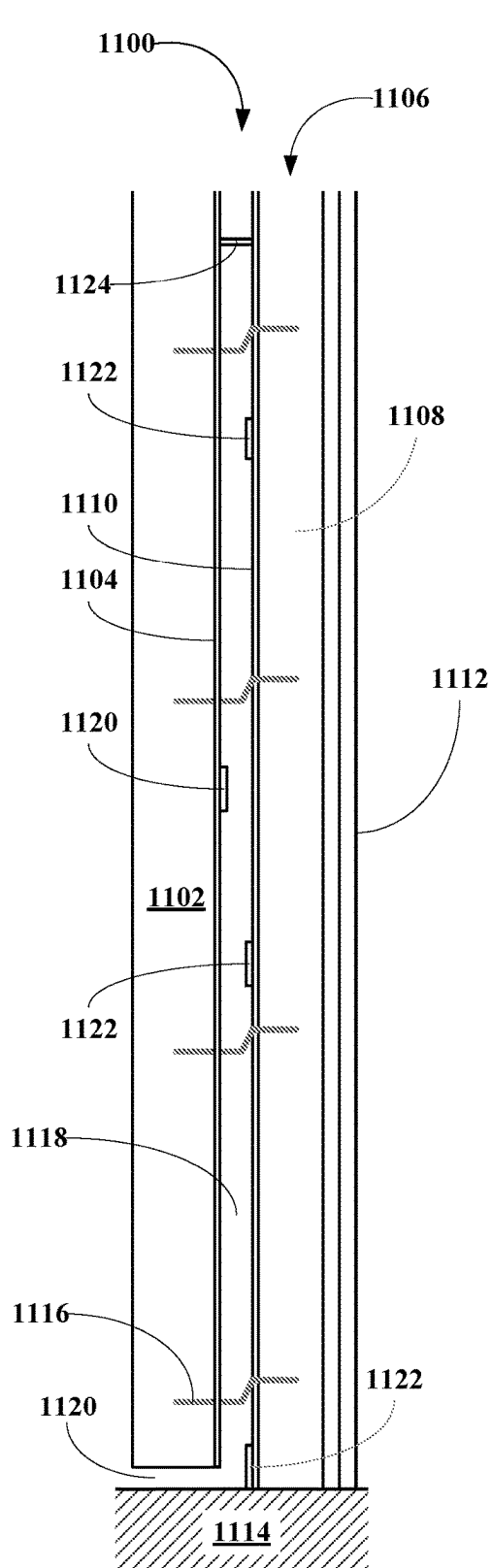


FIG. 11A

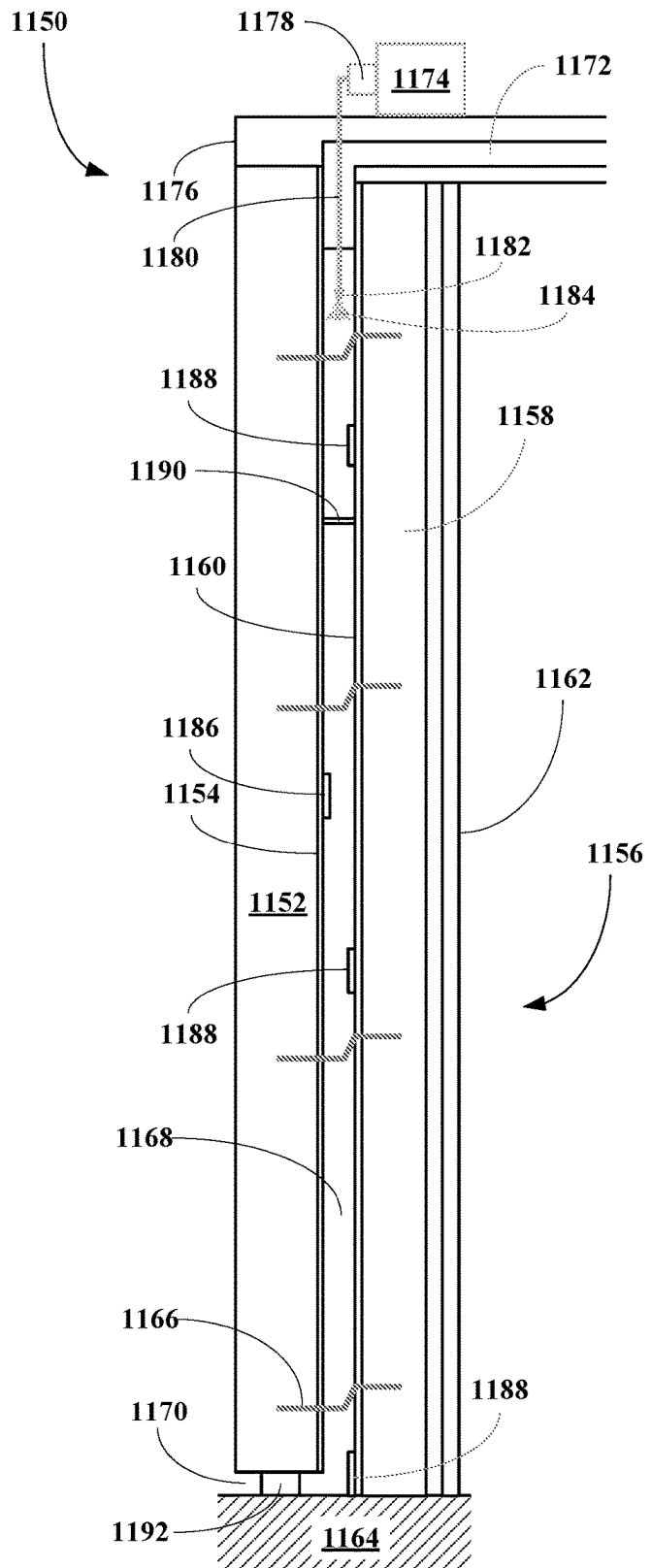


FIG. 11B

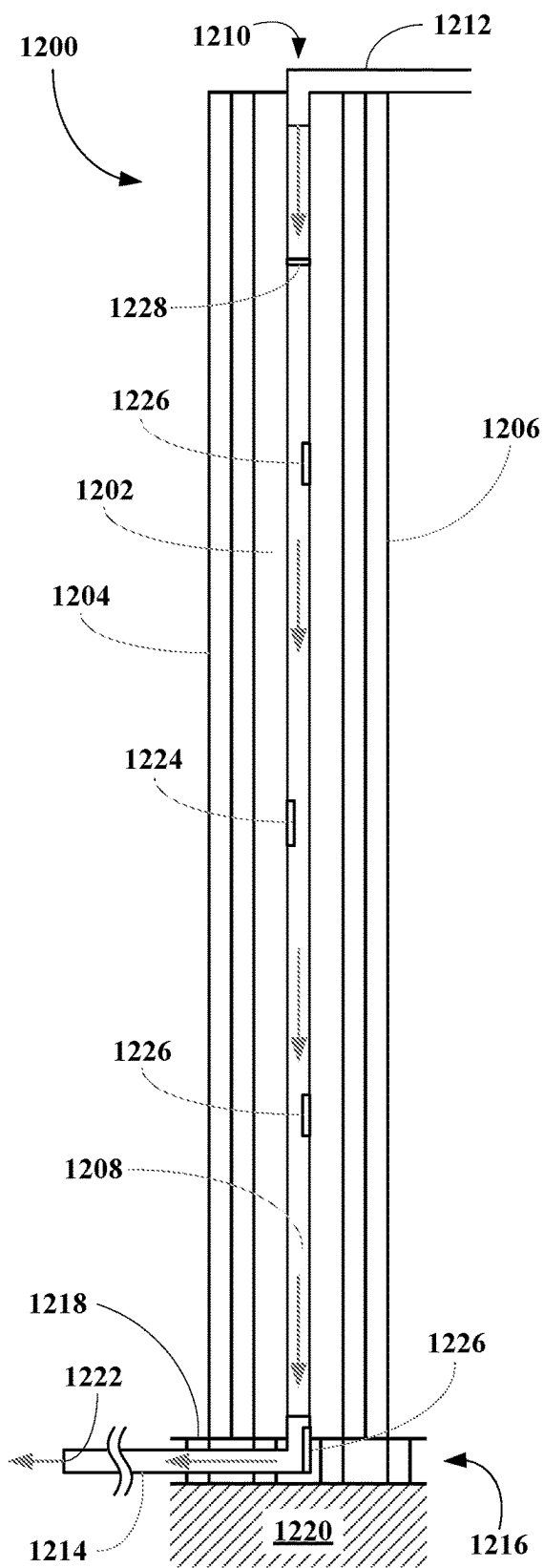


FIG. 12A

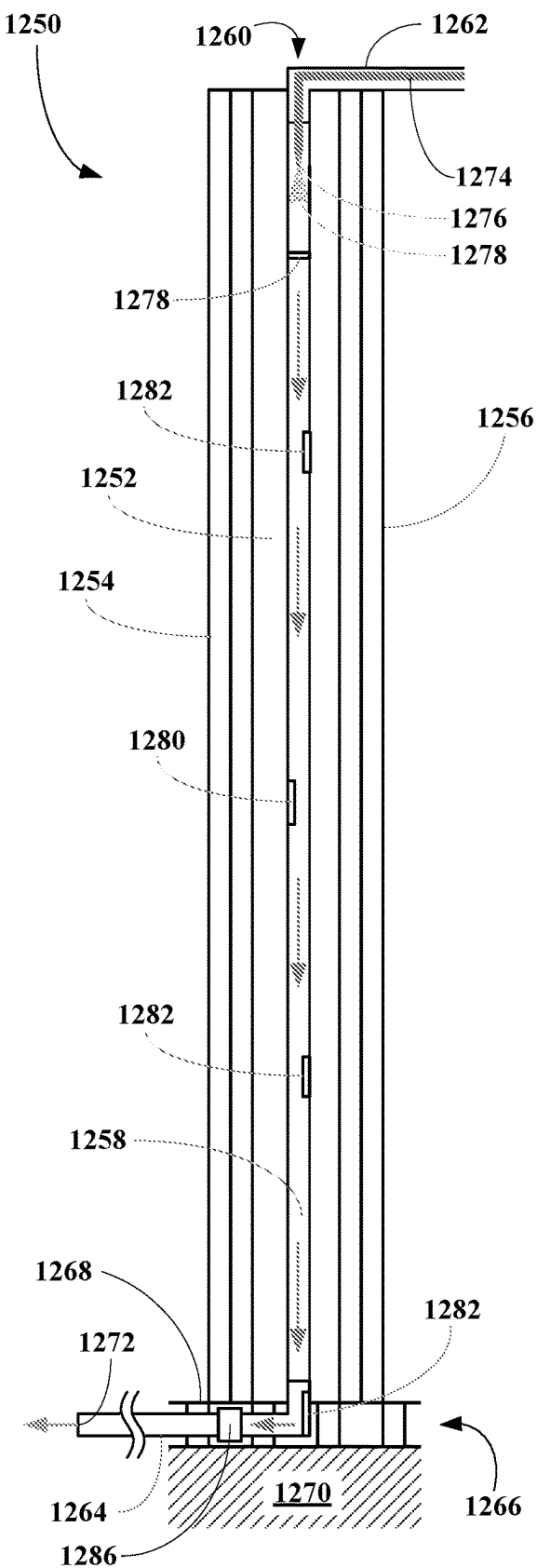


FIG. 12B

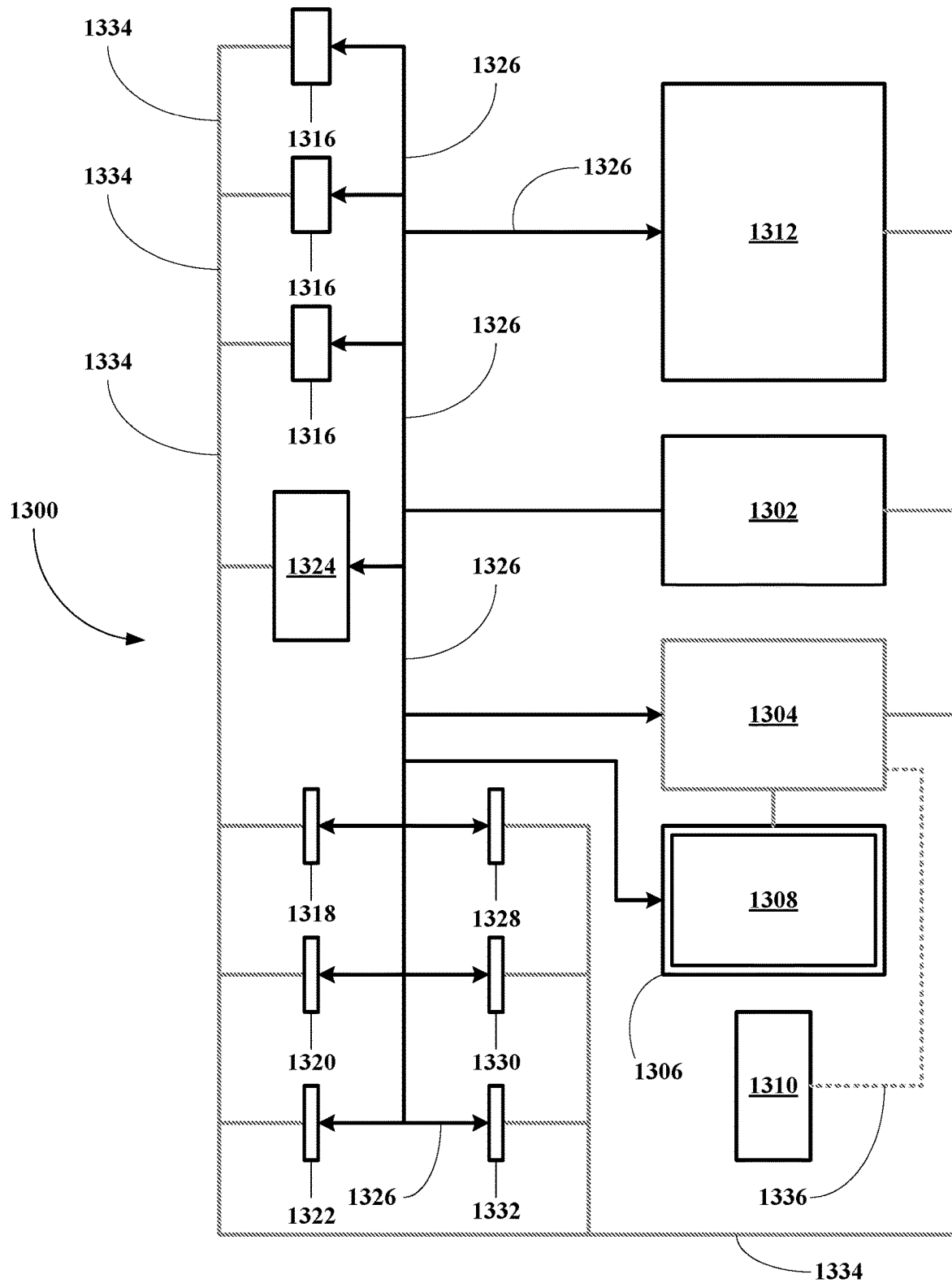
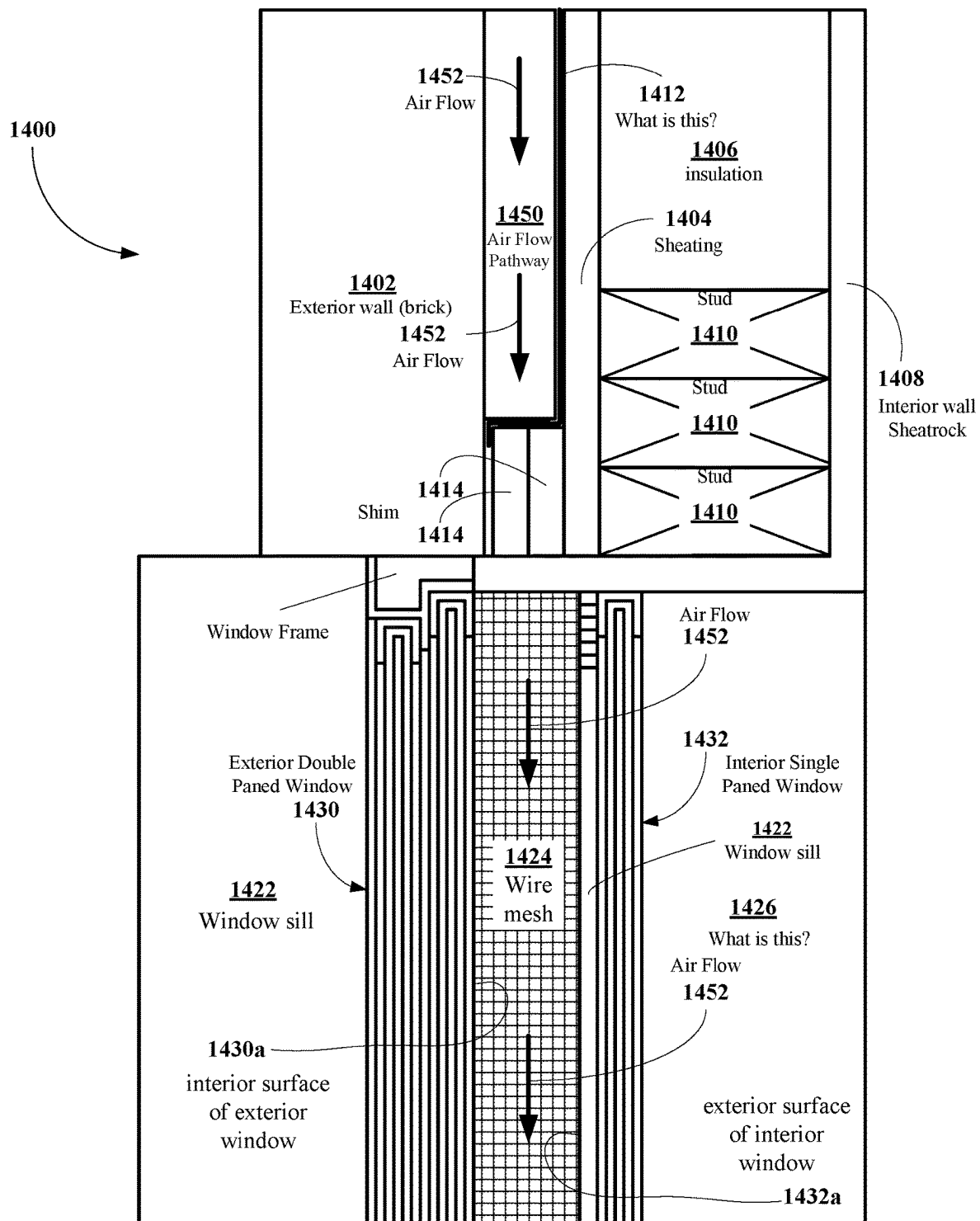


FIG. 13

**FIG. 14A**

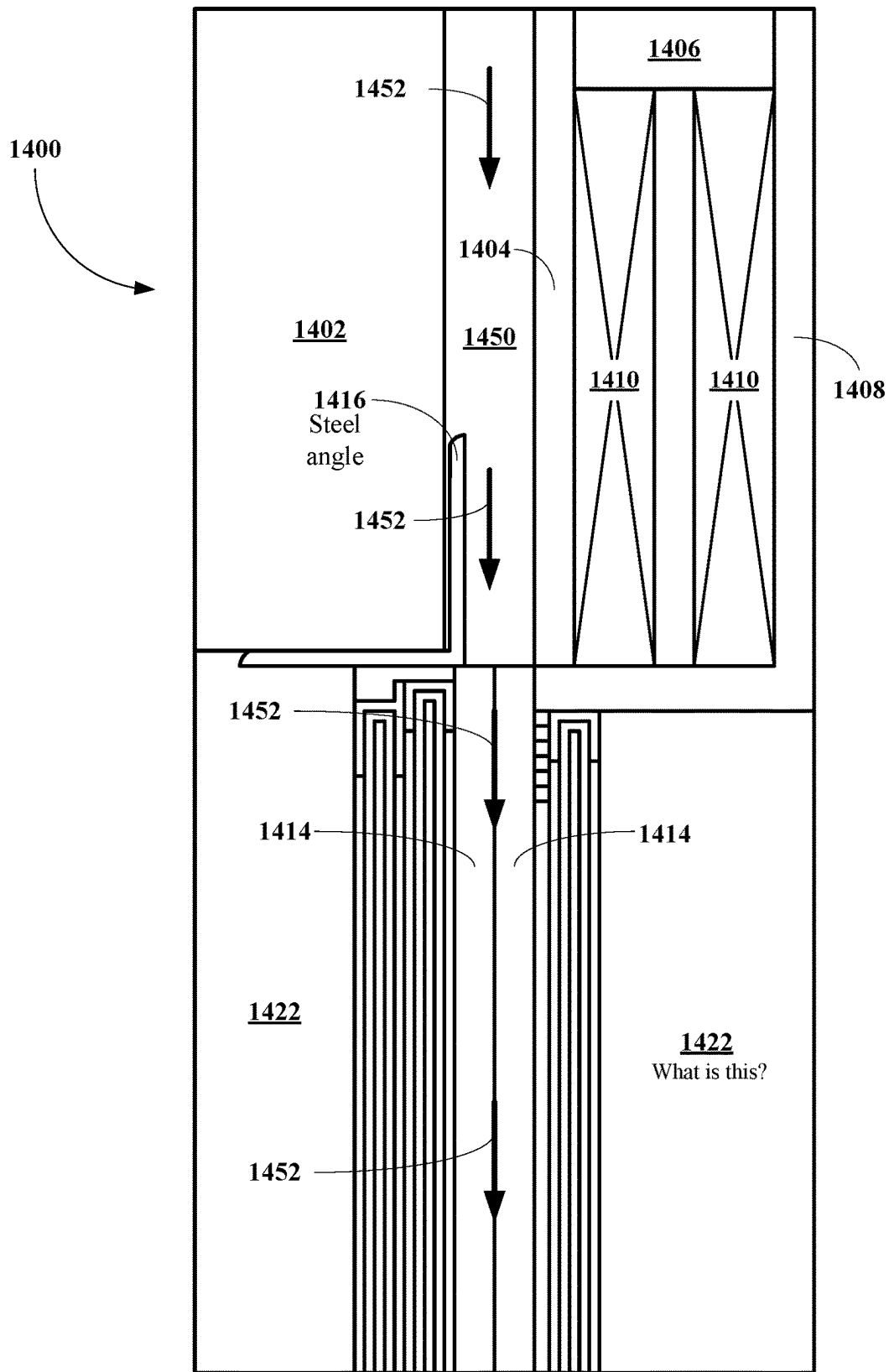


FIG. 14B

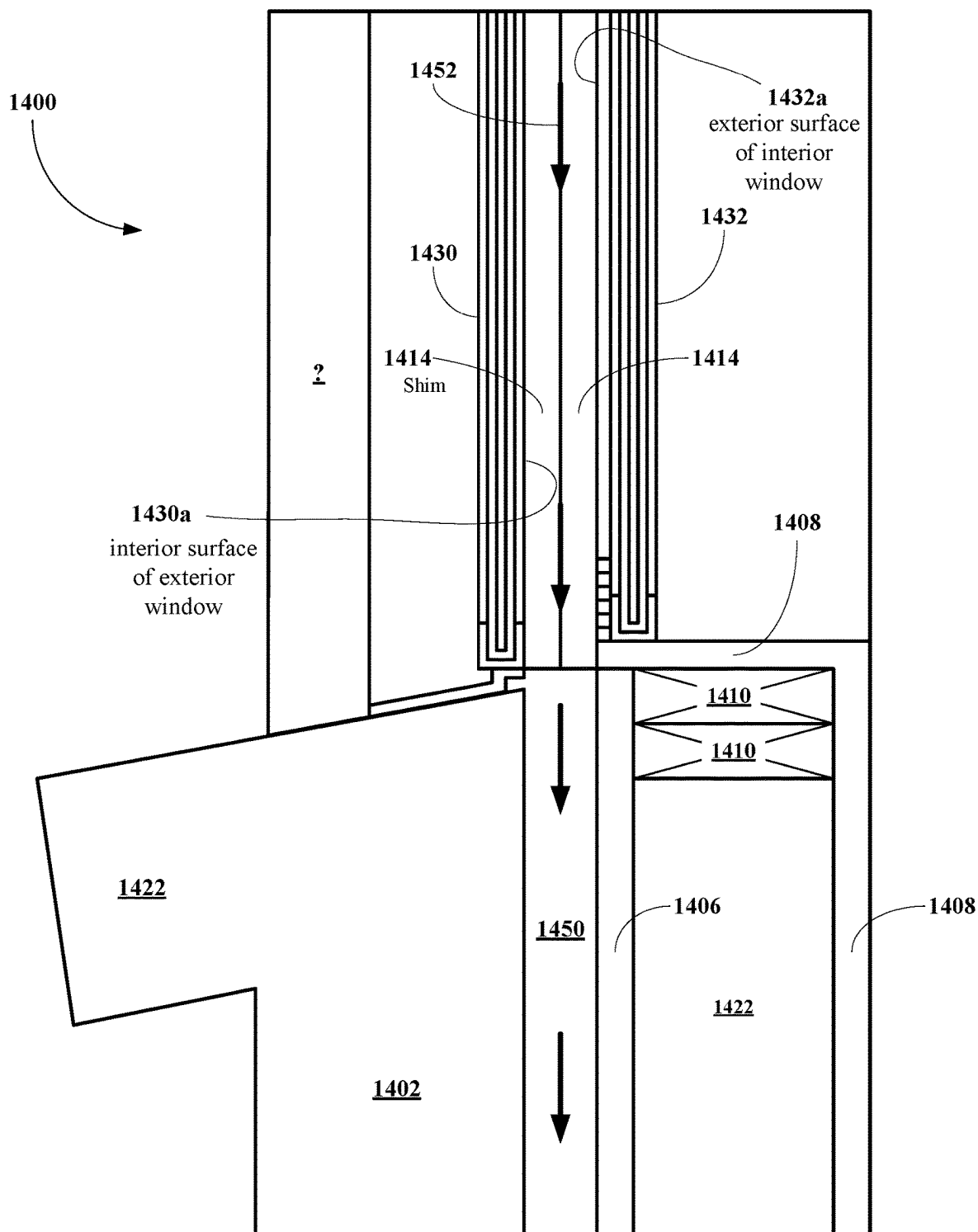


FIG. 14C

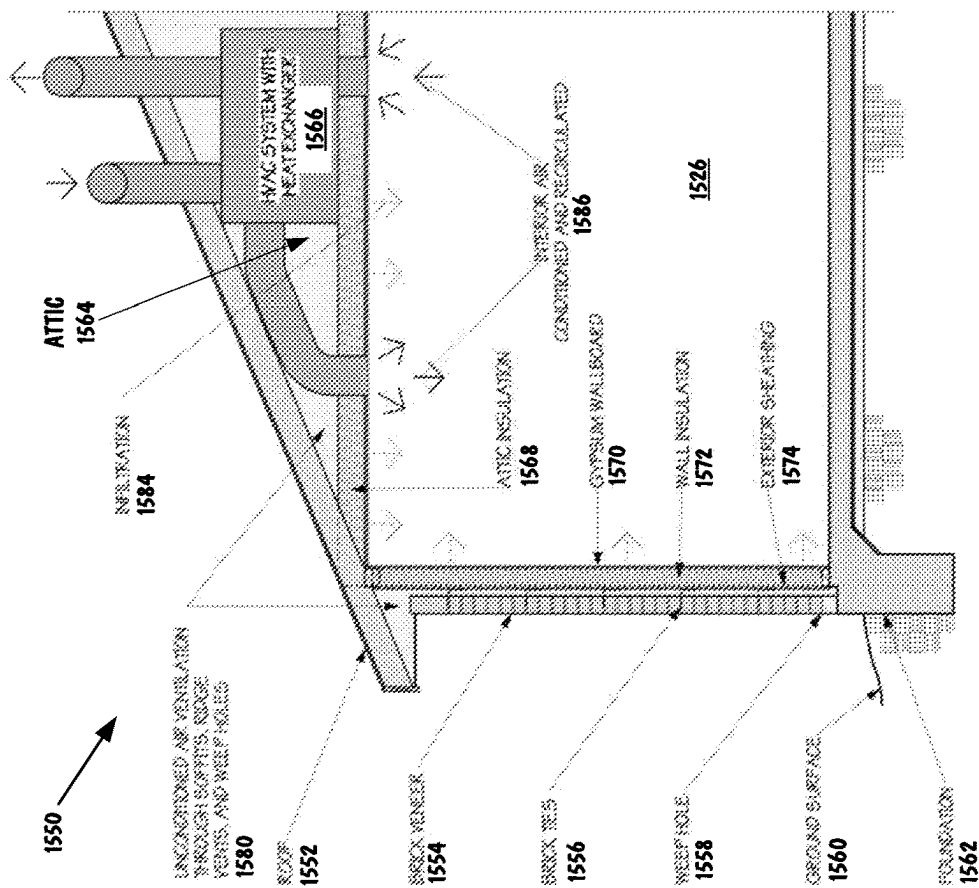


FIG. 15B

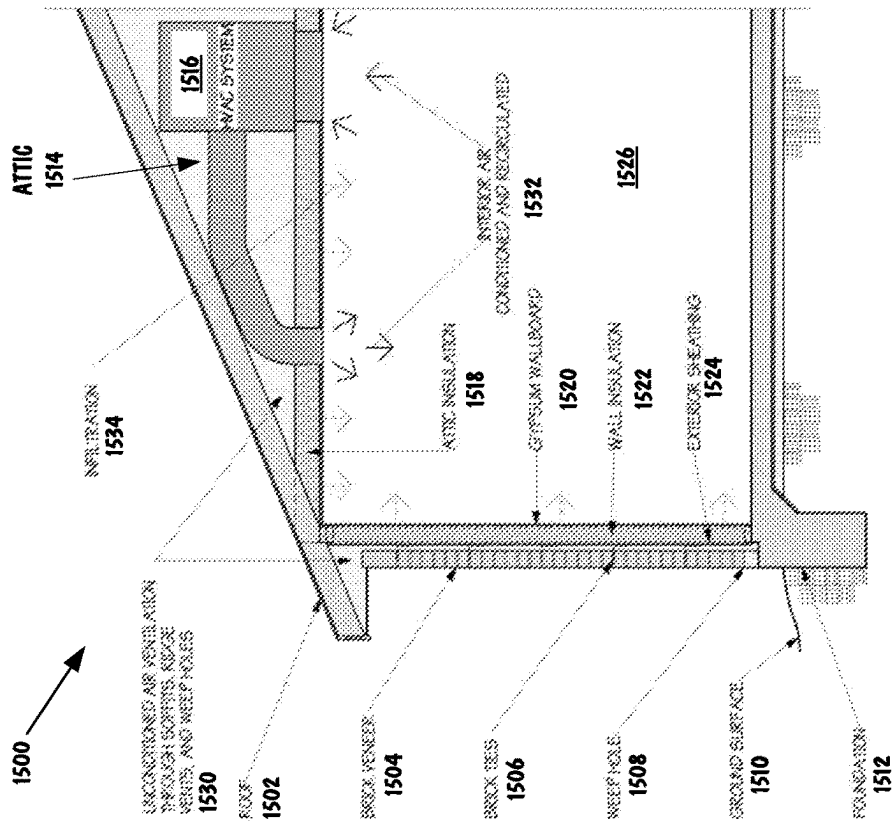


FIG. 15A

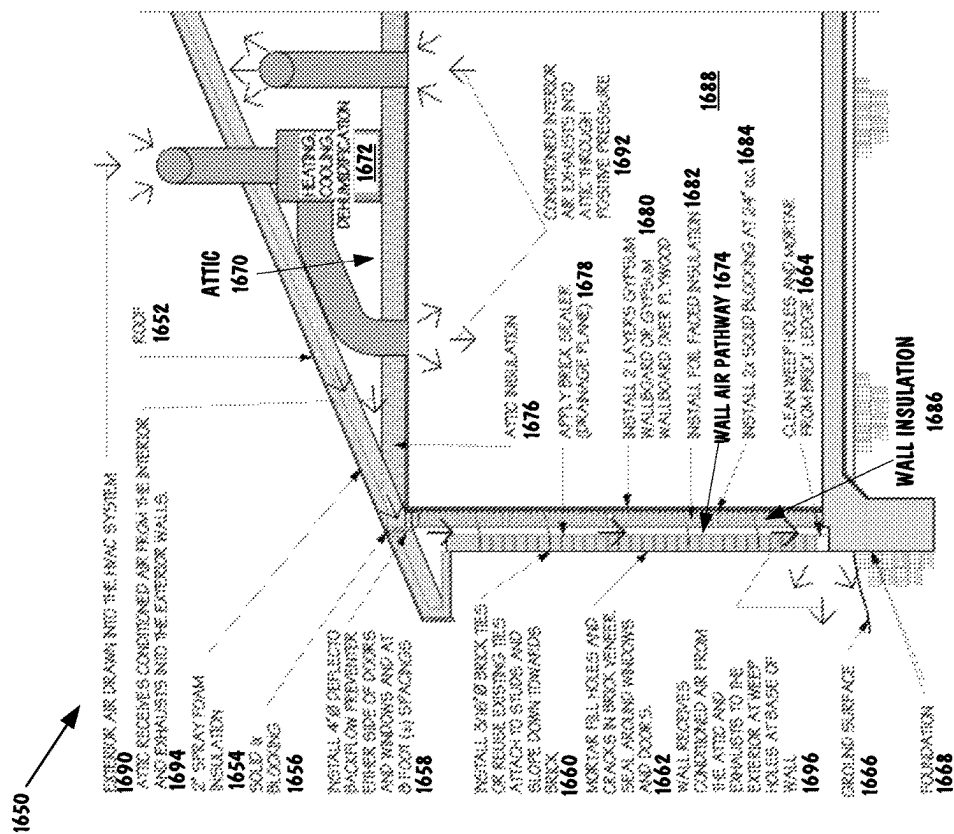


FIG. 16B

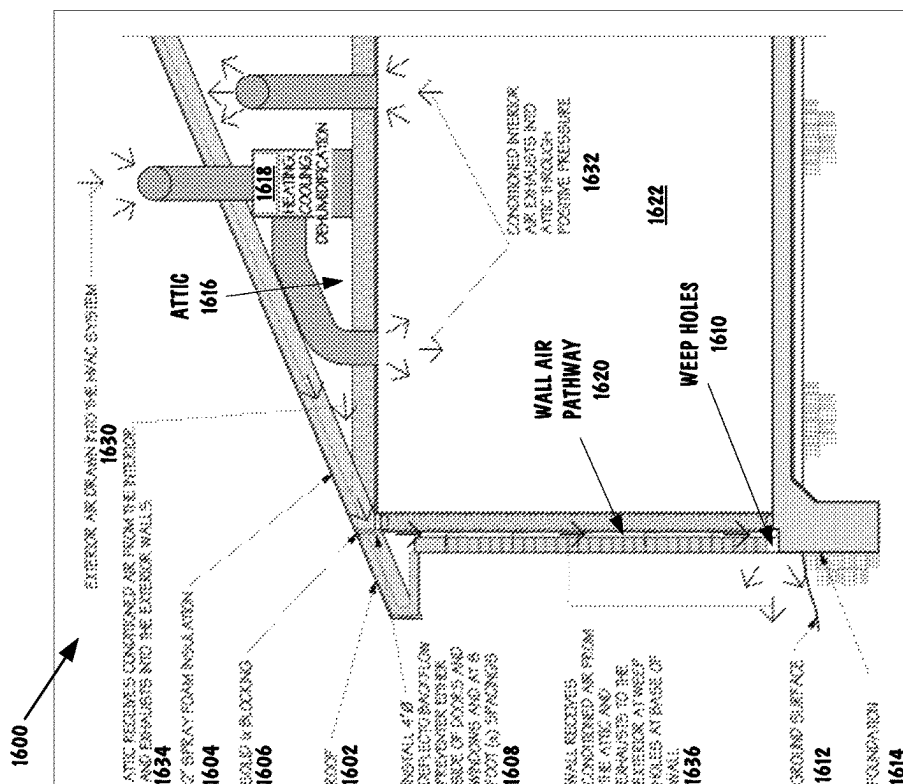


FIG. 16A

1700 ↗

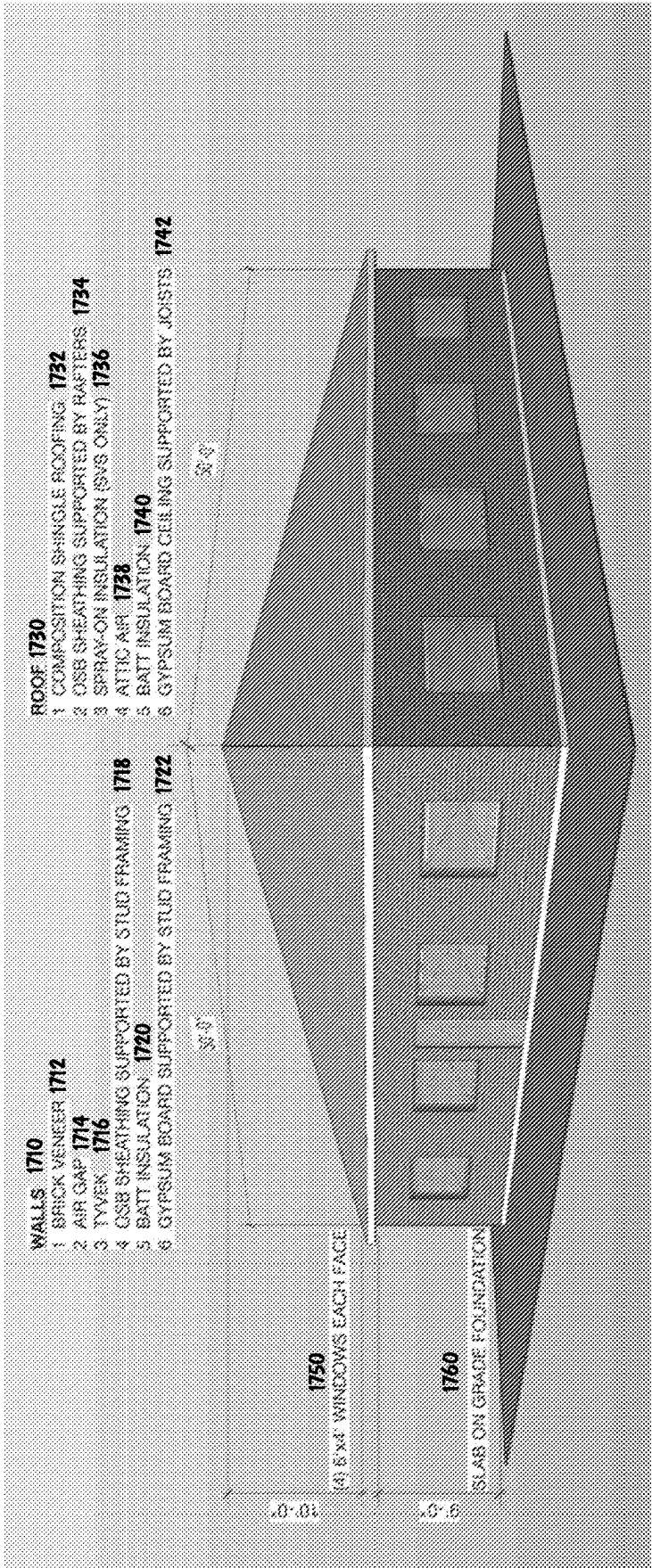


FIG. 17

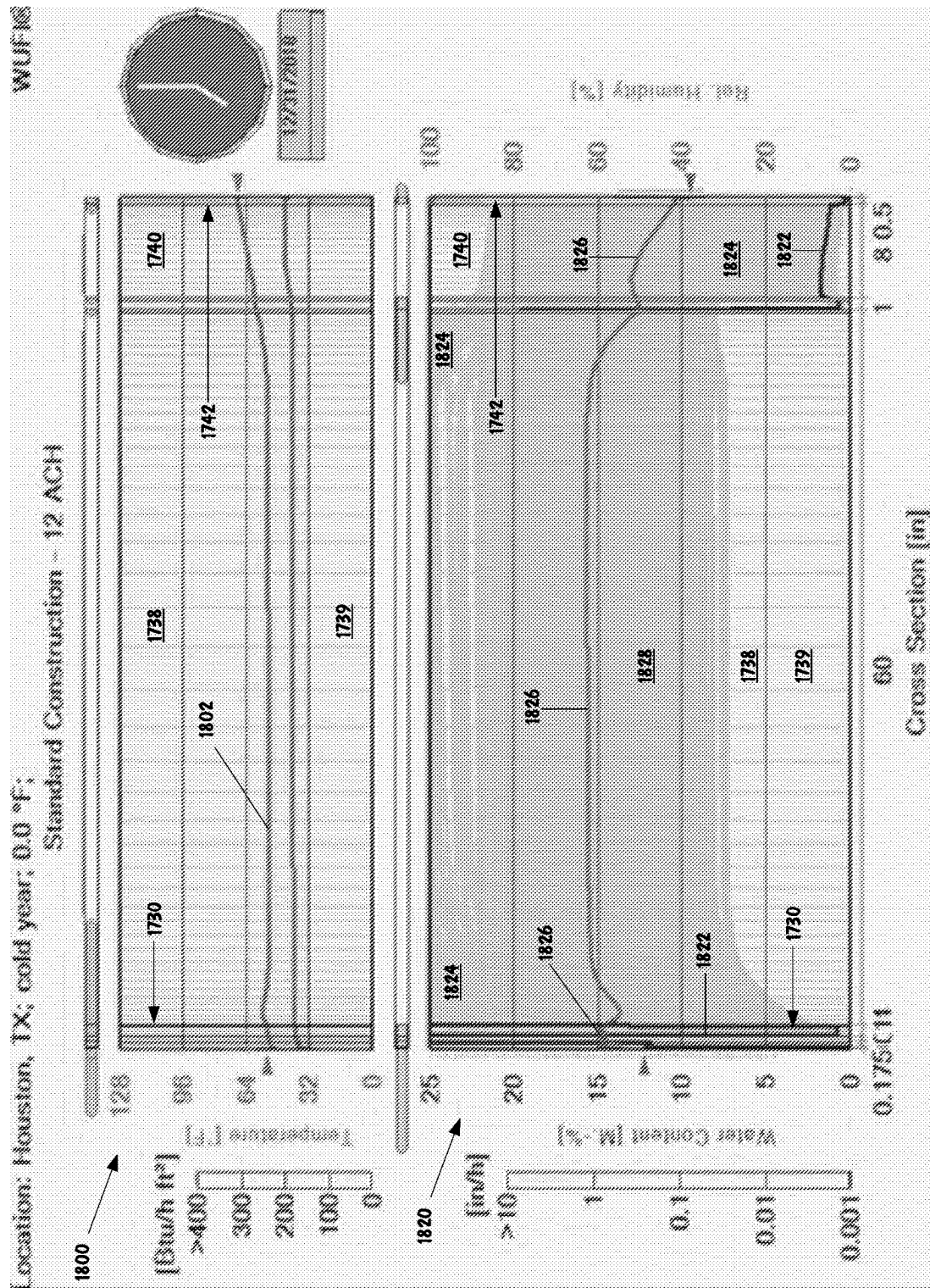
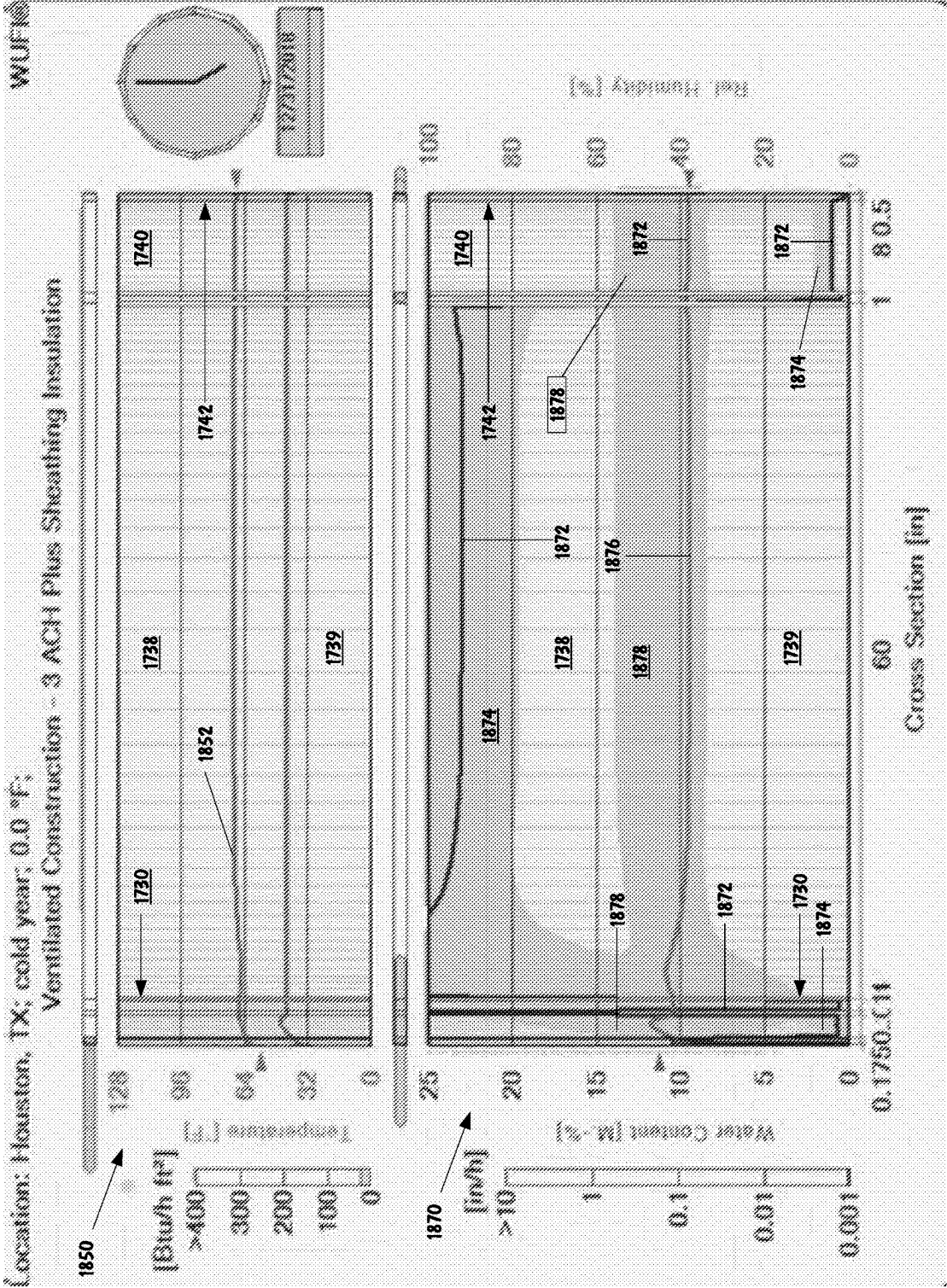
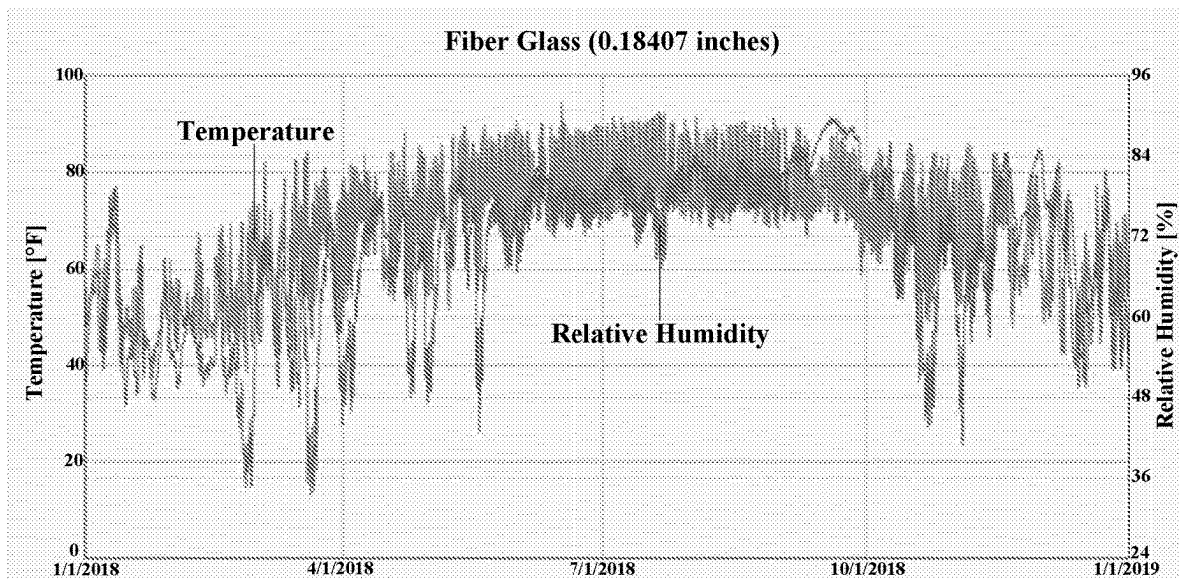
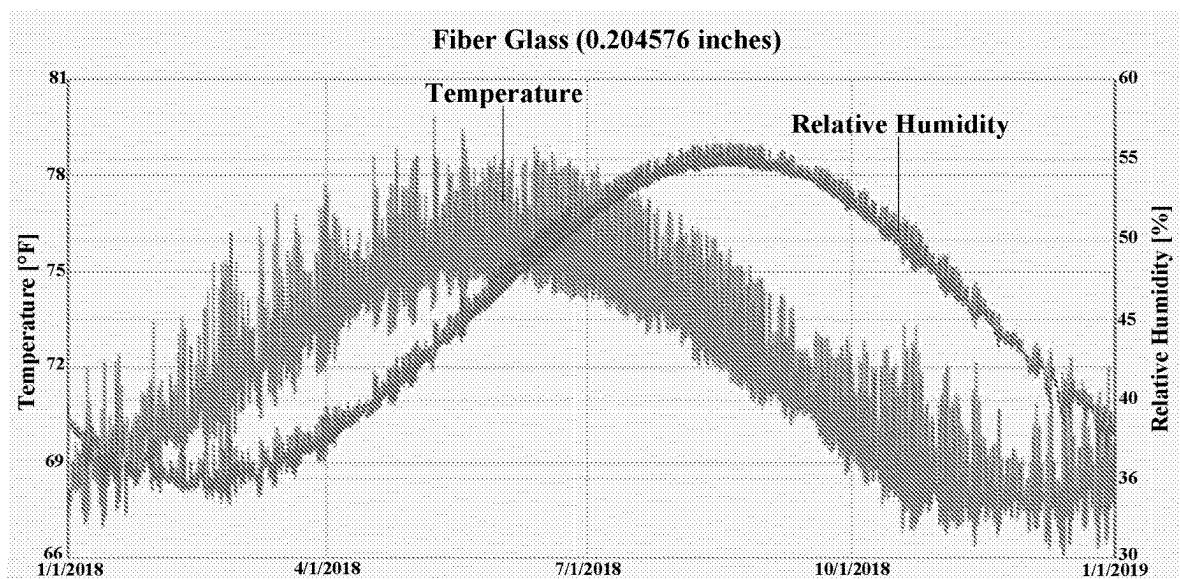


FIG. 18A



**FIG. 19A****FIG. 19B**

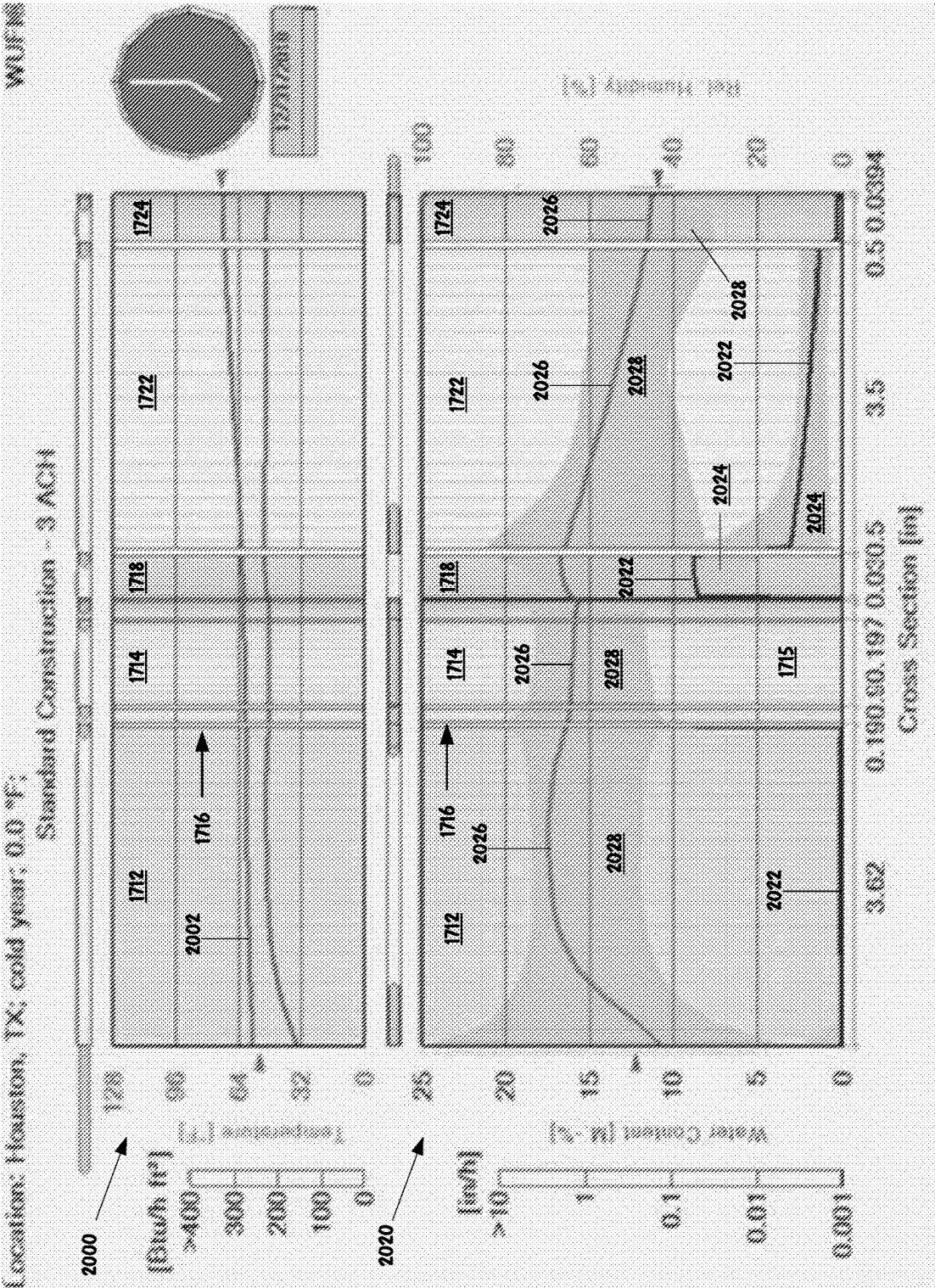


FIG. 20A

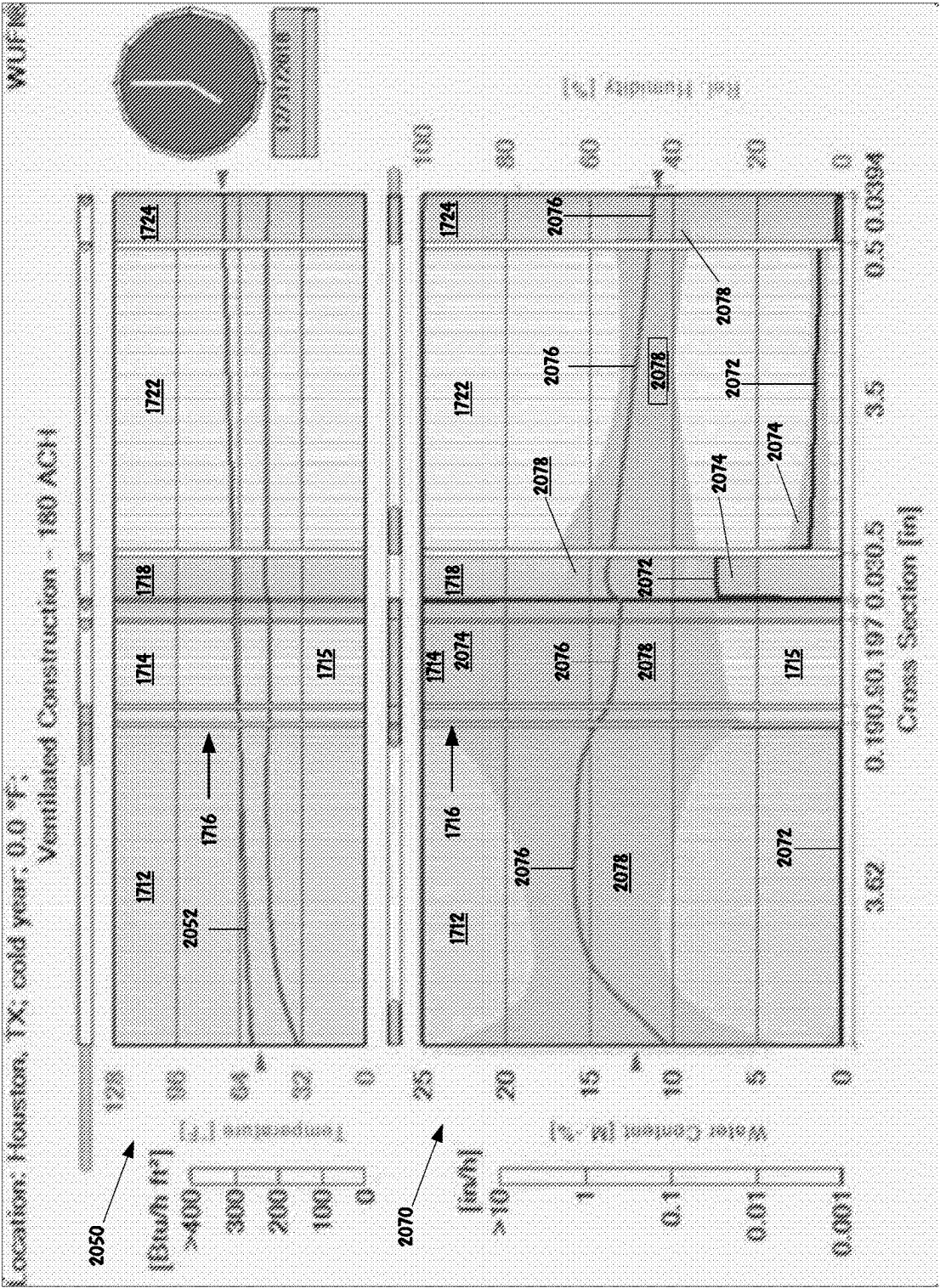
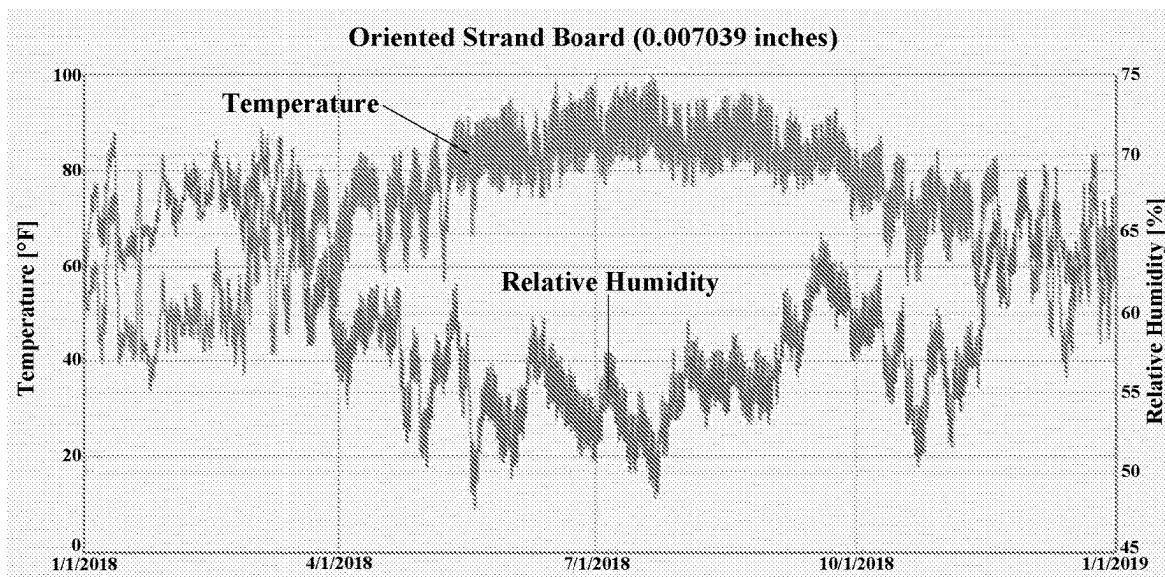
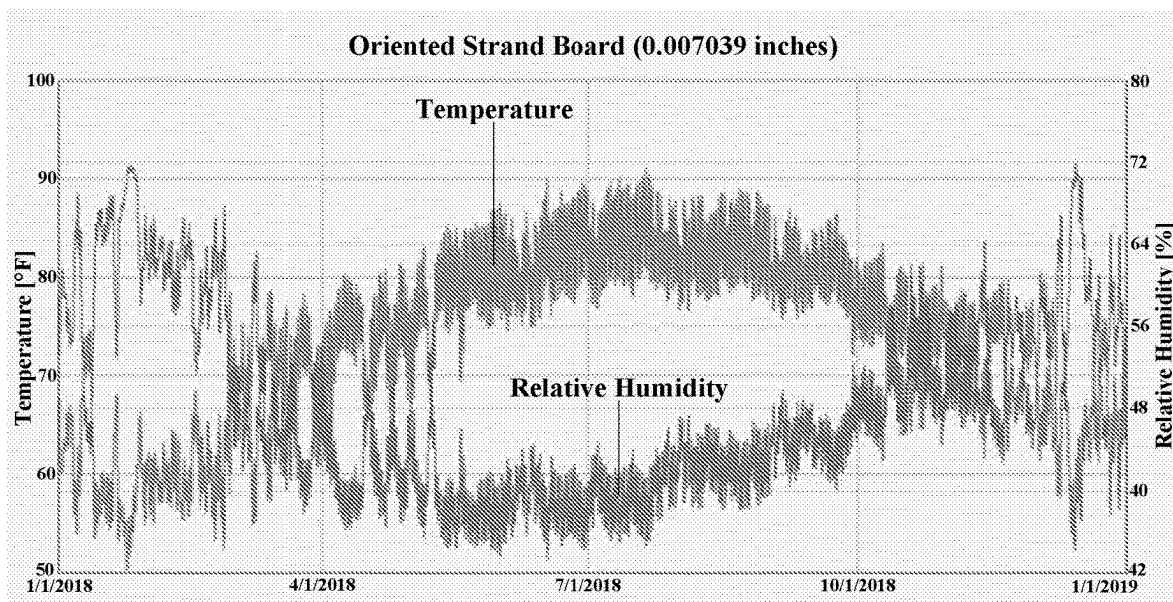


FIG. 20B

**FIG. 21A****FIG. 21B**

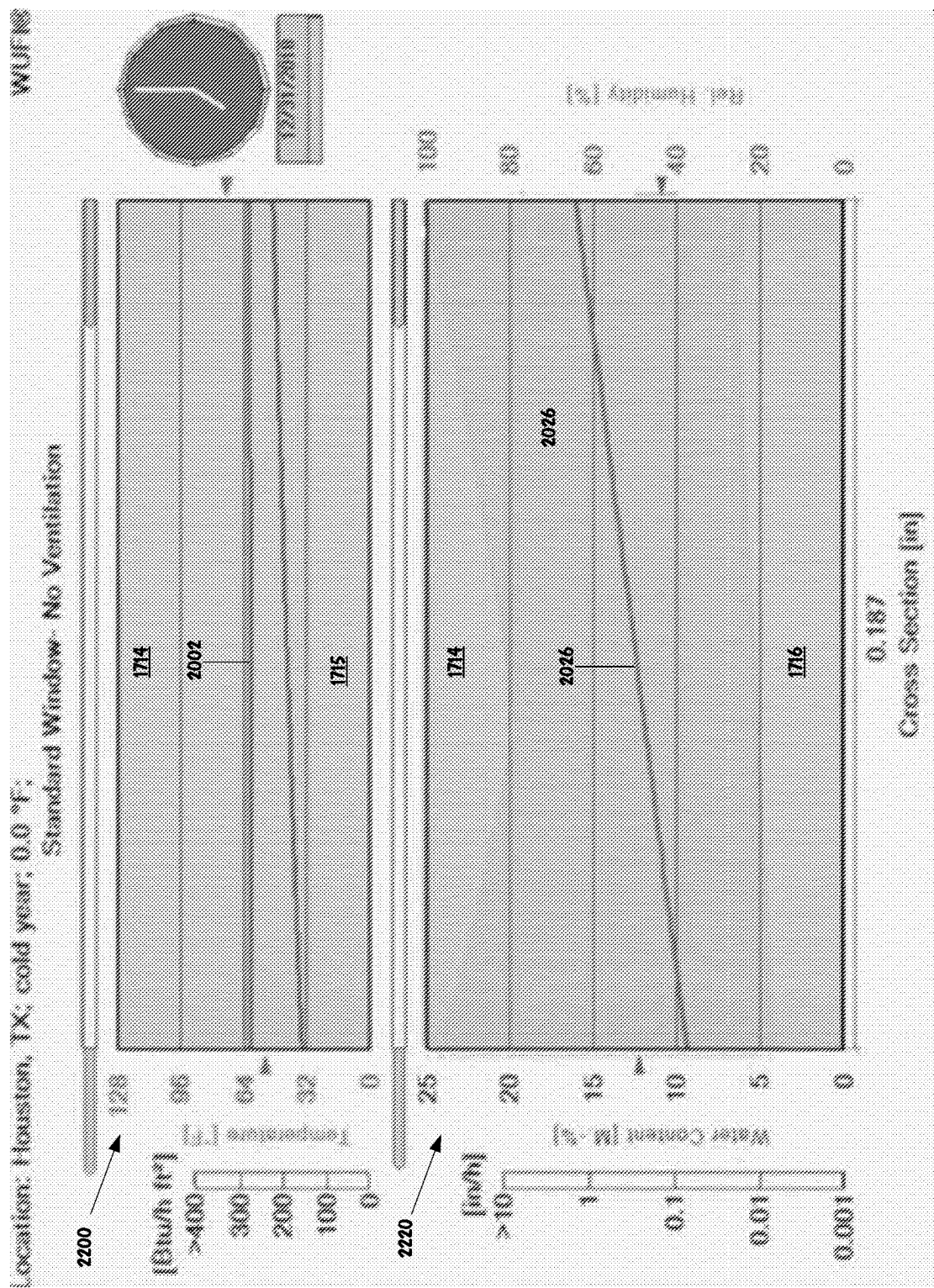


FIG. 22A

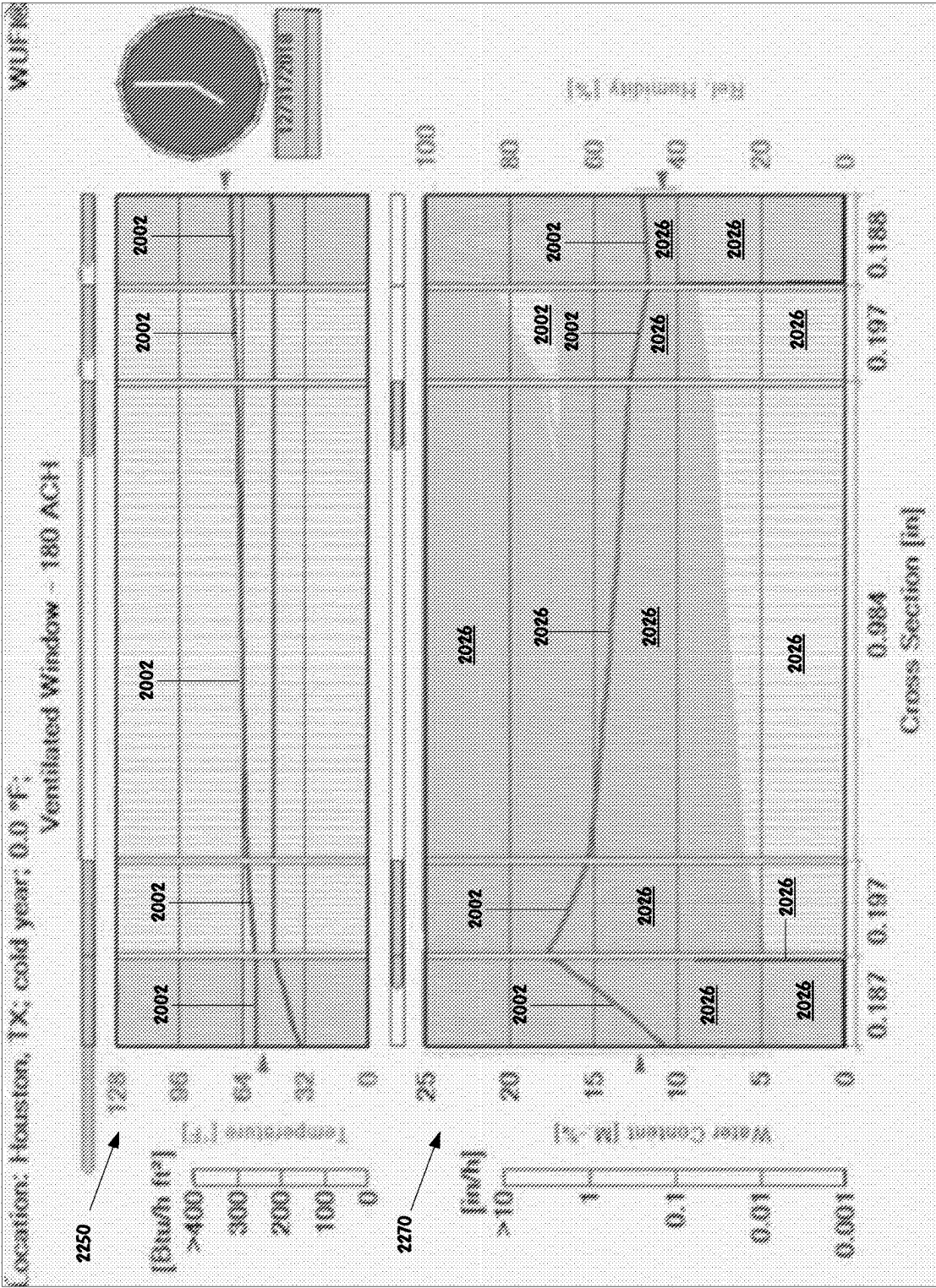


FIG. 22B

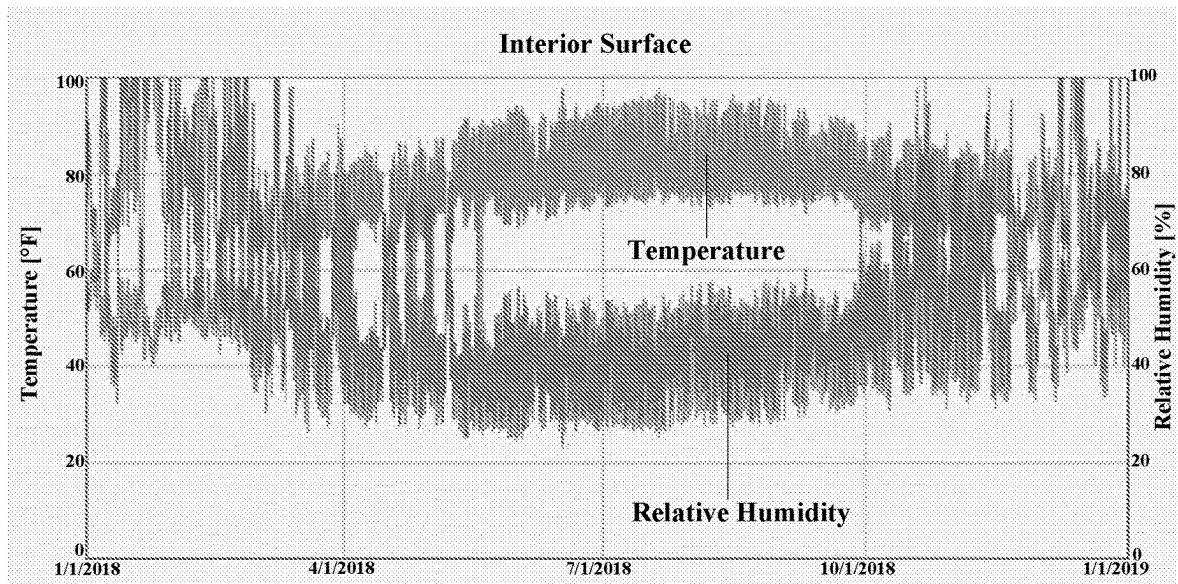


FIG. 23A

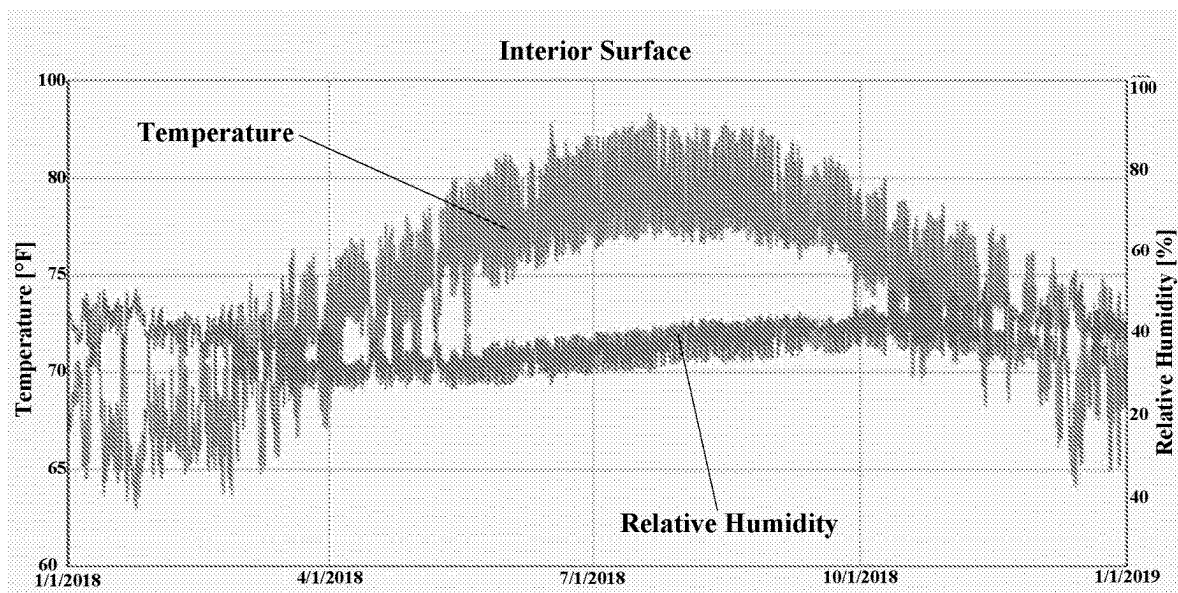


FIG. 23B

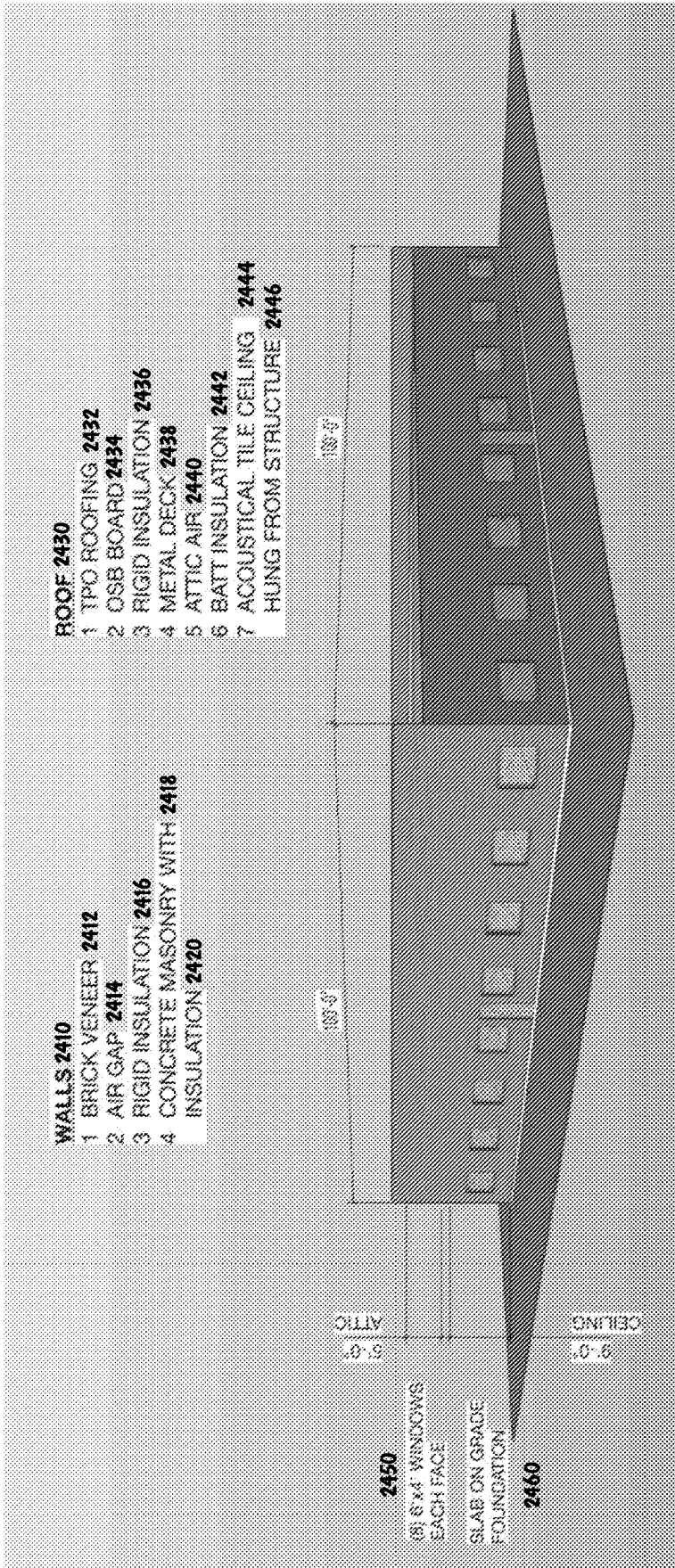
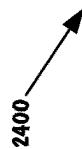


FIG. 24

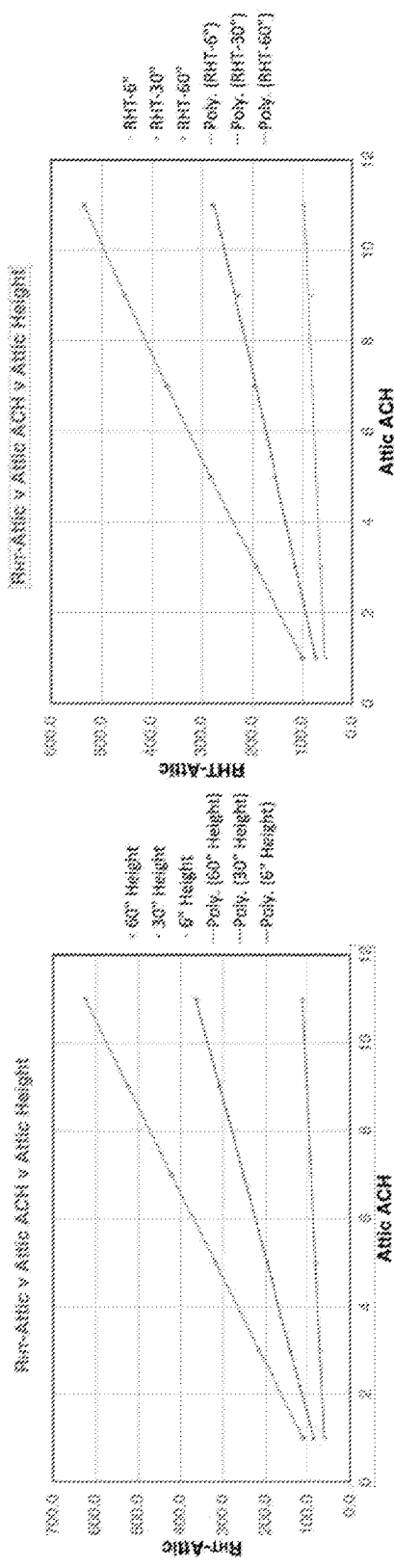


FIG. 25A

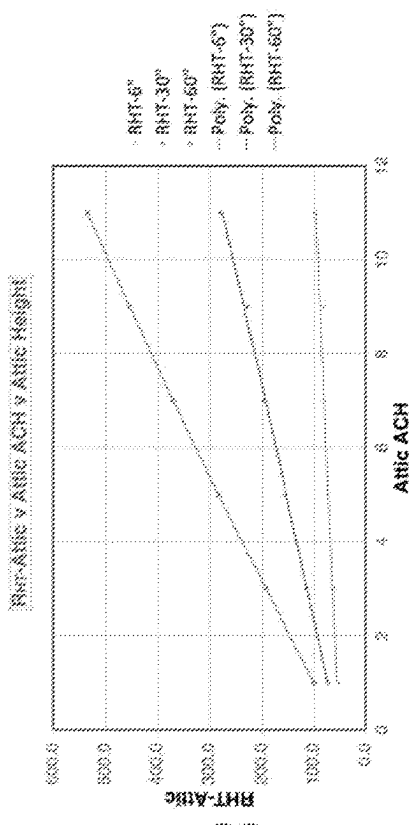


FIG. 25B

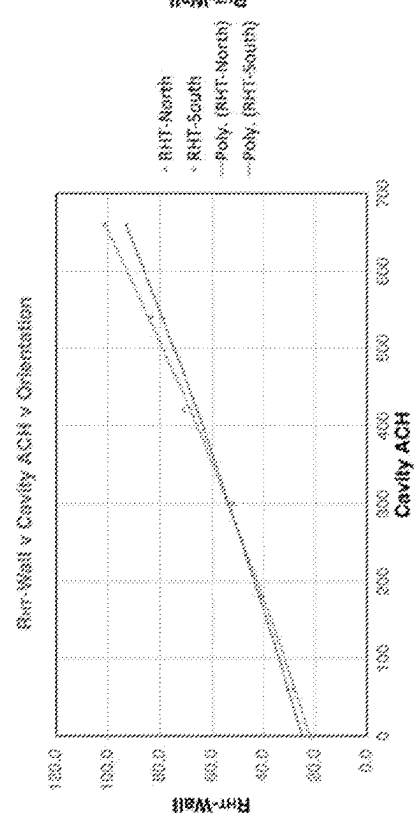


FIG. 25C

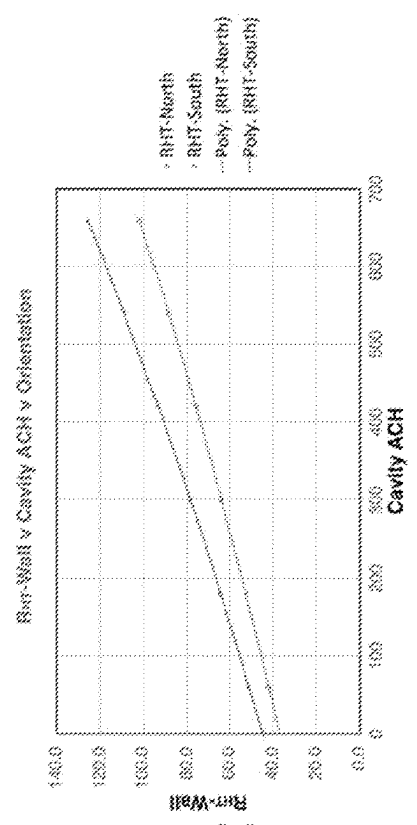
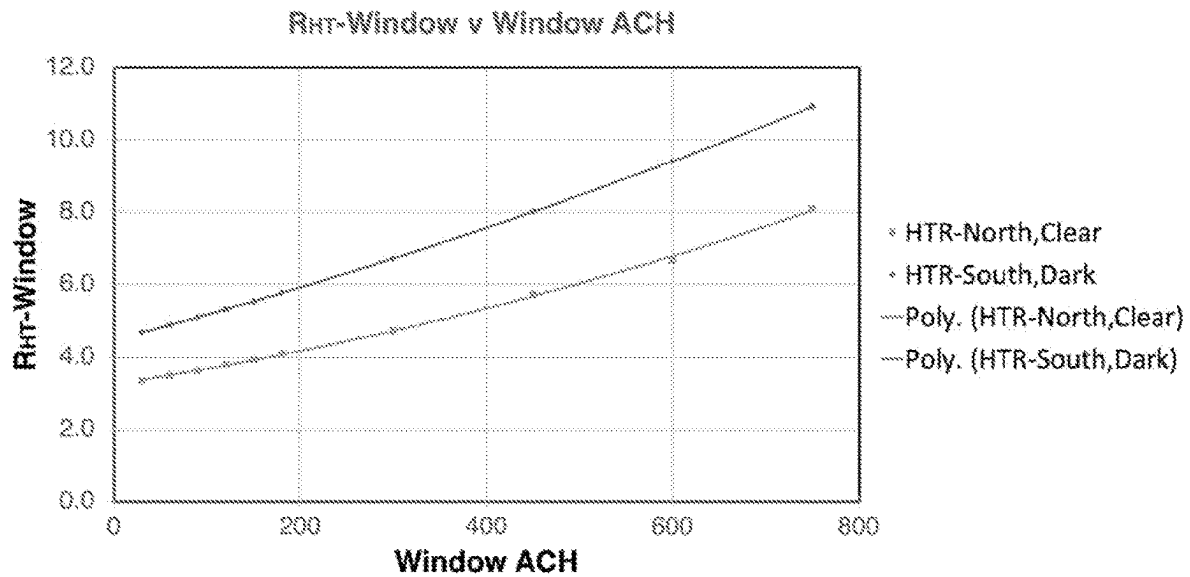


FIG. 25D

**FIG. 26**

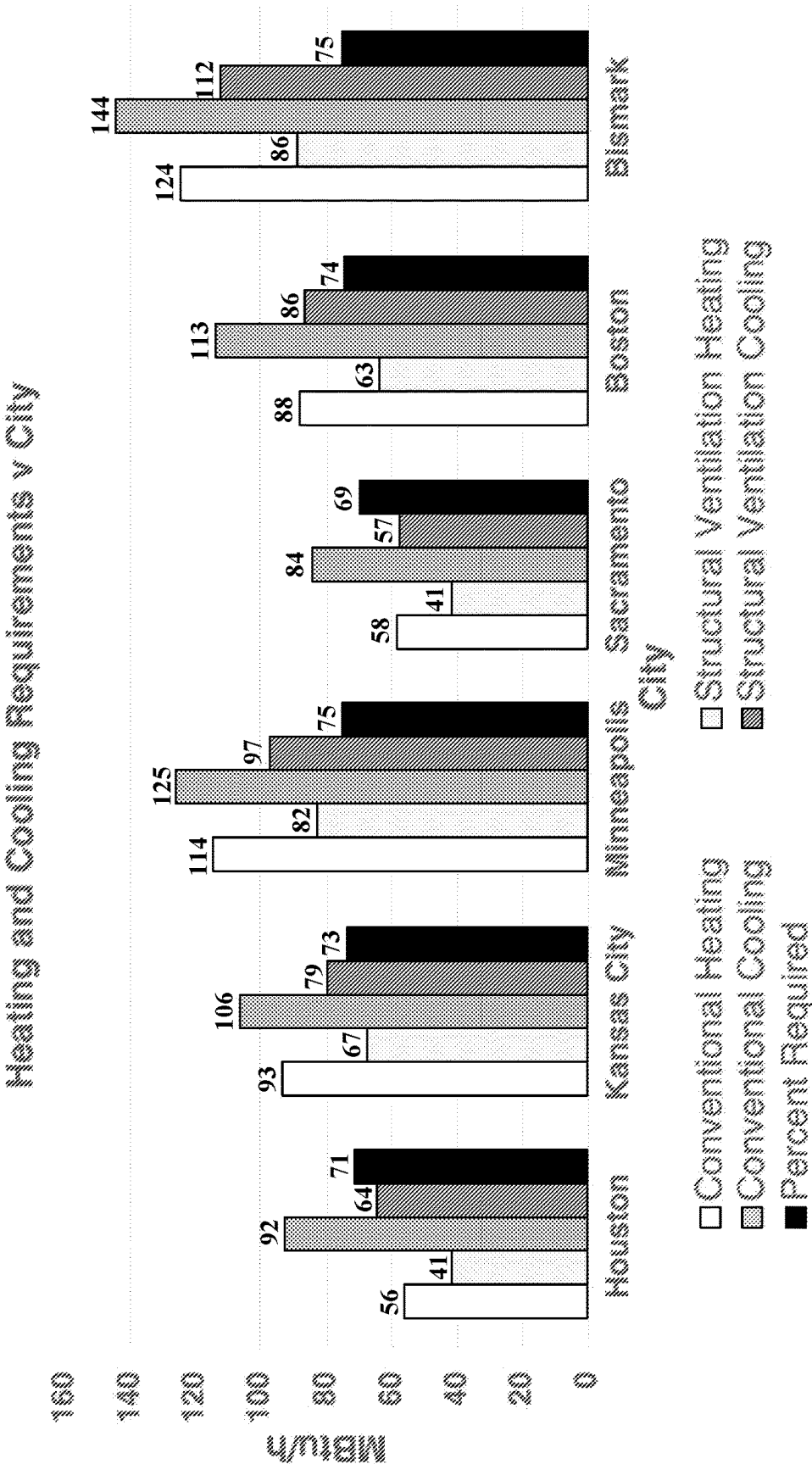
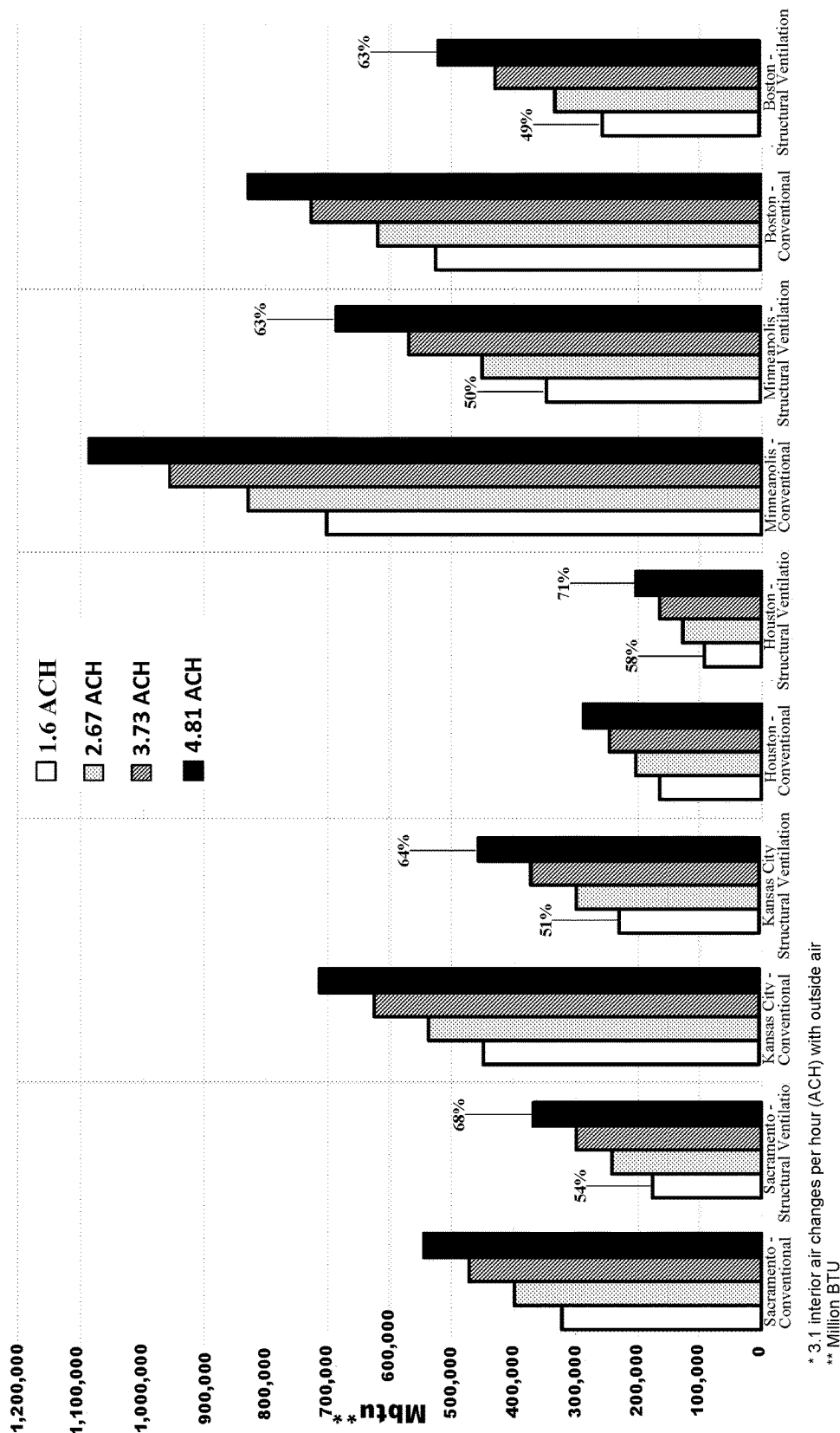


FIG. 27

Annual Heating Energy Usage v. Location v. Ventilation Systems v Building ACH*



* 3.1 interior air changes per hour (ACH) with outside air
** Million BTU

FIG. 28

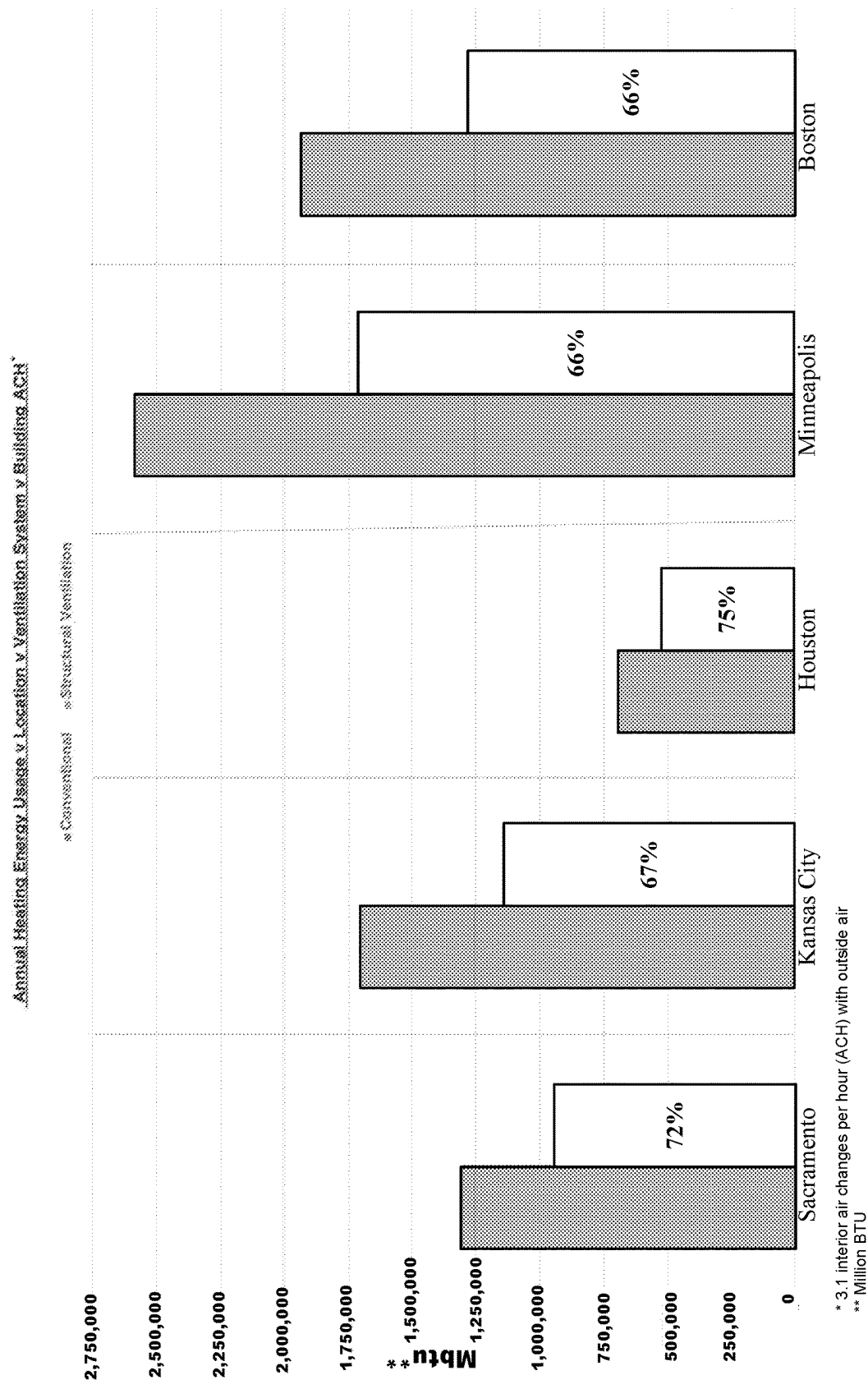


FIG. 29

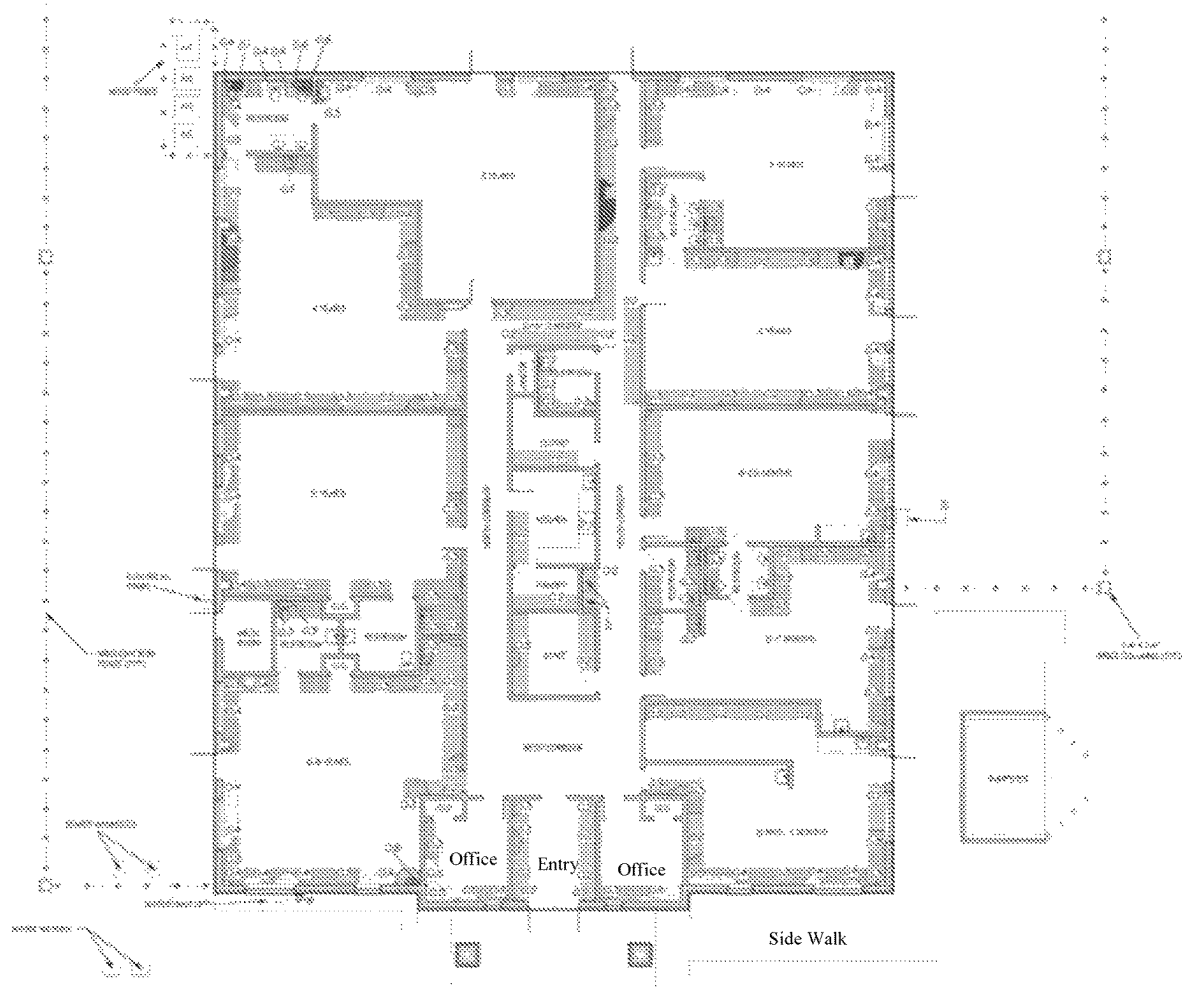


FIG. 30A

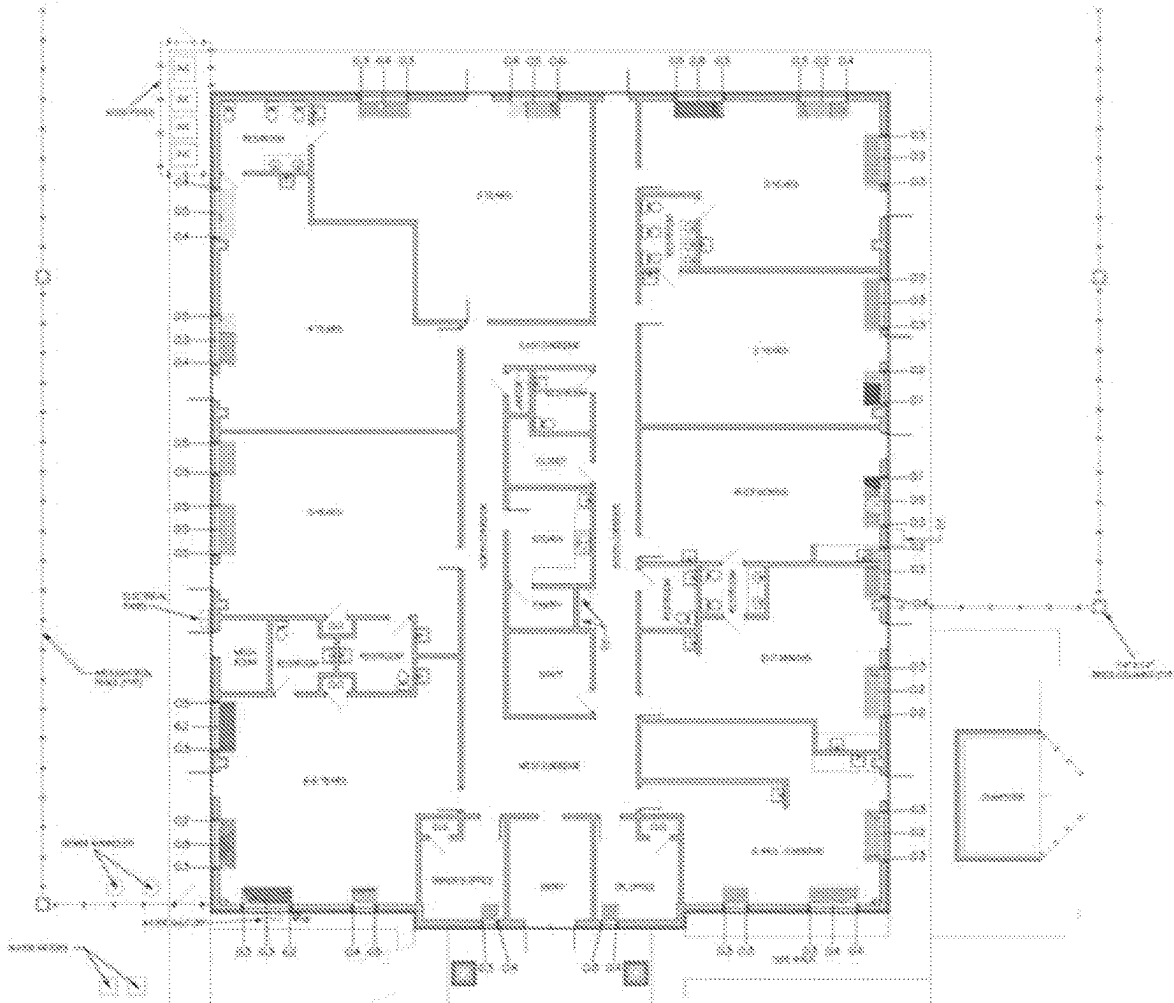
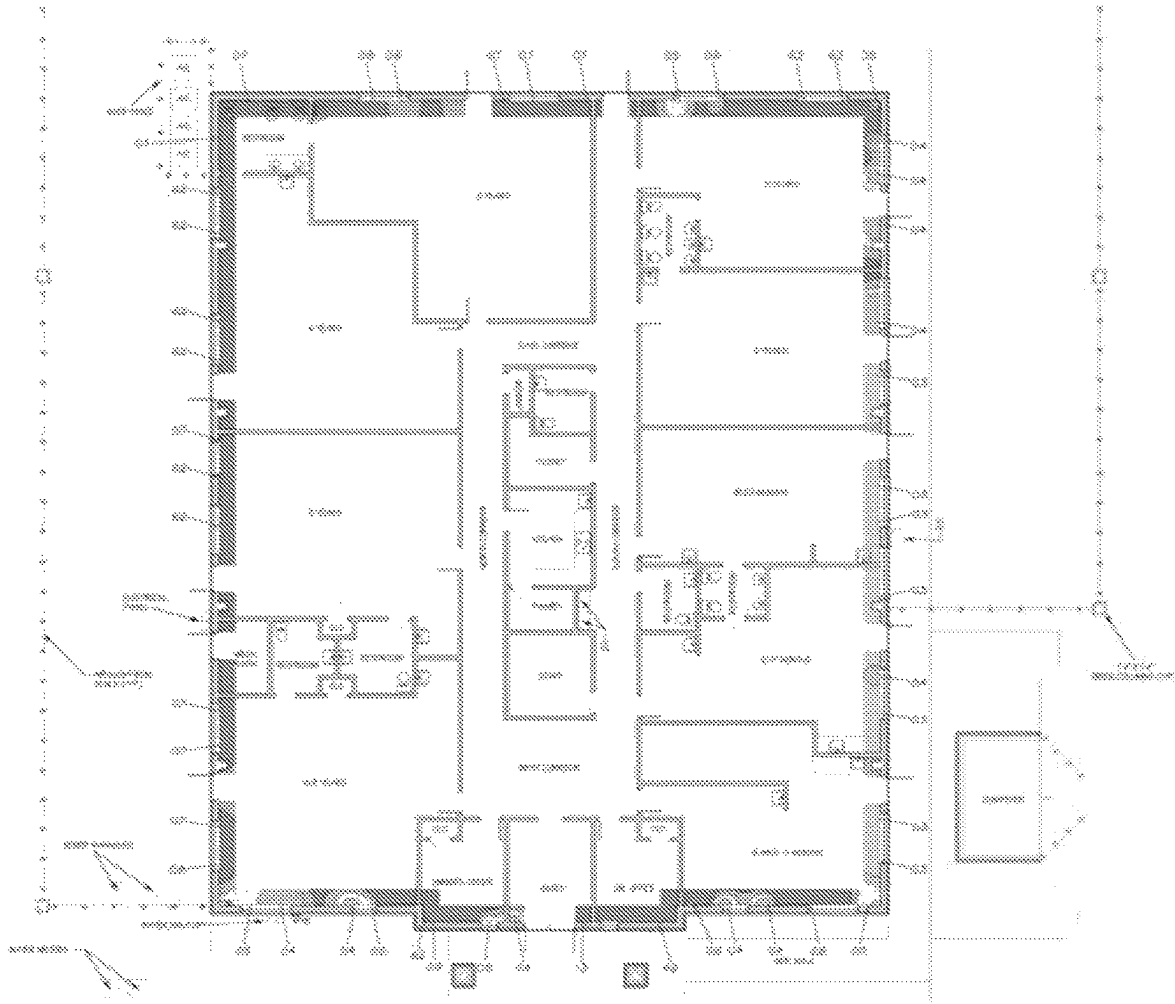
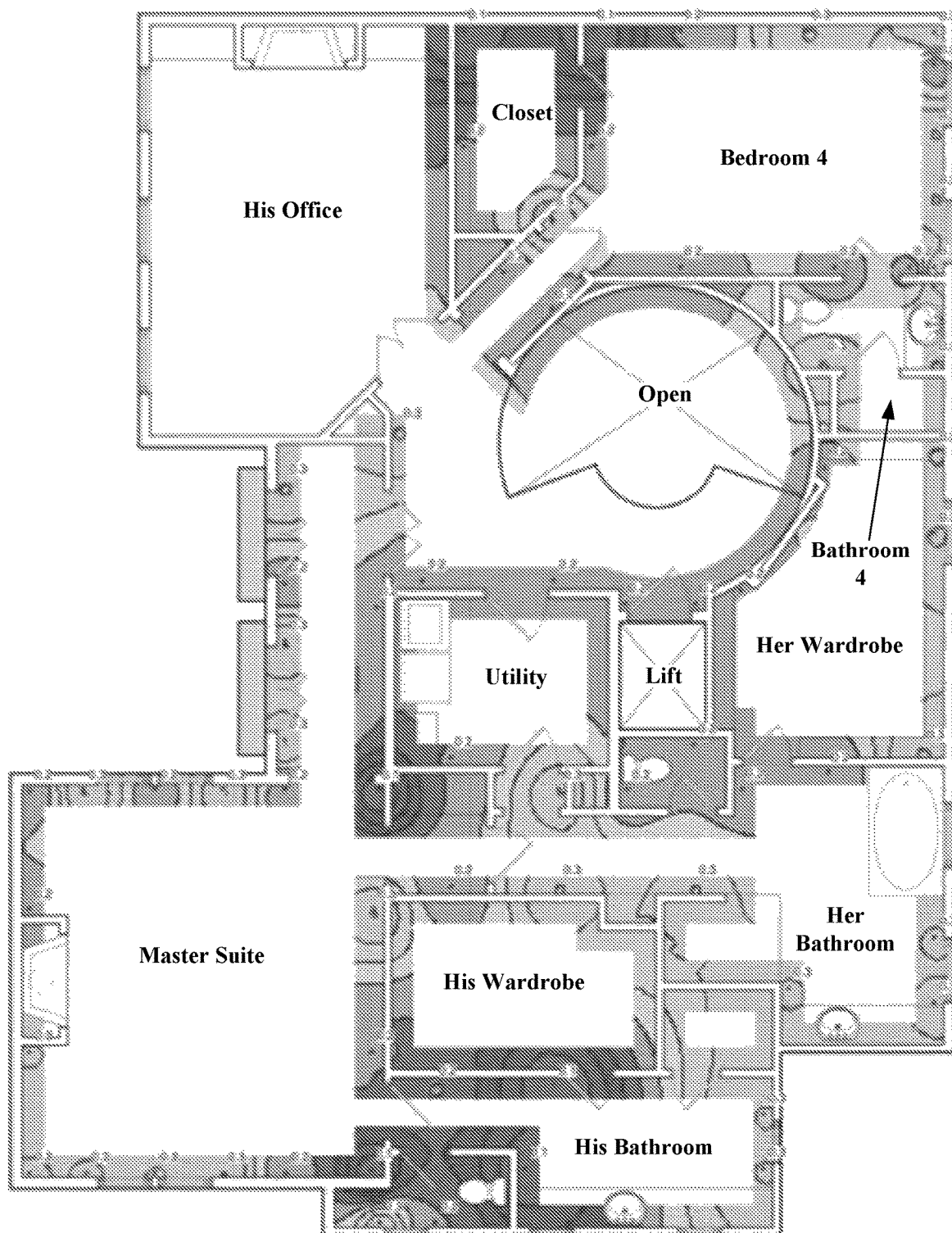
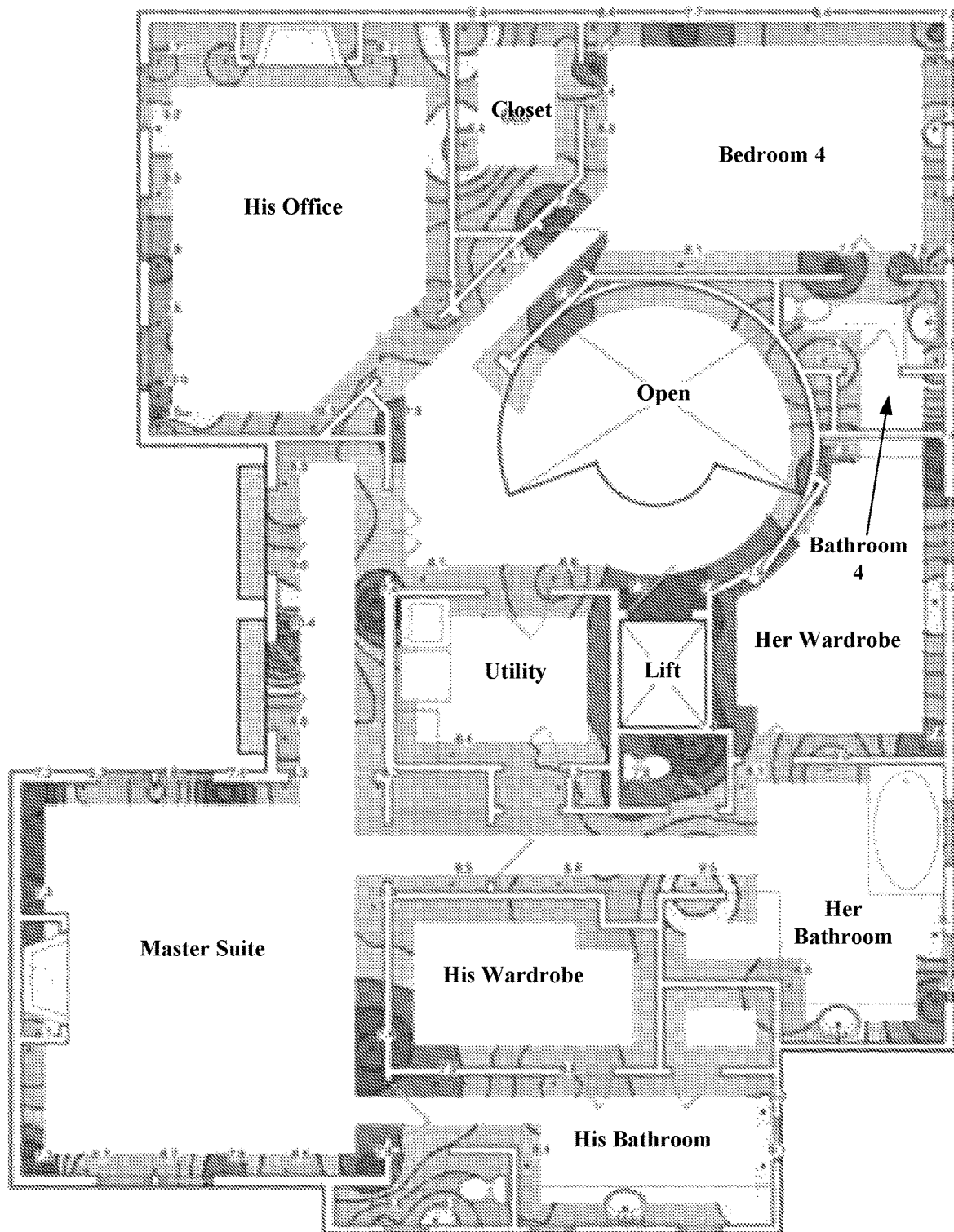
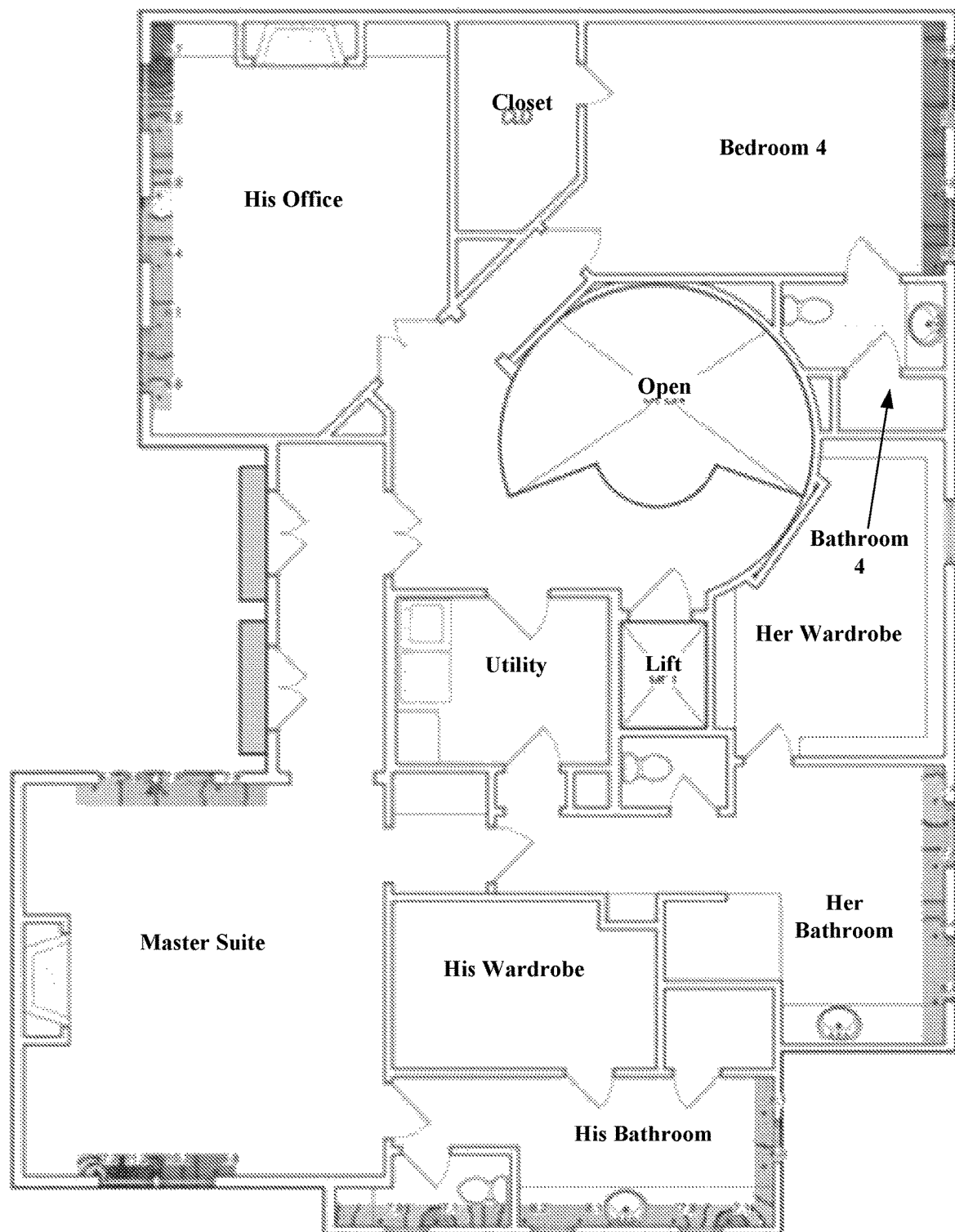


FIG. 30B

**FIG. 30C**

**FIG. 31A**

**FIG. 31B**

**FIG. 31C**

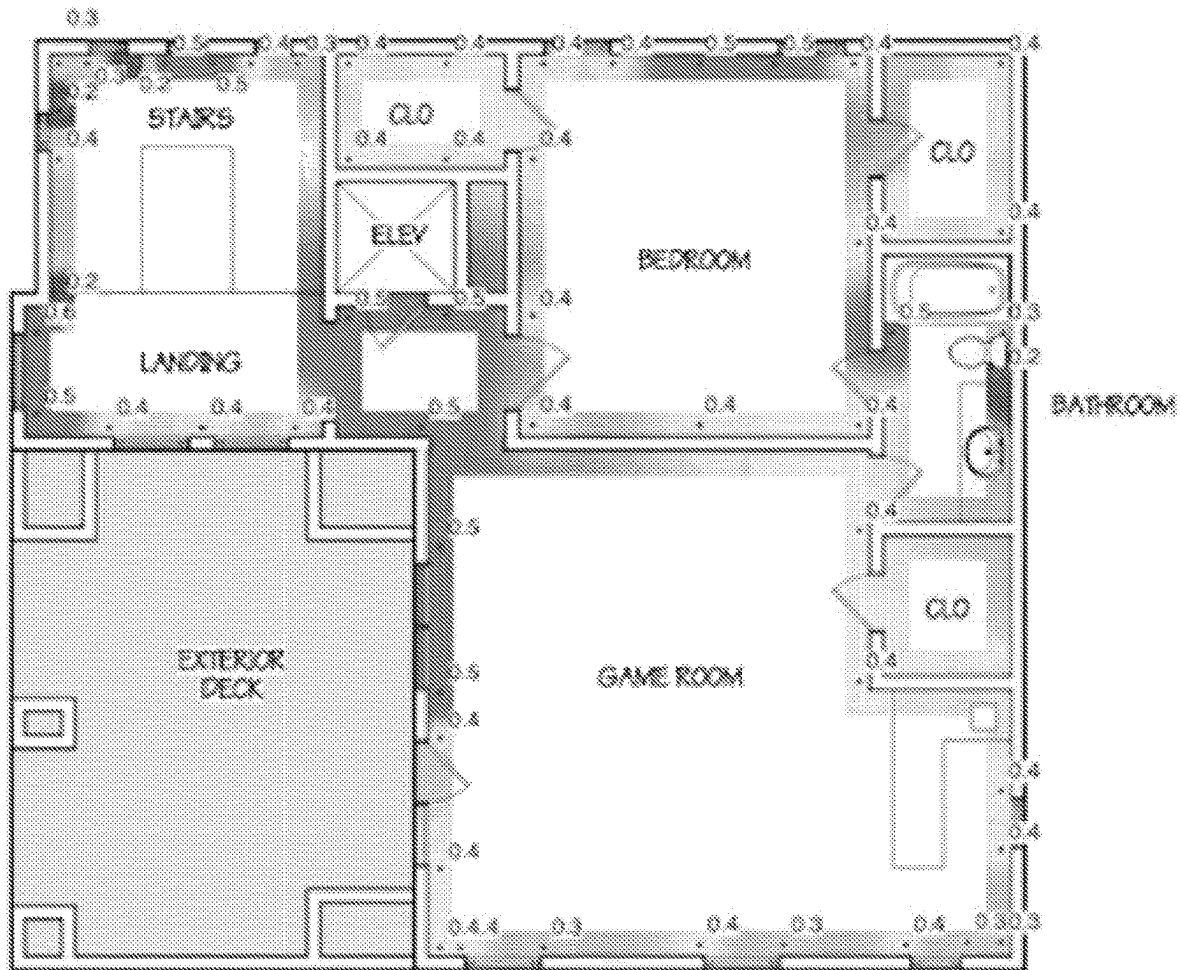
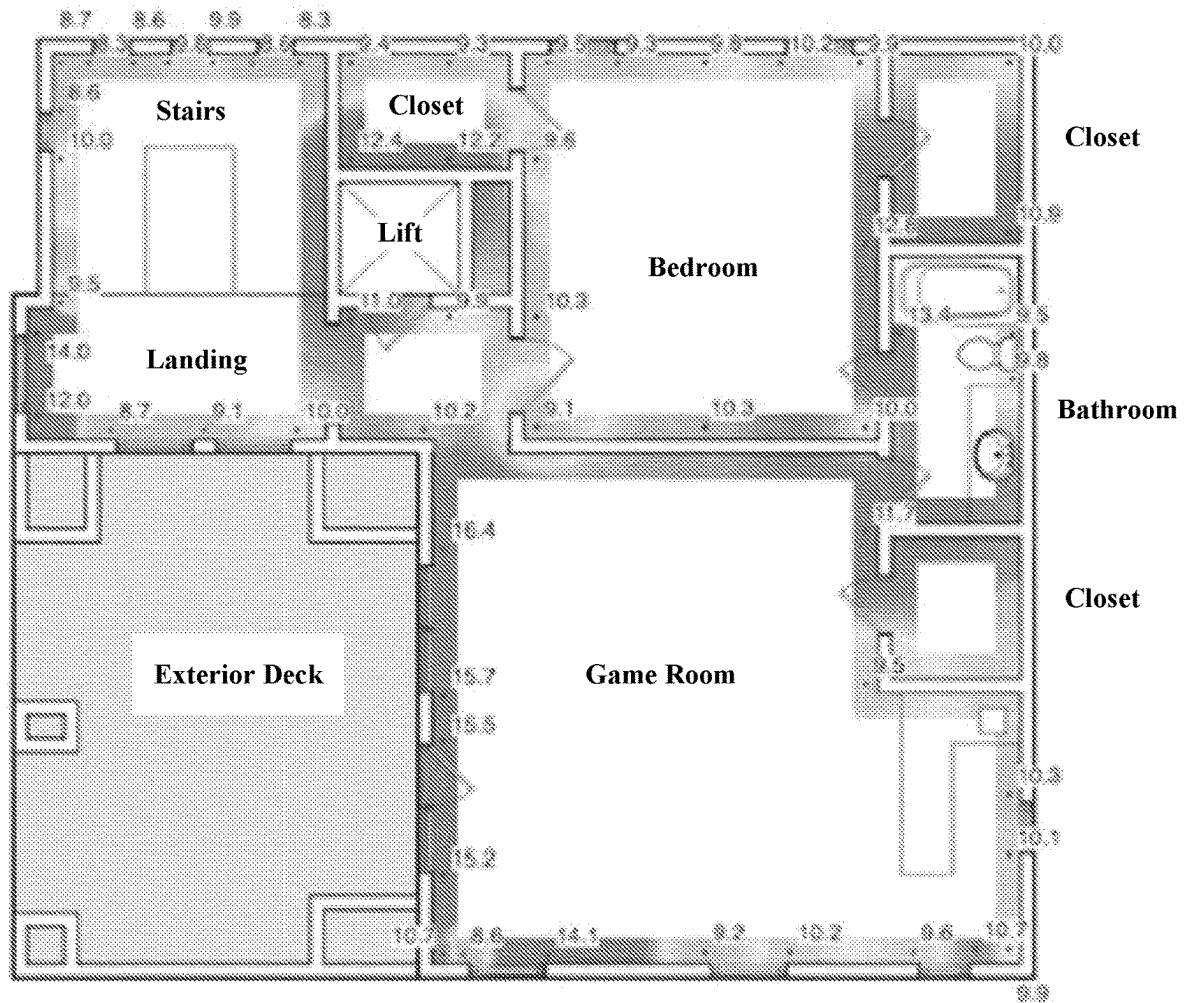
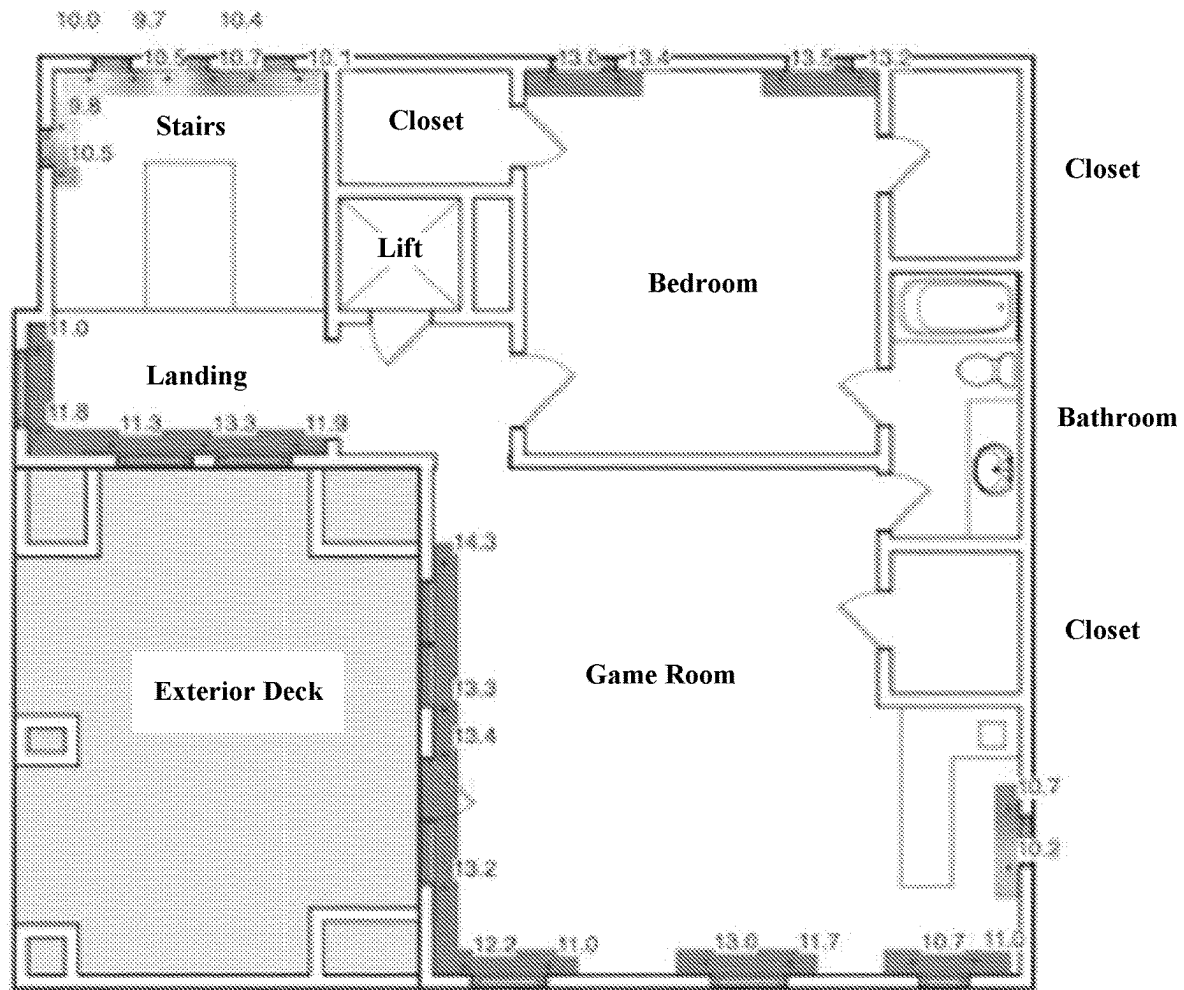
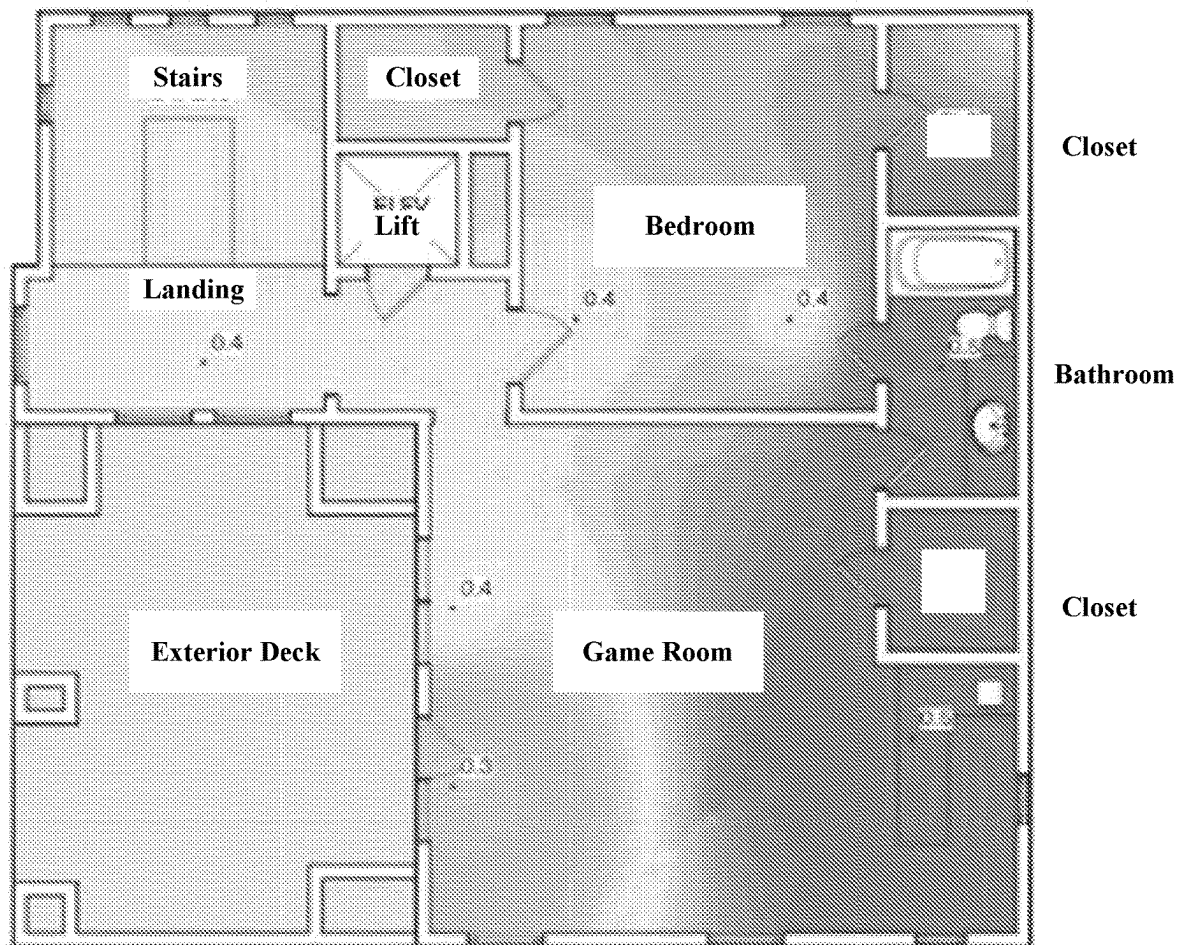


FIG. 32A

**FIG. 32B**

**FIG. 32C**

**FIG. 32D**

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**SYSTEMS AND METHODS FOR
CONTROLLING AIR PROPERTIES IN
STRUCTURES AND INHIBITING MOISTURE
ACCUMULATION AND MOLD
PROPAGATION IN STRUCTURES**

RELATED APPLICATIONS

This application is a Continuation-In-Part of U.S. patent application Ser. No. 17/556,527 filed 10 Dec. 2021 (Dec. 20, 2021), which is a continuation of U.S. patent application Ser. No. 16/273,344 filed 12 Feb. 2019 (Feb. 12, 2019) now U.S. Pat. No. 11,248,814 issued 15 Feb. 2022 (Feb. 15, 2022).

BACKGROUND OF THE DISCLOSURE

1. Field of the Disclosure

Embodiments of the present disclosure relate to systems, apparatuses, and methods for reducing, inhibiting, or eliminating water and/or moisture and/or their accumulation and/or mold growth in exterior walls or exterior and interior walls of habitable structures.

In particular, embodiments of the present disclosure relate to systems, apparatuses and methods for reducing, removing, inhibiting, or eliminating water and/or moisture and/or their accumulation and/or mold growth in exterior and interior walls of habitable structures, wherein the systems include air passageways or pathway supporting conditioned air through exterior walls or exterior and interior walls of habitable structures.

2. Description of the Related Art

Previously, I presented a system and method for inhibiting moisture and mold in an outer wall structure, U.S. Pat. No. 7,247,090B2. I now present improved systems, apparatuses, and methods for reducing, inhibiting, or eliminating water and/or moisture penetration and/or accumulation in internal and external walls of buildings and for drying buildings that have been subjected to flooding or other water and/or moisture penetration and/or accumulation events.

SUMMARY OF THE DISCLOSURE

Embodiments of present disclosure provide apparatuses and systems including exterior wall air passageways, optionally interior walls air passageways, and a heating, ventilation, and air conditioning unit (HVAC) that supplies conditioned air to the air passageways, valves, nozzles, and a size and a spacing of weep holes or air outlets associated with the walls and passageways or pathways to control the amount and direction of air flow into and through the passageways, wherein the conditioned air flow is designed to dry, to reduce moisture accumulation, and/or to inhibit mold growth in exterior walls or exterior and interior walls and/or to maintain a substantially uniform temperature in the structure throughout the day.

Embodiments of present disclosure provide methods of drying, inhibiting moisture accumulation, or controlling moisture in exterior and/or interior walls of a structure and/or controlling or maintaining temperature, air flow and air quality within the structure, wherein the methods include providing exterior walls or exterior and interior walls having air flow passageways therein and supplying conditioned air into the flow passageway by the HVAC unit controlled by valves and/or nozzles to dry or reduce or inhibit moisture

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accumulation in the walls, to maintain a certain moisture content in the walls, and/or to maintain a substantially uniform temperature in interior spaces in the structure as well as maintaining air flow and air quality in the interior spaces. The term air quality here means maintaining, in all interior spaces, air flow, moisture content, temperature, oxygen content, carbon dioxide content, carbon monoxide contents at desired levels. It should be recognized that the air passageways or pathways may include valves and nozzles for controlling air flow into the passageways or pathways and the sizes, number, and spacing of weep holes or air outlets associated with the passageways and pathways control the flow of air out of the passageways or pathways. It should also be recognized that the valves may be “active” or “passive”, wherein the term passive means that the valves are one-way valves allowing flow into the passageways and pathways, but not out of the passageways or pathways, and wherein the term active means that the valves actively control the flow of air into the passageways or pathways. Again, the size, number and spacing of the passageway or pathway outlets along with valves ultimately control the air flow rate through the passageways or pathways.

BRIEF DESCRIPTION OF THE DRAWINGS OF
THE DISCLOSURE

The disclosure may be better understood with reference to the following detailed description together with the appended illustrative drawings in which like elements are numbered the same:

FIG. 1 depicts a prior art wall or building structure.

FIG. 2A depicts a first embodiment of a wall or building structure of this disclosure.

FIG. 2B depicts the first embodiment of a wall or building structure of this disclosure with an attic exhaust assembly.

FIG. 3A depicts a second embodiment of a wall or building structure of this disclosure.

FIG. 3B depicts the second embodiment of a wall or building structure of this disclosure with an attic exhaust assembly.

FIGS. 4A&B depict expanded views of wall embodiments of this disclosure.

FIG. 5A depicts a third embodiment of a wall or building structure of this disclosure.

FIG. 5B depicts the third embodiment of a wall or building structure of this disclosure with an attic exhaust assembly.

FIG. 5C depicts a fourth embodiment of a wall or building structure of this disclosure.

FIG. 5D depicts the fourth embodiment of a wall or building structure of this disclosure with an attic exhaust assembly.

FIG. 6A depicts a first embodiment of a multistory building structure of this disclosure.

FIG. 6B depicts a second embodiment of a multistory building structure of this disclosure.

FIG. 6C depicts a third embodiment of a multistory building structure of this disclosure.

FIGS. 7A&B depict an embodiment of a peer and beam structure of this disclosure.

FIGS. 8A-D depict fourth embodiments of wall air distribution systems of this disclosure.

FIGS. 9A-C depict three other embodiments of wall air distribution systems of this disclosure.

FIG. 10A depicts a fifth embodiment of a wall or building structure of this disclosure.

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FIG. 10B depicts the fifth embodiment of a wall or building structure of this disclosure with an attic exhaust assembly.

FIGS. 11A&B depict expanded views of embodiments of outer wall assemblies of this disclosure, with sensors and a sprayer.

FIGS. 12A&B depict expanded views of embodiments of outer wall assemblies of this disclosure, with sensors and a sprayer.

FIG. 13 depicts a control system of this disclosure.

FIGS. 14A-C depict three view of a window installation having an air flow pathway between the exterior and interior window panes, whether single exterior and interior pane windows or double exterior and interior pane window.

FIG. 15A depicts a conventional HVAC system in a single family home.

FIG. 15b depicts a conventional HVAC system including a heat exchanger in a single family home.

FIG. 16A depicts an embodiment of a structural ventilation HVAC system of this disclosure.

FIG. 16B depicts the structural ventilation HVAC system of FIG. 16A in a flooded state.

FIG. 17 depicts an office building located in Houston, Texas that was subjected to analysis using the WUFI software program developed with assistance from Oak Ridge National Laboratory.

FIG. 18A depicts a WUFI analysis of an attic assembly of a conventional HVAC system.

FIG. 18B depicts a WUFI analysis of an attic assembly of an embodiment of a structural ventilation HVAC system of this disclosure.

FIG. 19A depicts attic assembly temperatures and relative humidities at the top surface of the ceiling insulation of a conventional HVAC system.

FIG. 19B depicts attic assembly temperatures and relative humidities at the top surface of the ceiling insulation of a structural ventilation HVAC system of this disclosure.

FIG. 20A depicts a WUFI analysis of a wall assembly WUFI analysis of a conventional HVAC system.

FIG. 20B depicts a WUFI analysis of a wall assembly WUFI analysis of a structural ventilation HVAC system of this disclosure.

FIG. 21A depicts wall assembly temperatures and relative humidities at the exterior surface of the OSB sheathing of a structural ventilation HVAC system of this disclosure.

FIG. 21B depicts wall assembly temperatures and relative humidities at the exterior surface of the OSB sheathing of a structural ventilation HVAC system of this disclosure.

FIG. 22A depicts a WUFI analysis of a window assembly WUFI analysis of a conventional HVAC system.

FIG. 22B depicts a WUFI analysis of a window assembly WUFI analysis of a structural ventilation HVAC system of this disclosure.

FIG. 23A depicts window assembly temperatures and relative humidities at the exterior surface of the OSB sheathing of a conventional HVAC system.

FIG. 23B depicts window assembly temperatures and relative humidities at the exterior surface of the OSB sheathing of a structural ventilation HVAC system of this disclosure.

FIG. 24 depicts a school building located in Houston, Texas that was subjected to analysis using the WUFI software program developed with assistance from Oak Ridge National Laboratory.

FIG. 25A depicts the R_{HT} of the attic assembly of the office building of FIG. 17.

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FIG. 25B depicts the R_{HT} of the attic assembly of the school building of FIG. 24.

FIG. 25C depicts the R_{HT} of the wall assembly of the office building of FIG. 17.

FIG. 25D depicts the R_{HT} of the wall assembly of the school building of FIG. 24.

FIG. 26 depicts the R_{HT} of the window assembly of the office building of FIG. 17 and school building of FIG. 24.

FIG. 27 depicts the heating and cooling requirements of the building in various cities comparing conventional HVAC systems and the structural ventilation HVAC system of this disclosure.

FIG. 28 depicts the annual heating energy usage of the office building at several cities comparing conventional HVAC systems and the structural ventilation HVAC system of this disclosure.

FIG. 29 depicts the annual heating energy usage of the school building at several cities comparing conventional HVAC systems and the structural ventilation HVAC system of this disclosure.

FIGS. 30A-C depict moisture content readings at the walls of a one-story, one-year old, daycare.

FIGS. 31A-C depicts moisture content readings at the third floor walls of a three-story, three-year old, house.

FIGS. 32A-D depict moisture content readings at the third floor walls and ceiling of a three-story, four-year old, house.

DEFINITIONS USED IN THE DISCLOSURE

The term “at least one” means one or more or one or a plurality, additionally, these three terms may be used interchangeably within this application. For example, at least one device means one or more devices or one device and a plurality of devices.

The term “one or a plurality” means one item or a plurality of items.

The term “about” means that a value of a given quantity is within $\pm 20\%$ of the stated value. In other embodiments, the value is within $\pm 15\%$ of the stated value. In other embodiments, the value is within $\pm 10\%$ of the stated value. In other embodiments, the value is within $\pm 5\%$ of the stated value. In other embodiments, the value is within $\pm 2.5\%$ of the stated value. In other embodiments, the value is within $\pm 1\%$ of the stated value.

The term “substantially” means that a value of a given quantity is within $\pm 5\%$ of the stated value. In other embodiments, the value is within $\pm 2.5\%$ of the stated value. In other embodiments, the value is within $\pm 2\%$ of the stated value. In other embodiments, the value is within $\pm 1\%$ of the stated value. In other embodiments, the value is within $\pm 0.1\%$ of the stated value.

The term “HVAC” means a heating, ventilation, and air conditioning unit, which generally includes a furnace, cooling coils, and blower along with the electronic and electrical components that control the unit.

DETAILED DESCRIPTION OF THE DISCLOSURE

The inventor has found that new systems, apparatuses, and methods may be constructed and implemented for reducing, inhibiting, or eliminating water and/or moisture and/or their accumulation and/or mold growth in exterior and interior walls of habitable structures by providing conditioned air pathways in the walls of the structures that are externally vented to force the removal of water and/or moisture from the walls and reduce, remove, inhibit or

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eliminate water and/or moisture and/or their accumulation and/or mold growth in exterior and interior walls.

The inventor has also found the new systems, apparatuses, and methods may include controlled structural ventilation of conditioned exhaust air from the attic, wherein the conditioned exhaust air comprises air that flow from the HVAC system into the interior of the structure and then is exhausted into the attic. A portion of the attic conditioned exhaust air may then distributed into the walls and between the window panes for energy saving and moisture remediation. In fact, the inventor surprisingly found that for existing structures using traditional HVAC units installing a controlled attic conditioned exhaust air vent. Such an attic control vent may afford up to about half the total energy saved that would result if a completed system were installed without the attic control vent designed to exhaust a portion of the conditioned exhaust air flowing into the attic from the interior of the existing structure into the surroundings. In addition to the energy savings, the cost of installation of the attic control vent is a fraction of the cost of an entire system. Thus, the installation of the attic control vent will unexpectedly result in energy savings for existing structures at a significant reduced installation cost compared to an entire system of this invention. Of course, it should be recognized that such an installation would not affect moisture reduction in exterior and/or interior walls and multi-planed windows.

Embodiments of present disclosure broadly related to apparatuses and systems including: exterior wall air passageways; optionally interior walls air passageways; a heating, ventilation, air conditioning unit (HVAC) to supply conditioned air, and valves to control an amount and/or direction of air flowing into and through the passageways, wherein the conditioned air flow may dry the walls, reduce moisture content in the walls, inhibit or control mold growth in the walls, and/or maintain a substantially uniform temperature in the structure throughout the day and night.

In certain embodiments, each exterior wall includes an internal wall section and an external wall section with the passageway disposed therebetween, wherein each passageway includes an inlet having a control valve or a unidirectional valve and an outlet, wherein the control valves or the unidirectional valves insure either a desired amount of air flow into the passageways or the air flow proceeds from the inlets to the outlets.

In other embodiments, each exterior wall includes an internal wall section and an external wall section with the passageway disposed therebetween, wherein each passageway includes an inlet having a control valve or a unidirectional valve and an outlet, and each interior wall includes a first wall section and a second wall section with the passageway disposed therebetween, wherein each passageway includes an inlet having a control valve or a unidirectional valve and an outlet, wherein the control valves or the unidirectional valves insure either a desired amount of air flow into the passageways or the air flow proceeds from the inlets to the outlets.

In other embodiments, the apparatuses and systems also include at least one sensor configured to sense moisture and generate a signal corresponding to the sensed moisture and a control unit configured to receive the signal from the at least one sensor and to control the HVAC and the valves to provide a conditioned air having properties to dry or reduce the sensed moisture or to maintain a certain moisture content in the walls due to the air flow into, through, and out of the passageways.

In other embodiments, the apparatuses and systems also include at least one sensor disposed in one, some, or all of

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the passageways and configured to sense moisture in the passageways and to generate a signal corresponding to the sensed moisture and a control unit configured to receive the signal from the at least one moisture sensor and to control the HVAC and optionally the valves to provide conditioned air and control an amount of the conditioned air into the air flow passageways, wherein the conditioned air has properties to dry the walls or to reduce the sensed moisture in the walls or to maintain a certain moisture content in the walls as the conditioned air flows into, through, and out of the passageways.

In other embodiments, the apparatuses and systems also include at least one sensor disposed in one, some, or all of the passageways and configured to sense temperature in the passageways and to generate a signal corresponding to the sensed temperature, and a control unit configured to receive the signals from the temperature sensors and to control the HVAC and optionally the valves to variably supply controlled amounts of conditioned air into the air flow passageways to maintain a constant and/or uniform temperature in the structure and/or to dry or reduce the moisture or to maintain a certain moisture content in the walls.

In other embodiments, the apparatuses and systems also include a supply unit for supplying mold control compositions or other compositions into the walls, wherein the supply unit is under the control of the control unit. In other embodiments, the apparatuses and systems also include fire suppressant units that may feed carbon dioxide or carbon dioxide foam or other fire retardant or suppression compositions.

In other embodiments, the apparatuses and systems are designed to permit conditioned air to be flow through the exterior and interior walls so that a uniform temperature and moisture content in the wall interiors may be maintained. During the day and night, the apparatuses and systems may continuously adjust rate and amount of conditioned air flowing into each passageway so that the temperature and moisture content within the interior of each wall is maintained at substantially uniform values that may be preset set or set as desired by the user. As the sun rises, one side of a structure will begin to heat differently than the non-illuminated side, the apparatuses and systems adjust the distribution of air flow so that the temperature and/or moisture content in the interior the walls and therefore throughout the entire structure may be more uniformly controlled.

Embodiments of present disclosure broadly relates to methods of drying water or moisture in exterior walls or exterior and interior walls of a structure, inhibiting moisture accumulation in the walls, or maintaining a substantially uniform temperature and/or moisture content in the wall, where the methods include providing exterior walls or exterior and interior walls having air flow passageways and supplying conditioned air into the flow passageway by an HVAC unit and valves to dry or reduce or inhibit moisture accumulation in the walls, to maintain a certain moisture content in the walls, and/or to maintain a substantially uniform temperature in the structure.

In certain embodiments, the methods further include receiving output from at least one sensor and controlling properties of the conditioned air in response thereto.

In other embodiments, the methods further include receiving output from at least one sensor, controlling properties of the conditioned air in response thereto and controlling an air flow rate into the passageways in response thereto.

In other embodiments, the methods further include supplying a treating composition into the passageways, wherein the treating composition is a mold control composition, a

pesticide, an insecticide, a fungicide, a bactericide, a poison for rodents, other control compositions, or mixtures and combinations thereof.

In other embodiments, the methods also include collecting temperature data, moisture data, air flow data, pressure data, air component data, and/or other data and adjusting the air flow rate and/or amount into each passageway so that the values of each measure property may be maintained at some desired level or some pre-set level. In other embodiments, the adjusting may be in conjunction with data indicating a fire or other emergency situation in which case the air flow and fire suppressants and/or retardants flow rate into the passageways may be used to impede progression of the fire or to produce fire breaks, wall as that do not support fire due to air flow and flame retardant flow rate into each passageway.

Again, it should be recognized that the air passageways or pathways may include valves and nozzles for controlling air flow into the passageways or pathways and the sizes, number, and spacing of weep holes or air outlets associated with the passageways and pathways control the flow of air out of the passageways or pathways. It should also be recognized that the valves may be “active” or “passive”, wherein the term passive means that the valves are one way valves allowing flow into the passageways and pathways, but not out of the passageways or pathways, and wherein the term active means that the valves actively control the flow of air into the passageways or pathways. Again, the size, number and spacing of the passageway or pathway outlets along with valves ultimately control the air flow rate through the passageways or pathways.

Embodiments of this disclosure broadly relate to structures including a floor or slab, a ceiling, an insulated roof, and an attic interposed between the ceiling and the roof. The structures further include one or a plurality of exterior walls, each of the exterior walls includes: an exterior wall outer portion, an exterior wall inner portion including a wall framing system, an exterior wall air flow passageway interposed therebetween, and at least one exterior wall exhaust conduit having: a passageway inlet, and a passage way outlet. The structures further include a plurality of interior walls and a plurality of interior spaces defined by the exterior and interior walls. The structures further include an HVAC unit, disposed in the attic, including: an HVAC air inlet conduit having: an HVAC inlet, and an HVAC inlet control valve configured to control a supply of unconditioned air or a mixture of unconditioned air and exhaust conditioned air to the HVAC unit; an HVAC air outlet conduit having: an HVAC outlet, and an HVAC outlet control valve configured to control a supply of conditioned air to the interior spaces; and at least one HVAC exhaust conduit having: an HVAC exhaust inlet, and an HVAC exhaust outlet configured to support an air flow of exhaust conditioned air into the attic. The structures further include at least one passageway control valve associated with the passageway inlets and configured to support passageway air flow into the passageway(s) and out of the exhaust conduit(s) into the surroundings and to stop air flow from the surroundings into the passageways, whereby the passageway air flow reduces, removes, and/or eliminates moisture in the exterior walls.

In certain embodiments, the at least one passageway control valve is a passive one-way or unidirectional valve, and the passageway air flow comprises attic exhaust conditioned air. In other embodiments, the at least one passageway control valve is part of the HVAC unit and is associated with a HVAC passageway outlet conduit connected to the HVAC outlet conduit and having a passageway outlet asso-

ciated with the passageway inlets, the at least one passageway control valve is a passive one-way or unidirectional valve or an active control valve, and the passageway air flow comprises conditioned air from the HVAC unit. In other embodiments, each of the interior walls includes: two interior wall outer portions, an interior wall inner portion having: an interior wall air flow passageway having: a passageway inlet and a passageway outlet; at least one interior wall exhaust conduit. In other embodiments, the exterior wall inner portion includes: an insulating layer, a weather-resistant layer disposed on an outer surface of the insulating layer; and/or a sheathing disposed on an inner surface of the insulating layer. In other embodiments, the exterior wall inner portion further includes: a liquid barrier and/or a first sheathing between the liquid barrier and the wall framing system. In other embodiments, the at least one HVAC exhaust conduit further includes: a second HVAC exhaust outlet configured to support an air flow of exhaust conditioned air into the surrounding and having: an HVAC exhaust control valve configured to control amounts of exhaust conditioned air exhausted into the attic and into the surrounds.

In other embodiments, the structure further comprising: an air flow control system including: a processing unit; and plurality of sensors; wherein one, some, or all of the control valves are active control valves, wherein the processing unit is in communication with the active control valves, and wherein the processing unit is configured to: receive input from the sensors, and control the HVAC unit and the active control valves thereby controlling air flow rates into the interior spaces and into the passageways. In other embodiments, the sensors are selected from the group consisting of temperature sensors, water/moisture sensors, air flow sensors, chemical sensors, other physical sensors, and combinations thereof. In other embodiments, at least one of the sensors is disposed in: the air flow passageways or the air flow passageways and the attic and/or the interior spaces. In other embodiments, all of the control valves are active control valves, and during a 24 hour period of time, the processing unit is further configured: to adjust the air flow into each of the interior spaces and into each passageway so that one or more interior space properties and/or one or more passageway properties are maintained at desired values. In other embodiments, the one or more properties include moisture concentrations in the interior spaces and/or passageways, temperature in the interior spaces and/or passageways, or mixtures and combinations thereof. In other embodiments, the one or more properties further include pressures in the interior spaces and/or passageways, air flow rates in the interior spaces and/or passageways, sensed chemical concentrations in the interior spaces and/or passageways, and/or microbial concentrations in the interior spaces and/or passageways.

In other embodiments, the structures further comprises: a dispensing assembly associated with one, some, or all of the passageways and including: a reservoir containing a treating composition; a dispensing supply conduit having: a dispensing assembly control valve, a dispensing head at its distal end, wherein the dispensing assembly control valve is configured to control an amount of treating solution or gas introduced into one, some, or all of the passageways. In other embodiments, the treating composition is selected from the group consisting of a mold control composition, a pesticide, an insecticide, a fungicide, a bactericide, a rodent poison, other control compositions, a gas, a fire retardant compositions, a foam, and mixtures or combinations thereof, and the gas is selected from the group consisting of nitrogen,

nitrogen enriched air, argon, haloalkanes and mixtures thereof, the haloalkanes are selected from the group consisting of iodomethane (Halon 10001); bromomethane (Halon 1001), bromochloromethane, CH_2BrCl (Halon 1011), carbon tetrachloride (Halon 104), tribromofluoromethane (Halon 1103), dichlorofluoromethane (Halon 112), bromodifluoromethane (Halon 1201), dibromodifluoromethane (Halon 1202), bromochlorodifluoromethane, CF_2ClBr (Halon 1211), dichlorodifluoromethane (Halon 122), bromotrifluoromethane, CBrF_3 (Halon 1301), tetrafluoromethane (Halon 14), 1,2-dichlorotetrafluoroethane (Halon 242), dibromotetrafluoroethane, $\text{C}_2\text{Br}_2\text{F}_4$ (Halon 2402), hexafluoroethane (Halon 2600), and mixtures thereof.

Embodiments of this disclosure multistory structures comprising: a plurality of stories, each of the stories including: a floor, a ceiling, and a dead spaces interposed therebetween. The multi-story structures further include: a plurality of exterior walls, each of the exterior walls includes: an exterior wall outer portion; an exterior wall inner portion; an exterior wall framing system; an exterior wall air flow passageway interposed between the exterior wall outer portion and the exterior wall inner portion; and at least one exterior wall exhaust conduit. The multi-story further include: a plurality of interior walls; a plurality of interior spaces defined by the exterior walls and interior walls; and an air flow control system including: a processing unit; plurality of sensors. The multi-story structures further include: an air handling system comprising: an HVAC unit; an HVAC inlet conduit including: an HVAC surroundings inlet conduit having: a surrounding inlet; and a surrounding inlet control valve configured to supply unconditioned air to the HVAC unit; a return conditioned air conduit having: a return conduit control valve configured to control an amount of exhausted conditioned air to the HVAC unit; a plurality of floor return conduits, each of the floor return conduits includes: an inlet disposed in the each of the floor dead spaces; a floor return conduit control valve configured to control an amount of exhaust conditioned air supplied to the return conduit from each of the floor dead spaces; and a main HVAC outlet conduit including: a plurality of floor outlet conduits, each of the floor outlet conduits includes: a floor outlet conduit control valve configured to control an amount of conditioned air supplied to each of the floors; a plurality of interior space outlet conduits, each of the interior space outlet conduits having: a plurality of interior space outlets, each of the interior space outlets; and an interior space outlet conduit control valve configured to control an amount of conditioned air supplied to each of the interior space outlets, wherein one, some, or all of the control valves are active control valves, wherein at one or more of the sensors is disposed in each of the interior spaces and one or more of the sensors is disposed in the passageways, wherein the processing unit is in communication with the active control valves, the HVAC unit, and the sensors, and wherein the processing unit is configured to: receive input from the sensors, and control the HVAC unit and the active control valves based on the sensor inputs, thereby controlling unconditioned air flow rates into the HVAC, interior space air flow rates from the HVAC, return air flow rates into the HVAC, exhaust air flow rate to the surroundings, and passageway air flow rates through the passageways and into the surroundings.

In certain embodiments, each of the floor return conduits further includes: at least one return air control valve configured to control an amount of the return conditioned air to the return conduit.

Embodiments of the disclosure broadly relates to methods for controlling air flow in a single story structures, wherein the method comprises: receiving, via the processing unit, output from the sensors, and adjusting unconditioned air flow rates into the HVAC, interior space air flow rates from the HVAC, return air flow rates into the HVAC, exhaust air flow rate to the surroundings, and passageway air flow rates through the passageways and into the surroundings to maintain properties within the interior spaces and/or passageways at desired values.

In certain embodiments, in the adjusting steps, the properties include temperature and relative humidity in each of the interior spaces and moisture concentration in the passageways.

Embodiments of the disclosure broadly relates to methods for controlling air flow in a multi-story structures, wherein the method comprises: receiving, via the processing unit, output from the sensors, and adjusting unconditioned air flow rates into the HVAC, interior space air flow rates from the HVAC, return air flow rates into the HVAC, exhaust air flow rate to the surroundings, and passageway air flow rates through the passageways and into the surroundings to maintain properties within the interior spaces and/or passageways at desired values.

In certain embodiments, in the adjusting steps, the properties include temperature and relative humidity in each of the interior spaces and moisture concentration in the passageways.

Suitable Components for Use in the Disclosure

Suitable treating compositions include, without limitation, mold control compositions, pesticides, insecticides, fungicides, bactericides, rodent poisons, other control compositions, gases, fire retardant compositions, foams, and mixtures or combinations thereof.

Suitable gases include, without limitation, air, nitrogen enriched air, nitrogen (all grades), argon, carbon dioxide, haloalkanes and mixtures thereof.

Suitable haloalkanes include, without limitation, iodomethane (Halon 10001); bromomethane (Halon 1001), bromochloromethane, CH_2BrCl (Halon 1011), carbon tetrachloride (Halon 104), tribromofluoromethane (Halon 1103), dichlorofluoromethane (Halon 112), bromodifluoromethane (Halon 1201), (Halon 1202), bromochlorodifluoromethane, CF_2ClBr (Halon 1211), dichlorodifluoromethane (Halon 122), bromotrifluoromethane, CBrF_3 (Halon 1301), tetrafluoromethane (Halon 14), 1,2-dichlorotetrafluoroethane (Halon 242), dibromotetrafluoroethane, $\text{C}_2\text{Br}_2\text{F}_4$ (Halon 2402), hexafluoroethane (Halon 2600), and mixtures thereof.

Suitable foams include, without limitation, any of the gases and compositions introduced as a foam using a foaming agent or the co-introduction of a foaming agent and a gas or treating compositions. Suitable foaming agents include all surfactants and any other material the forms a foam when combined with a gas.

Suitable flow control valves or dampers for use herein include, without limitation, control valves or dampers manufactured by Johnson Controls, Belimo, Honeywell, Siemens, Greenheck, KMC Controls, or other similar manufacturers.

Suitable sensor for use in the disclosure include, without limitation, relative humidity sensors, moisture sensors, water sensors, physical sensors such as temperature sensors, pressure sensors, air flow sensors, etc., chemical sensors such as carbon dioxide sensors, carbon monoxide sensors, oxygen sensors, etc., microbial sensors such as mold sen-

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sors, fungus sensors, bacterial sensors, viral sensors, etc., other sensors, and combinations thereof.

Suitable air flow sensors for use herein include, without limitation, E+E Elektronik, Honeywell, Siemens, IFM Electronics, Air Logic, SDT, EGE, Delta OHM, Weber, EC, or other similar manufacturers.

Suitable temperature sensors for use herein include, without limitation, Analog Devices, Mitsumi, Melexis, Envirodata, Geokon, IST, SBE, Andigilog, Aquistar, Envco, STMicroelectronics, ZMD, Measurement Specialties, Microship, MicroDAQ, Smartec, Dallas Semiconductor, Slop Indicator, Honsberg, Maxim, ReGMet, GF Signet, Philips, Texas Instruments, FTDI Chip, Thermometrics, Spica Technology, or other similar manufacturers.

Suitable water/moisture sensors for use herein include, without limitation, Converge, Edyn, Oval Digital, Sengen, Galltec, E+E Elektronik, TE Connectivity, Kurabe Industrial Co., Ltd., Scale-Tron, Measurement Specialties, Michell Instruments, EMS Brno, Hoperf Electronics, GE Sensing, Hygrosens Instruments, Sensirion, Gefran, U.P.S.I., Precon, EME Systems, FTDI Chip, IST, Blue Earth, Vishay, or other similar manufacturers.

Suitable chemical sensors for use herein include, without limitation, carbon dioxide sensors, carbon monoxide sensors, mold sensors, or other similar chemical and microbial sensors.

Suitable physical sensors for use herein include, without limitation, pressure sensors, air compositional sensors, partial pressure sensors, or other physical sensors.

Suitable processors or processing units for use herein include, without limitation, without limitation, digital processing units (DPUs), analog processing units (APUs), any other technology that can receive motion sensor output and generate command and/or control functions for objects under the control of the processing unit, or mixtures and combinations thereof.

Suitable digital processing units (DPUs) include, without limitation, any digital processing unit capable of accepting input from a plurality of devices and converting at least some of the input into output designed to select and/or control attributes of one or more of the devices. Exemplary examples of such DPUs include, without limitation, microprocessor, microcontrollers, or the like manufactured by Intel, Motorola, Ericsson, HP, Samsung, Hitachi, NRC, Applied Materials, AMD, Cyrix, Sun Microsystems, Philips, National Semiconductor, Qualcomm, or any other manufacture of microprocessors or microcontrollers.

Suitable analog processing units (APUs) include, without limitation, any analog processing unit capable of accepting input from a plurality of devices and converting at least some of the input into output designed to control attributes of one or more of the devices. Such analog devices are available from manufacturers such as Analog Devices Inc.

Suitable interfaces and interface components for use herein include, without limitation, manufactured by Microsoft, Samsung, Apple, Linex, or other similar GUI or user interfaces.

Suitable HVAC unit and associated components for use herein include, without limitation, AAON, Airedale International Air Conditioning, Amana Corporation, American Standard Companies, ASI Controls, Baxi, BDR Thermea, BELIMO Holding AG, Buffalo Forge Company, CaptiveAire Systems, Carrier Corporation, Chunlan Group, CIAT Group, Climaveneta, Coleman Heating & Air Conditioning, Comfort Systems USA, Cylon Controls Ltd., Dae-woo Electronics, Daikin, Daikin Applied Americas, De Dietrich Remeha, DeLonghi, DeLclima, Electronic Tem-

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perature Instruments, Embraco, Fegyver-és Gépgyár, Fuji Electric, Fujitsu, Galanz, General Airconditioners, GlenDimplex, Goodman Global, Gree Electric, Haier, Hart & Cooley, Help-Link, Hisense, Hisense Kelon, Hitachi, J & E Hall, Jasun Filtration, Johnson Controls, Jøtul, Kenstar, KMC Controls, Lennox International, LG Electronics, Marinaire, Mirgor, Mitsubishi Electric, MPF Industry Group, Panasonic, PEC Power Electric Cooperation, Petra Engineering Industries Company, Rettig ICC, Rheem, Robert Bosch GmbH, Roberts-Gordon LLC, Sampo Corporation, Sanden Corporation, Sanhua, Sanyo, Shinco, Siragon, Stelrad, Taikisha, Tatung Company, Temperzone, Thermax, Thermo King, Toshiba, Trane, Vaillant Group, Viega, Viessmann, Voltas, Walton Group, Wellman Group, Westaflex, Whirlpool Corporation, Worcester, Bosch Group, Wrightsoft, Yilida, York International, or other similar manufacturers.

DETAILED DESCRIPTION OF THE DRAWINGS OF THE DISCLOSURE

Prior Art

Referring now to FIG. 1, a prior art structure, generally **100**, is shown to include a slab **102** situated on the ground **104** with a water resistant layer **106** on a bottom surface **108** of the slab **102**. The structure **100** also includes an outer wall **110** (shown here as a masonry and mortar wall) anchored to an inner wall **112** via ties **114**. The outer wall **110** includes weep holes **116**. The inner wall **112** includes studs **118**, a sheathing layer **120**, and a drywall layer **122** (e.g., gypsum board or plywood) and insulation.

The structure **100** also includes a roof **140**, a ceiling **142** supported by framing **144**, an attic **146**, and an interior **148**. The structure **100** also includes an HVAC unit **150** situated in the attic **146**. The HVAC unit **150** includes an inlet conduit **152** having an inlet **154** for pulling air **156** from the interior **148** into the HVAC unit **150**. The HVAC unit **150** also includes an outlet conduit **158** having an outlet **160** adapted to push conditioned air **162** into the interior **148**.

The structure **100** is ill equip to reduce, prevent, inhibit, or eliminate water, moisture and/or mold intrusion into exterior walls and as shown later herein, interior walls.

In U.S. Pat. No. 7,247,090B2, the inventor disclosed systems and methods for reducing water and moisture accumulation in external walls by providing a path way for conditioned air pass through the external walls and exhaust out through weep holes in the external walls. The present invention describes improved systems, apparatuses, and methods for reducing, removing, inhibiting, or eliminating water and/or moisture and/or their accumulation and/or mold growth in exterior and interior wall of habitable structures.

First Embodiment

Referring now to FIG. 2A, an embodiment of a structure of this disclosure, generally **200**, is shown to include a slab **202** situated on the ground **204** with a water resistant layer **206** on a bottom surface **208** of the slab **202**. It should be noted that the water resistant layer **206** may be disposed on the top surface of the slab **202** or both surfaces may include a water resistant layer **206**. The structure **200** also includes an outer wall **210** (shown here as a masonry and mortar wall) anchored to an inner wall **212** via ties **214**. The outer wall **210** includes a masonry sealer layer **216** and weep holes **218**. The inner wall **212** includes studs **220**, bracings between

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studs 222, a sheathing and/or drywall layer 224, and an optional drywall layer 226, which may be insulating or non-insulating. The ties 214 are angled up and affixed to the studs 220.

The structure 200 also includes an insulated roof 240, a ceiling 242 supported by framing 244, an attic 246, and an interior 248. The structure 200 also include an HVAC unit 250 situated in the attic 246. The HVAC unit 250 includes an inlet conduit 252 having an inlet 254, wherein the inlet 254 pulls unconditioned air 256 from the surrounding 258 into the HVAC unit 250. The HVAC unit 250 also includes an outlet conduit 260 having an outlet 262 adapted to push conditioned air 264 into the interior 248. The structure 200 also includes an outlet conduit 266 having an inlet 268 and an outlet 270, wherein the inlet 268 receives interior exhaust conditioned air 272 and the outlet 268 exhausts the interior exhaust conditioned air 272 into the attic 246. The structure 200 also includes a unidirectional valve 274 so that the exhausted conditioned air 272 flows (indicated by arrows) from the attic 246 into an air flow path 276 between the outer wall 210 and the inner wall 212 and out the weep holes 218 exhausting spent conditioned air 278 into the surrounding 258. The flow of the conditioned air from the attic through the pathways and out of the weep holes may be used to push water and/or moisture out of the walls to facilitate wall drying or to decrease/reduce or inhibit water/moisture accumulation or to decrease/reduce or inhibit mold growth in the outer wall and/or to control the moisture content in the walls.

Referring now to FIG. 2B, the structure of FIG. 2A further including an attic outlet conduit 280 having an inlet 282 and an outlet 284 for exhausting the attic exhaust conditioned air 286 from the attic 246. The attic outlet 280 may also include a control valve 288 so that a portion of the attic exhausted conditioned air 282 flows (indicated by arrows) from the attic 246 into the surrounding 258.

Second Embodiment

Referring now to FIG. 3A, another embodiment of a structure of this disclosure, generally 300, is shown to include a slab 302 situated on the ground 304 with a water resistant layer 306 on a bottom surface 308 of the slab 302. Again it should be noted that the water resistant layer 306 may be disposed on the top surface of the slab 302 or both surfaces may include a water resistant layer 306. The structure 300 also includes an outer wall 310 (shown here as a masonry and mortar wall) anchored to an inner wall 312 via ties 314. The outer wall 310 includes a masonry sealer layer 316 and weep holes 318. The inner wall 312 includes studs 320, bracings between studs 322, a sheathing and/or drywall layer 324, and an optional drywall layer 326, which may be insulating or non-insulating. The ties 314 are angled up and affixed to the studs 320.

The structure 300 also includes an insulated roof 340, a ceiling 342 supported by framing 344, an attic 346, and an interior 348. The structure 300 also includes an HVAC unit 350 situated in the attic 346. The HVAC unit 350 includes an inlet conduit 352 having an inlet 354, wherein the inlet 354 pulls unconditioned air 356 from the surrounding 358 into the HVAC unit 350. The HVAC unit 350 also includes an outlet conduit 360 having an outlet 362 adapted to push conditioned air 364 into the interior 348. The structure 300 also includes an outlet conduit 366 having an inlet 368, an outlet 370, and a control valve 371, wherein the inlet 368 receives interior exhaust conditioned air 372 and the outlet 370 exhausts the interior exhaust conditioned air 372 into the surrounding 358 and wherein the control valve 371

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controls the amount of air being exhausted through the outlet 370. The structure 300 may also includes a control valve 374 so that the exhausted conditioned air 370 flows (indicated by arrows) from the attic 346 into an air flow path or passageway 376 between the outer wall 310 and the inner wall 312 and out the weep holes 318 exhausting spent conditioned air 378 into the surrounding 358. However, in certain embodiments, the control valve 371 will be sufficient to direct the desired amount of conditioned air to the passageways, unless there are more than one passageways, then each passageway would include a control valve 371 that would control the air flow into each of the passageways/pathways. The flow of the conditioned air from the attic through the pathways or passageways and out of the weep holes may be used to push water and/or moisture out of the walls to facilitate wall drying or to decrease/reduce or inhibit water/moisture accumulation or to decrease/reduce or inhibit mold growth in the outer wall and/or to control the moisture content in the walls.

Referring now to FIG. 3B, the structure of FIG. 3A further including an attic outlet conduit 380 having an inlet 382 and an outlet 384 for exhausting the attic exhaust conditioned air 386 from the attic 346. The attic outlet 380 may also include a control valve 388 so that a portion of the attic exhausted conditioned air 382 flows (indicated by arrows) from the attic 346 into the surrounding 358.

Expanded Views of Walls

Referring now to FIGS. 4A&B, expanded views of embodiments of an exterior wall of a structure of this disclosure, generally 400, are shown. FIG. 4A depicts an expanded cross-sectional view of the walls, while FIG. 4B depicts successive cut away elevation view of the exterior wall.

Looking at FIG. 4A, the exterior wall 400 includes an outer wall assembly 402 comprising an outer wall 404 and a sealer layer 406 and an inner wall assembly 408 comprising studs 410, bracings between studs 412, and insulation 414 (shown only in FIG. 4B), an thermal and/or moisture barrier layer 416, and sheathing and/or drywall layer 418, which may be insulating or non-insulating. The exterior wall 400 also includes ties 420 anchoring the outer wall 404 to the studs 410 or the bracings between studs 412 of the inner wall assembly 408 and an air pathway or passageway 422 disposed between the outer wall assembly 402 and the inner wall assembly 408.

Looking at FIG. 4B, the wall 400 depicts successive cutaways showing each layer of the wall 400. The wall 400 includes the outer wall 404, then the sealer layer 406, then the air pathway 422, and then the studs 410, the bracings between studs 412, and the insulation 414. The wall 400 then include the barrier layer 416, and then the layer 418.

Third Embodiment

Referring now to FIG. 5A, another embodiment of a structure of this disclosure, generally 500, is shown to include an outer wall 502, an inner wall 504, an air flow control valve 506, an air flow pathway 508 interposed between the outer wall 502 and the inner wall 504, a roof 510, an attic 512, and a slab 514. The structure 500 includes a ceiling 516 and support framing 518.

The structure 500 also includes an air circulation assembly 520 including an HVAC unit 522 including an air intake conduit 524 having an inlet 526 and a control valve 527 for supplying unconditioned air 528 from an outside 530 to the HVAC unit 522 and an attic air conduit 532 having an air control valve 534 and an inlet 536 for supplying exhaust

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conditioned air 538 to the HVAC unit 522 from the attic 512. The circulation assembly 520 also includes an outlet conduit 540 having a control valve 542 and an outlet 544 for sending conditioned air 546 into an interior 548 of the structure 500. It should be recognized that the structure 500 may include a plurality of outlets 544 associated with the outlet conduit 540 for sending the conditioned air 546 into the interior 548. The circulation assembly 520 also includes an exhaust conduit 550 having an inlet 552 that withdraws exhaust conditioned air 554 from the interior 548 and an outlet 556 through which the exhaust conditioned air 554 is discharged into the attic 512. Some of the attic air 554 then exits through the control valve 506 and through the air flow pathway 508 shown by the dark grey arrows and out through weep holes or channels 558 into the outside 530.

Referring now to FIG. 5B, the structure of FIG. 5A further including an attic outlet conduit 560 having an inlet 562 and an outlet 564 for exhausting the attic exhaust conditioned air 566 from the attic 512. The attic outlet 560 may also include a control valve 568 so that a portion of the attic exhausted conditioned air 566 flows (indicated by arrows) from the attic 512 into the surrounding 530.

Fourth Embodiment

Referring now to FIG. 5C, another embodiment of a structure of this disclosure, generally 500, is shown to include an outer wall 502, an inner wall 504, an air flow control valve 506, an air flow pathway 508 interposed between the outer wall 502 and the inner wall 504, a roof 510, an attic 512, and a floor 514. The structure 500 includes a ceiling 516 and support framing 518.

The structure 500 also includes an air circulation assembly 520 including an HVAC unit 522 including an air intake conduit 524 having an inlet 526 and a control valve 527 for supplying unconditioned air 528 from an outside 530 to the HVAC unit 522 and an attic air conduit 532 having an air control valve 534 and an inlet 536 for supplying exhaust conditioned air 538 to the HVAC unit 522 from the attic 512. The circulation assembly 520 also includes an outlet conduit 540 having a control valve 542 and an outlet 544 for sending conditioned air 546 into an interior 548 of the structure 500. It should be recognized that the structure 500 may include a plurality of conduits 540, control valves 542, and outlets 544 for sending the conditioned air 546 into the interior 548. The circulation assembly 520 also includes an exhaust conduit 550 having an inlet 552 that withdraws exhaust conditioned air 554 from the interior 548 and an outlet 556 through which the exhaust conditioned air 554 is discharged into the attic 512. It should be recognized that the structure 500 may also include a plurality of conduits 550, inlets 552, and outlets 556 for withdrawing exhaust air 554 from the interior 548. The system 520 also includes a wall conditioned air conduit 560 having a control valve 562 through which conditioned air shown by the dark grey arrows is discharged into the air flow pathway 508 via an outlet 564 into the pathway 508, through the pathway 508, and out through weep holes or channels 558 into the outside 530. Again, it should be recognized that the structure 500 may also include a plurality of conduits 560, valves 562, and outlets 564 associated with a plurality of pathways 508 associated with exterior or interior walls.

Referring now to FIG. 5B, the structure of FIG. 5A further including an attic outlet conduit 566 having an inlet 568 and an outlet 570 for exhausting the attic exhaust conditioned air 572 from the attic 512. The attic outlet 560 may also include a control valve 574 so that a portion of the attic exhausted

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conditioned air 566 flows (indicated by arrows) from the attic 512 into the surrounding 530.

First Multistory Building Embodiment

Referring now to FIG. 6A, an embodiment of a multistory structure of this disclosure, generally 600, is shown to include a lower floor 602, outer/exterior wall or walls 604, inner walls 606, and air flow pathways 608 interposed between the outer wall or walls 604 and the inner walls 606, wherein the pathways or passageways 608 include inlets 610 having control valves 612 and outlet channels 614. The structure 600 also includes floors 616, ceilings 618, dead spaces 620 interposed between the intermediate floors 616 and the ceiling 618, and a roof 622. It should be recognized that the outer walls 604 may be transparent (made of glass) and the air flow passageways 608 comprise a gap between two transparent layer or panes of glass through which air may flow.

The structure 600 also includes an air circulation assembly or system 630 including an HVAC unit 632 including an unconditioned air intake conduit 634 having an inlet 636 and a control valve 638 for supplying unconditioned air 640 from an outside 642 to the HVAC unit 632 and optionally an exhaust conditioned air intake conduit 644 having an inlet 646 and a control valve 648 for supplying exhaust conditioned air 670 from the dead spaces 620 to the HVAC unit 632 via the conduits 644 and 634.

The system 630 also includes a main conditioned air conduit 652 from which floor conditioned air conduits 654 branch. Each of the floor conduits 654 include a flow control valve 656 and terminate in an outlet 658 passing through their respective ceilings 618 (only one per floor shown here, but of course, each floor will generally have a plurality of outlets 658). The floor conduits 654 discharge conditioned air 660 into interiors 662 between the floors 602 and 616 and the ceilings 618.

The system 630 also includes floor outlet conduits 664 having inlets 666 and outlets 668 for discharging exhaust conditioned air 670 into the dead spaces 620, which then progresses through the control valves 612 through the pathways 608 and out of the channels 614 into the outside 642 or through the conduits 644 and 634 to the HVAC unit 632.

The system 630 also includes a dead space return conduit 672 configured to allow exhaust conditioned air to flow from each dead space 620 with inlets 674 and an outlet 676 in the last floor dead space, wherein the return conduit 672 is configured to supply return exhausted conditioned air to the HVAC unit via the conduit 644. The return conduit 672 may include a control valve at each floor (not shown) to control the amount of air introduced into the return conduit 672 from each of the dead spaces 620.

Second Multistory Building Embodiment

Referring now to FIG. 6B, another embodiment of a multistory structure of this disclosure, generally 600, is shown to include a lower floor 602, outer/exterior wall or walls 604, inner walls 606 and air flow pathways 608 interposed between the outer wall or walls 604 and the inner walls 606 and exiting through exit channels 610. The structure 600 also includes floors 616, ceilings 618, dead spaces 620 interposed between the intermediate floors 610 and the ceiling 612, and a roof 622. Again, it should be recognized that the outer walls 604 may be transparent (made of glass) and the air flow passageways 608 comprise a gap between two transparent layer or panes of glass through which air may flow.

The structure 600 also includes an air circulation assembly 630 including an HVAC unit 632 including an unconditioned air intake conduit 634 having an inlet 636 and a

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control valve **638** for supplying unconditioned air **640** from an outside **642** to the HVAC unit **632** and optionally an exhaust conditioned air intake conduit **644** having an inlet **646** and a control valve **648** for supplying exhaust conditioned air **670** from the dead spaces **620** to the HVAC unit **632** via the conduit **634**.

The system **630** also include a main conditioned air conduit **652** from which floor conditioned air conduits **654** branch. Each of the floor conduits **654** include flow controllers **656** and terminate in outlets **658** passing through their respective ceilings **618** (only one per floor shown here, but of course, each floor will generally have a plurality of outlets **658**). The floor conduits **654** discharge conditioned air **660** into interiors **662** between the floors **616** and the ceilings **618**. The system **630** also includes pathway floor outlet conduits **664** having inlets **666** and outlets **668** for discharging exhaust conditioned air **670** into the dead spaces **620**.

The system **630** also includes conditioned air pathway conduits **672** including a control valves **674** and outlets **676** through which conditioned air shown in the dark gray arrows flows into the pathways **608** and then out through the channels **610**.

The system **630** may also include an exhaust conduit **678** having an outlet **680** and a control valve **682** for exhausting the air **670** from the dead spaces **620** to the outside **642**.
Third Multistory Building Embodiment

Referring now to FIG. **6C**, another embodiment of a multistory structure of this disclosure, generally **600**, is shown to include in addition to the other features disclosed above, exhaust channels **610** at each floor. Of course, the exhaust channels may be also include control valves to control the rate of air being exhausted from the passageways. Additionally, the passageways may extend between floors so that the flow rate of exhausted air may be controlled floor by floor. It should be recognized that all of the multi-story structures will include a return system for returning exhaust conditioned air to the HVAC unit. Additionally, all structures may include an air circulation systems that monitors carbon dioxide, carbon monoxide, oxygen levels, humidity, and temperature, and to control the air flow into and out of the interior spaces and into and out of the passageways to maintain these properties at desirable levels.
Pier and Beam Building Embodiment

Referring now to FIGS. **7A&B**, embodiment of a pier and beam structure of this disclosure, generally **700**, is shown to include a pier and beam facade **702** having air vents **704** and solid portions **706** and piers **708** extending from the ground or below the ground **710** to a floor **712** and a dead space **714** between the floor **712** and the ground **710**. The structure **700** also includes outer walls **716**, inner walls **718**, and air flow pathways or passageways **720**. The structure **700** also includes interiors **722**. It should be recognized that the pier and beam structure **700** may include any or all of the circulation systems, control systems, and sensor systems set forth herein. It should be recognized that there does not have to be an air vent **704** between every pair of piers **708** and the number of air vents **704** are generally a matter of design.
Embodiments Wall Air Distribution Systems

Referring now to FIGS. **8A-D**, four embodiments of wall air distribution systems of this disclosure, generally **800**, **830**, **850**, and **870**, respectfully, are shown.

Looking at FIG. **8A**, the wall air distribution system **800** includes an inlet **802**, an air distribution conduit **804** having an outlet **806** that passes through a ceiling **808** and ends within an air pathway **810** disposed between an outer wall section and an inner wall section (not shown) of an interior or exterior wall **812**. Conditioned air proceeds from an

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HVAC unit (not shown) and flows into the inlet **802** through the distribution conduit **804** and out the outlet **806** depicted by the dark gray arrows and then through the pathway **810** and out through weep holes or outlet channels (not shown), but fully described herein.

Looking at FIG. **8B**, the wall air distribution system **830** includes a plurality of inlets **832**, a distribution conduit **834**, and a plurality of outlets **836**. Each of the outlets **836** passes through a ceiling **838** and ends within one of a plurality of air pathways **840** formed by a plurality of pathway dividers **842** within an interior or exterior wall **844**. The pathways **840** are disposed between an outer wall section and an inner wall section (not shown). Conditioned air proceeds from an HVAC unit (not shown) and flows into the inlets **832** through the distribution conduit **834** and out the outlets **836** and then through the pathways **840** as depicted by the dark gray arrows and out through weep holes or outlet channels (not shown).

Looking at FIG. **8C**, the wall air distribution system **850** includes an inlet **852**, an air distribution conduit **854** having a plurality of flow control valves **856** and a plurality of outlets **858**. Each of the outlets **858** passes through a ceiling **860** and ends within a plurality of air pathways **862** formed by a plurality of pathway dividers **864** within an interior or exterior wall **866**. The pathways **862** are disposed between an outer wall section and an inner wall section (not shown). Conditioned air proceeds from an HVAC unit (not shown) and flows into the inlet **852** through the distribution conduit **854** and out the outlets **858** as depicted by the dark gray arrows and then through the pathways **862** and out through weep holes (not shown), where the air flow through each of the pathways **862** is controlled by the control valves **856**.

Looking at FIG. **8D**, the wall air distribution system **870** includes a plurality of inlets **872**, a plurality of distribution conduits **874**, a plurality of flow control valves **876**, and a plurality of outlets **878**. Each of the outlets **878** passes through a ceiling **880** and ends within one of a plurality of air pathways **882** formed by a plurality of pathway dividers **884** within an interior or exterior wall **886**. The pathways **882** are disposed between an outer wall and an inner wall of an interior or exterior wall (not shown). Conditioned air proceeds from an HVAC unit (not shown) and flows into the inlets **872** through the distribution conduits **874** and out the outlets **878** depicted by the dark gray arrows and then through the pathways **882** and out through weep holes (not shown), where the air flow through each of the pathways **882** is controlled by the control valves **876**.

Embodiment Wall Air Distribution Systems

Referring now to FIG. **9A**, a structure, generally **900**, is shown to include an outer wall **902** and an inner wall **904** having a thermal and/or moisture barrier layer **906**, studs **908**, a sheathing and/or drywall layer **910**, which may be insulating or non-insulating, and an air flow pathway **912** between an inner surface of the outer wall **902** and the barrier layer **906**. The structure **900** also includes a ceiling **914** upon which is disposed an air conditioning and distribution assembly **920**.

The assembly **920** includes an air inlet **922**, an air inlet conduit **924** connected to an HVAC unit **928**. The HVAC unit **928** includes an outlet **930** and an outlet conduit **932**. The outlet conduit **932** splits into a main conduit **934** having a main control valve **936** and a secondary conduit **938** having a secondary control valve **940**. The main conduit **934** supplies conditioned air to outlets **942** associated with registers **944** for heating or cooling an interior (not shown) of the structure **900**, while the secondary conduit **938** supplies conditioned air to the air flow pathway **912** so that

conditioned air may flow down the pathway **912** pushing out water and/or moisture through weep holes or drainage channels and drying the outer wall **902** and/or the inner wall **904**. Of course, the same wall assembly may be used to dry out interior walls, but with different drainage and exhaust channels as described herein. The assembly **920** also includes an outlet **946** associated with an outlet conduit (not shown) for exhausting air from the interior of the structure **900** to the surroundings or to an attic or similar space associated with the structure **900**.

Looking at FIG. 9B, generally **900**, is shown to include an outer wall **902** and an inner wall **904** having a thermal and/or moisture barrier layer **906**, studs **908**, a sheathing and/or drywall layer **910**, which may be insulating or non-insulating, and air flow pathways **912a-d** between an inner surface of the outer wall **902** and the barrier layer **906**. The structure **900** also includes a ceiling **914** upon which is disposed an air conditioning and distribution assembly **920**. The pathways **912** are isolated by corner dividers **916** that prevent air exchange between the pathways **912a-d**.

The assembly **920** includes an air inlet **922**, an air inlet conduit **924** connected to an HVAC unit **928**. The HVAC unit **928** includes an outlet **930** and an outlet conduit **932**. The outlet conduit **932** splits into a main conduit **934** having a main control valve **936** and a secondary conduit **938** having a secondary control valve **940**. The main conduit **934** supplies conditioned air to outlets **942** associated with registers **944** for heating or cooling an interior of the structure **900** (not shown). The assembly **920** also includes an outlet **946** associated with an outlet conduit (not shown) for withdrawing air from the interior of the structure and exhausting either to the surroundings or to an attic or similar space associated with the structure **900**.

The secondary conduit **938** terminated in a distribution unit **948** that supplies conditioned air to four wall conduits **950a-d**, where the conduits **950a-d** include flow control valves **952a-d**. The four conduits **950a-d** supply conditioned air to the four pathways **912a-d** so that the conditioned air may flow down the pathway **912a-d** pushing out water and moisture through weep holes or drainage channels and drying the outer wall **902** and the inner wall **904**. The control valve **952a-d** controls the amount of conditioned air that flows into each pathway **912a-d** so that each wall section may be dehydrated at different rates depending on water, moisture and/or mold content.

Of course, the same wall assembly may be used to dry out interior walls, but with different drainage and exhaust channels as described herein.

Looking at FIG. 9C, generally **900**, is shown to include an outer wall **902** and an inner wall **904** having a thermal and/or moisture barrier layer **906**, studs **908**, a sheathing and/or drywall layer **910**, which may be insulating or non-insulating, and sixteen air flow pathways **912a-p** between an inner surface of the outer wall **902** and the barrier layer **906**. The structure **900** also includes a ceiling **914** upon which is disposed an air conditioning and distribution assembly **920**. The pathways **912a-p** are isolated by dividers **916** that prevent air exchange between the pathways **912a-p**.

The assembly **920** includes an air inlet **922**, an air inlet conduit **924** connected to an HVAC unit **928**. The HVAC unit **928** includes an outlet **930** and an outlet conduit **932**. The outlet conduit **932** splits into a main conduit **934** having a main control valve **936** and a secondary conduit **938** having a secondary control valve **940**. The main conduit **934** supplies conditioned air to outlets **942** associated with registers **944** for heating or cooling an interior of the structure **900** (not shown). The assembly **920** also includes

an outlet **946** associated with an outlet conduit (not shown) for withdrawing air from the interior of the structure and exhausting either to the surroundings or to an attic or similar space associated with the structure **900**.

The secondary conduit **938** terminated in a distribution unit **948** that supplies conditioned air to four main wall conduits **950a-d**. The four main wall conduits **950a-d** end in sixteen flow control valves **952a-p** and feed conditioned air to sixteen air distribution conduits **954a-p** that in turn supply conditioned air to the sixteen pathways **912a-p** so that the conditioned air may flow down the pathway **912a-d** pushing out water and moisture through weep holes or drainage channels and drying the outer wall **902** and the inner wall **904**. The control valve **952a-p** controls the amount of conditioned air that flows into each pathway **912a-p** so that each wall section may be dehydrated at different rates depending on water, moisture and/or mold content.

Of course, the same wall assembly may be used to dry out interior walls, but with different drainage and exhaust channels as described herein.

Fifth Embodiment

Referring now to FIG. 10A, an embodiment of a structure of this disclosure, generally **1000**, is shown to include a slab **1002** situated on the ground **1004**. The structure **1000** also includes an outer wall **1006** and an inner wall **1008**. The outer wall **1006** and the inner wall **1008** may be of any type or construction. The structure **1000** also includes an outside HVAC unit **1010** disposed on an HVAC slab **1012** and including an air intake **1014** having a screen **1016** for pulling air **1018** from the surroundings into the HVAC unit **1010**. The structure **1000** also includes an outlet **1020** and an outlet conduit **1022** leading through a roof **1024**, through an attic space **1026**, and through a ceiling **1028** supported by framing **1030** to an interior inlet **1032** from which conditioned air **1034** is discharged into an interior **1036** of the structure **1000**. The structure **1000** also includes an interior outlet **1038** associated with an outlet conduit **1040** having an exhaust outlet **1042** for withdrawing air from the interior **1036** and discharging exhaust air **1044** into the attic **1026** or surrounding. The HVAC unit **1010** also includes a conditioned air wall discharge conduit **1046** including a control valve **1048** for controlling a rate and amount of conditioned air discharged into a wall pathway **1050** flows through the pathway **1048** and out exhaust holes **1052**, which may be weep holes if the outer wall is masonry and mortar or just discharge channels if the outer wall is of some other construction. It should be recognized that in multistory building, the space into which the air is exhausted and that may be used to dry out walls may be any dead space including the space between floors or any other dead space that would allow air to flow through the walls and out the outlet channels. Of course, it should also be recognized that some structures the air flow may be from the bottom of the wall up especially in multistory buildings.

Referring now to FIG. 10B, the structure of FIG. 10A further including an attic outlet conduit **1054** having an inlet **1056** and an outlet **1058** for exhausting the attic exhaust conditioned air **1060** from the attic **1026**. The attic outlet conduit **1054** may also include a control valve **1062** so that a portion of the attic exhausted conditioned air **1060** flows (indicated by arrows) from the attic **1026** into the outside. Expanded Views of Exterior Walls with Sensors

Referring now to FIG. 11A, an expanded view of an embodiment of an outer wall assembly of a structure of this disclosure, generally **1100**, is shown to an outer wall **1102**

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and a sealer layer 1104 and an inner wall assembly 1106 including studs 1108, an moisture and/or vapor barrier layer 1110, an insulating sheathing and/or drywall layer 1112 situated on a slab 1114. The wall assembly 1100 also includes ties 1116 anchoring the outer wall 1102 to the studs 1108 of the inner wall assembly 1106. The wall assembly 1100 also includes an air pathway 1118 disposed between the outer wall 1102 and the barrier layer 1110. The wall assembly 1100 also includes exhaust ports, holes, or channels 1120 through which conditioned air is exits the pathway 1118.

The wall assembly 1100 also includes wall temperature sensors 1122, moisture/water sensors 1124 and an air flow rate sensor 1126. The sensors and control valves are all controlled by a controller describe more fully in FIG. 13.

Looking at FIG. 11B, an expanded view of another embodiment of an outer wall assembly of a structure of this disclosure, generally 1150, is shown to an outer wall 1152 and a sealer layer 1154 and an inner wall assembly 1156 including studs 1158, an moisture and/or vapor barrier layer 1160, an insulating sheathing and/or drywall layer 1162 situated on a slab 1164. The wall assembly 1150 also includes ties 1166 anchoring the outer wall 1152 to the studs 1158 of the inner wall assembly 1156. The wall assembly 1150 also includes an air pathway 1168 disposed between the outer wall 1152 and the barrier layer 1160. The wall assembly 1150 also includes exhaust ports, holes, or channels 1170 through which conditioned air is exits the pathway 1168. The wall assembly 1156 also includes an wall air conduit 1172 for supplying conditioned air to the pathway 1168 and a treating solution reservoir 1174 disposed on a beam 1176, a treating solution treating dispensing unit 1178, a dispensing conduit 1180 ending in a nozzle 1182 for spraying the treating solution 1184 into the pathway 1168.

The wall assembly 1100 also includes wall temperature sensors 1186, moisture/water sensors 1188 and an air flow rate sensor 1190. The sensors and control valves are all controlled by a controller describe more fully in FIG. 13.

The wall assembly 1100 may also include a control valve 1192 associated with the channels 1170 so that the passageways or pathways 1168 may be pressurized or cut off from air flow. Thus, the passageways or pathways 1168 may be flooded with a fire retardant gas or foam and pressurized to help impede the spread of a fire.

Expanded Views of Interior Walls with Sensors and Dispensing Assemblies

Referring now to FIG. 12A, an expanded view of an embodiment of an inner wall assembly of a structure of this disclosure, generally 1200, is shown to include studs 1202, a first insulating sheathing and/or drywall layer 1204 and a second insulating sheathing and/or drywall layer 1206. The inner wall assembly 1200 also includes an air pathway 1208 disposed in a center 1210 of the assembly 1200. The assembly 1200 also includes a conditioner air conduit 1212 that supplies conditioned air into the pathway 1208 and an exhaust conduit 1214 that exhaust air, water, and/or moisture from the interior of the wall assembly 1200 disposed in a sub floor 1216 supported by support members 1218 on a slab 1220 and is vented out of the structure 1200 via an outlet 1222, where conditioned air flow is shown by the dark gray arrows. Of course, the pathway 1208 may require holes or other opening to be provided through the inner wall cross members (not shown) so that the conditioned is capable of reducing, decreasing, and/or eliminating water, moisture or mold accumulation in the wall assembly 1200 or facilitate drying the wall assembly 1200 after a water, moisture or mold event.

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The wall assembly 1200 also includes wall temperature sensors 1224, moisture/water sensors 1226, and an air flow rate sensor 1228. The sensors and control valves are all controlled by a controller describe more fully in FIG. 13.

Looking at FIG. 12B, an expanded view of another embodiment of an inner wall assembly of a structure of this disclosure, generally 1250, is shown to include studs 1252, a first insulating sheathing and/or drywall layer 1254 and a second insulating sheathing and/or drywall layer 1256. The inner wall assembly 1250 also includes an air pathway 1258 disposed in a center 1260 of the assembly 1250. The assembly 1250 also includes a conditioner air conduit 1262 that supplies conditioned air into the pathway 1258 and an exhaust conduit 1264 that exhaust air, water, and/or moisture from the interior of the wall assembly 1250 disposed in a sub floor 1266 supported by support members 1268 on a slab 1270 and is vented out of the structure 1250 via an outlet 1272, where conditioned air flow is shown by the dark gray arrows. The assembly 1250 also includes a treating solution dispensing conduit 1274 ending in a nozzle 1276 for spraying a treating solution 1278 into the pathway 1258. The treating solution reservoir and dispensing mechanism are not shown, but may be as shown in FIG. 11B or of any other form used in the art to dispense a liquid, gas, or foam treating composition.

Of course, the pathway 1258 may require holes or other opening to be provided through the inner wall cross members (not shown) so that the conditioned air is capable of reducing, decreasing, and/or eliminating water, moisture or mold accumulation in the wall assembly 1250 or facilitate drying the wall assembly 1250 after a water, moisture or mold event.

The wall assembly 1250 also includes wall temperature sensors 1280, moisture/water sensors 1282, and an air flow rate sensor 1284. The sensors and control valves are all controlled by a controller describe more fully in FIG. 13.

The wall assembly 1200 may also include a control valve 1286 associated with the exhaust conduit 1264 so that the passageways or pathways 1258 may be pressurized or cut off from air flow. Thus, the passageways or pathways 1258 may be flooded with a fire retardant gas and pressurized to help impede the fire.

While FIGS. 11B and 12B demonstrate that introduction of compositions into the passageways to control pests, insects, molds, fungi, bacteria, and/or other pests, the compositions being dispensed may be fire suppressants and/or retardants that may be used to retard or impede the progression of fire through a structure.

Control System Embodiment

Referring now to FIG. 13, an embodiment of a control system, generally 1300, of a structure of this disclosure is shown to include a power supply 1302 (which may of course be the a power outlet in the structure or a dedicated power supply or power conditioning unit), a processor or processing unit 1304, one or a plurality of user interface or panel units 1306 including touch screen 1308 (or other interactive user interface), and optionally one or a plurality of remote control units 1310 (which may be a cell phone, a lap top, or desk top computer). The interface or panel units 1306 may be associated with each interior space or a plurality of interior spaces in communication with the processor 1304. By setting air properties in each of the interior spaces via the panels 1306, the air properties in each of the interior spaces or coupled interior spaces may be separately and independently controlled by the processor 1304. Additionally, air properties in each of the passageways may be entered into the panels 1306 communicated to the processor 1304 so that

the air properties in each passageway may be independently and separately controlled. Of course, the processor 1304 may be programmed so that air properties in each of the interior spaces may be collectively controlled based on the number of people and heat generating equipment in each of the interior spaces. The processor 1304 include memory, communication hardware and software, and firmware or other software for controlling HVAC units, sensors, control valves, and dispensing units that dispense treating solutions. It should be recognized that passive air control valves are not be controlled by the processing unit 1304, but may be controlled by the air flow.

The power supply 1302 supplies power to the processor 1304, the interface 1306, an HVAC unit 1312, control valves 1316, temperature sensors 1318, water/moisture sensors 1320, air flow sensors 1322, and treating units 1324 via power supply wires 1326. While I have disclosed using temperature sensors, water/moisture sensors, and air flow sensors, the systems of this invention may also include other sensors such as chemical sensors 1328 (e.g., carbon dioxide, carbon monoxide, mold, smoke, etc.) or other physical sensors 1330 (e.g., pressure, etc.) and any other sensors 1332 that may assist in reducing, removing, inhibiting and/or eliminating water, moisture, and/or mold or drying walls exposed to a water, moisture, and/or mold event.

The processor 1304 is in two way communication with the interface 1306, the remote control 1310, the HVAC unit 1312, the sensors 1318, 1320, 1322, 1328, 1330, and 1332, and the treating units 1324 via communication pathways 1334 (shown here as wires for the interface 1306, the HVAC unit 1312, the blower 1314, the sensors 1318, 1320, 1322, 1328, 1330, and 1332, and the treating units 1320 or a communication pathway 1336 (shown here as wireless for the remote control 1310). Of course, all of the communication pathways 1324 may also be wireless or a combination of wired and wireless communication pathways. Additionally, the interface 1306, the remote control 1310, the HVAC unit 1312, the blower 1314, the sensors 1316 and 1318 all include communication hardware and software for communication with the processor 1304, whether via wired or wireless communication pathways. Additionally, the remote control unit 1310 includes its own power supply, memory, processor, interface, touch screen or the like and human cognizable output devices such as speakers.

Window Systems

Referring now to FIGS. 14A-C, three views of an embodiment of a window system, generally 1400, are shown. Looking at FIG. 14A, the window system 1400 includes an exterior wall assembly 1410, an interior wall assembly 1430, and a window assembly 1450.

The exterior wall assembly 1410 includes an exterior wall 1412 having an interior 1414, an outer surface 1416, and an inner surface 1418. The exterior wall assembly 1410 may also include a moisture and/or air barrier layer 1420 having an inner surface 1422 and an outer surface 1424. The exterior wall 1412 may be constructed out of brick, stucco, planks, any other exterior building material and mixtures and combinations thereof.

interior wall assembly 1430 includes an air/moisture barrier layer 1432 having an inner surface 1434 an outer surface 1436. The interior wall assembly 1430 also includes a sheathing layer 1438 having an inner surface 1440 and an outer surface 1442. The interior wall assembly 1430 also includes an interior 1444. The interior 1444 includes an insulation layer 1446 having an inner surface 1448 and an outer surface 1450, and studs 1452. The interior wall assembly 1430 also includes an interior wall 1454 having an inner

surface 1456 an outer surface 1458. The interior wall assembly 1430 also includes a steel angle member 1460 and a wall air flow pathway 1462. The air flow pathway 1462 is disposed between the inner surface 1418 of the exterior wall 1412 or an inner surface 1422 of the optional moisture and/or air barrier layer 1420.

The window assembly 1470 includes shim members 1472, a window frame member 1474, a wire mesh member 1476, and a window sill 1478. The window assembly 1470 also includes an outer single, double, or triple pane window 1480 having an inner surface 1482 and an outer surface 1484. The window assembly 1470 also includes an interior single, double, or triple pane window 1486 having an inner surface 1488 and an outer surface 1490. The window assembly 1470 also includes a window air flow pathway 1492 interposed between the outer surface 1484 of the outer window 1480 and an inner surface 1488 of the inner window 1486. The wall air flow pathway 1462 is connected to the window air flow pathway 1466. These two air flow pathways 1462 and 1492 provide cooling or heating to the walls and windows and provide dry air to remove water, moisture, and/or contaminants in the walls or windows.

Experiments of the Disclosure

Buildings are a necessary part of our society to provide a safe, habitable, and protected environment. Typically, the more safe, habitable, and protected the environment; the higher the cost to operate the building. Outside air necessary to diminish indoor air contamination (COVID-19, mold, radon, etc.) results in increased energy costs. Structural Ventilation of the Building Envelope is a system that introduces continuously conditioned positively pressured outdoor air into the interior or habitable spaces of a structure to be exhausted into the building envelope and subsequently to the exterior. Heat gain and loss through the building envelope are significantly reduced. Multiple studies of existing buildings have been performed that indicate that moisture accumulation and damage, air contamination, and inadequate ventilation simultaneously occur in whole or in part. WUFI analyzes the hygrothermal properties of the building envelope. Analyses indicate that the reduced moisture and regulated temperatures within the building envelope, transmitted by the supply of clean outside air into the living space of the structure, diminishes mold growth and structural deterioration of the building envelope and provides significant hygrothermal resistances of the building envelope resulting in reduced HVAC equipment sizing and operating costs. Structural Ventilation of the Building Envelope is applicable to residential, institutional, medical, retail, and commercial buildings; either new construction or retrofit. Installation is performed by the current building construction workforce with currently available materials; which result in lower installation costs. Structural Ventilation of the Building Envelope is the next generation of building ventilation. A simple solution for a complex society.

The Issues

In the mid 1970's, the oil embargo resulted in rising energy costs. In an attempt to reduce these costs, the minimum building outside air intake was reduced to 2.5 L/s (5 cfm) per person. This minimum still exists today. The recent onset of COVID-19 brought the realization that more fresh outside air was necessary to dilute and remove the indoor air contamination. The EPA and others have stated this recommendation. Sick Building Syndrome, also identified as Building Related Illness, a condition where individuals exhibit sickness when occupying a particular building,

has been linked to airborne mold contamination. Other airborne contaminants are also blamed. More fresh outside air, necessary to dilute and remove the indoor air contamination, as well as moisture and humidity control, are required. The EPA and others have stated this recommendation. Radon Contamination, prevalent over more than half of the United States, is the leading cause of lung cancer following tobacco smoke. Radon is an odorless radioactive gas that is emitted from the earth and collects within structures. The best solutions are to ventilate beneath the building, if accessible, and to increase fresh outside air intake to dilute and remove the indoor air contamination. The EPA and others have stated this recommendation. Clearly, proper ventilation is the solution.

The Conflict

Buildings are a necessary part of our society to provide a safe, habitable, and protected environment. Typically, the more safe, habitable, and protected the environment; the higher the cost to operate the building. The conflict arises between this environmental goal and the costs necessary to maintain this goal. One viewpoint is "The best solution to breathe the contaminant free indoor air is to inject clean outdoor air into the building to dilute the contamination. The worst solution is to recirculate and filter indoor air". The opposing viewpoint is "The worst solution to minimize HVAC operating costs is to inject clean outdoor air into the building to dilute the contamination. The best solution is to recirculate and filter indoor air". Structural Ventilation of the Building Envelope simultaneously accomplishes both goals by minimizing indoor air contamination and reducing energy costs.

Conventional Systems

The majority of buildings are temperature controlled and humidity adjusted insulated enclosures. Referring to FIG. 15A a typical example of a HVAC system recirculates air within the building, air is drawn from within the space, conditioned, and circulated back into the space. The attic and cavity walls are unconditioned but ventilated with outdoor air. Interior air is temperature adjusted through recirculation but humidity is inadequately maintained. Outside air is introduced through infiltration through doors, windows, the perimeter walls, and attic with variable results. The building envelope collects moisture and offers conventional thermal resistance.

The building, generally 1500 of FIG. 15A is shown to include a roof 1502, a brick veneer 1504, brick ties 1506, and weep holes 1508. The building 1500 is situated on a ground surface 1510 and supported by a foundation 1512. The building 1500 also include an attic 1514, a HVAC system 1516, attic insulation 1518, gypsum wall board 1520, wall insulation 1522, and exterior sheathing 1524. Unconditioned air enters through soffit vents, ridge vents, and/or the weep holes 1508. The HVAC system 1516 cools the unconditioned air and recirculates it back into an interior of the building 1526 through vents not shown here. Additionally, unconditioned air enters the interior 1526 via infiltration through the attic insulation 1518 and wall insulation 1522.

Some buildings have heat exchangers, as shown in FIG. 15B, where some fresh outside air is introduced into the HVAC system and some indoor air is exhausted out of the building through the HVAC system. The heat exchanger works with the passage of outdoor air next to indoor air, allowing the outdoor air to be tempered closer to the indoor air temperature.

The building, generally 1550 of FIG. 15B is shown to include a roof 1552, a brick veneer 1554, brick ties 1556, and weep holes 1558. The building 1500 is situated on a ground surface 1560 and supported by a foundation 1562.

The building 1550 also include an attic 1564, a HVAC/heat exchange system 1566, attic insulation 1568, gypsum wall board 1570, wall insulation 1572, and exterior sheathing 1574. Unconditioned air enters through soffit vents, ridge vents, and/or the weep holes 1558. The HVAC system 1566 cools the unconditioned air and recirculates it back into an interior of the building 1576 through vents not shown here. Additionally, unconditioned air enters the interior 1576 via infiltration through the attic insulation 1568 and wall insulation 1572.

Other than small energy savings from the heat exchanger, the building characteristics are identical to FIG. 15A. This leads us the need for Structural Ventilation (SV) of the Building Envelope.

Structural Ventilation of the Building Envelope

The system was developed in the early 2000's and patented in 2007 and 2022. The system development began with the aid of the MOIST software and elaborate spreadsheets filled with hygrothermal equations. Technology has since improved. Structural Ventilation of the Building Envelope (SV) is a system that introduces continuously conditioned, positively pressured, outdoor air into the interior habitable spaces of a structure where the outdoor air is cleaned, temperature adjusted, and humidity adjusted by a Dedicated Outdoor Air System (DOAS) before entering the habitable spaces of a structure. The positively pressured indoor air is exhausted into the building envelope and subsequently to the exterior.

Part A—The Mechanical System

The volume of outside air entering the structure is as specified by ASHRAE and is based upon the building occupancy and use.

Outside air is cleaned, filtered, and temperature and humidity adjusted through the DOAS for heating and cooling. This low volume/high velocity air is continuously injected into the building to regulate building temperature and humidity. Indoor air contamination is reduced since indoor air is not recirculated; rather exhausted out of the occupied spaces as recommended by ASHRAE. State of the art mechanical and ducting systems are currently available to economically achieve these goals. Further discussion of these mechanical and ducting systems are beyond the scope of this paper.

Part B—The Building Envelope Ventilation

SV indoor air is exhausted out of the occupied spaces and into the building envelope to regulate the temperature and humidity of the building envelope.

Referring to FIG. 16A a typical example of SV is shown. The HVAC system draws outside air into the unit where it is cleaned and temperature and humidity adjusted. This air is circulated within the building, exhausted into the attic, exhausted into the exterior walls, and released into the exterior environment. The attic and cavity walls are conditioned and ventilated with indoor air.

The building, generally 1600 of FIG. 16A is shown to include a roof 1602, a 2" spray foam insulation 1604, a solid 1x blocking 1606, a 4" zero deflecto backflow preventer 1608, and weep holes 1610. The building 1600 is situated on a ground surface 1612 and supported by a foundation 1614. The building 1600 also include an attic 1616, a heating, cooling, dehumidification (HCD) system 1618, and a wall air pathway 1620. Unconditioned air 1630 enters through the HCD system 1618. The HCD system 1618 cools or heats and dehumidifies unconditioned air 1630 and circulates conditioned air 1632 into an interior 1622 of the building 1600. A portion of the conditioned air 1634 into the attic 1616, flows through the preventer 1608, flows through the wall air

pathway **1620** as conditioned wall air **1636**, and exits out of the weep holes **1610** allowing moisture reduction in the walls and improved heating and cooling via the conditioned air **1636** flowing through the wall pathways **1620** and out the weep holes **1610**.

Outside air is conditioned prior to distribution through the structure. Infiltration is eliminated. The building envelope is conditioned to near interior conditions. Differences for building retrofit or new construction include the absence of ventilation openings to the exterior (soffit vents, ridge vents, turtle vents) an updated HCC system and ducting (high velocity/low volume), the use of insulation beneath the roofing, the installation of a backflow preventer at top of wall, and the use of additional sealants at material terminations of the building envelope.

Referring now to FIG. **16B**, a flooded structure equipt with SV is shown. The sheathing behind the brick veneer, when wet, can become contaminated and soft. The sheathing is a part of the shear walls to keep the building upright. In order to remove and replace the moisture damaged sheathing and conform to existing construction, removal and replacement of the brick veneer is necessary. This SV repair method allows the brick veneer to remain intact, provides drying of the structure for this and future events, and minimizes repair costs.

The building, generally **1650**, of FIG. **16B** is shown to include a roof **1652**, a 2" spray foam insulation **1654**, a solid 1x blocking **1656**, a 4" zero deflecto backflow preventer **1658**, brick ties **1660**, mortar fill holes and cracks in the brick veneer **1662**, and weep holes **1664**. The building **1650** is situated on a ground surface **1666** and supported by a foundation **1668**. The building **1650** also include an attic **1670**, a HCD system **1672**, and a wall air pathway **1674**. The building **1650** also includes attic insulation **1676**, brick scaler **1678**, wall layers **1680**, a foil faced insulation layer **1682**, 2x solid blocking **1684**, and wall insulation **1686**. The HCD system **1672** cools or heats and dehumidifies unconditioned air **1690** and circulates conditioned air **1692** into an interior **1688** of the building **1650**. A portion of the conditioned air **1694** into the attic **1670**, flows through the preventer **1658**, flows through the wall air pathway **1620** as conditioned wall air **1696**, and exits out of the weep holes **1664** allowing moisture reduction in the walls and improved heating and cooling via the conditioned air **1696** flowing through the wall air pathways **1674** and out the weep holes **1664**.

Moisture Limitations

Many structural components are compromised by moisture. For example, mold can occur when the surface relative humidity of a building component exceeds 70%(±) and temperatures are above 4° C. (40° F.). Wood rot occurs at moisture contents of 30% or greater and with temperatures ranging between 10° C. (50° F.) and 40° C. (100° F.). Nail corrosion occurs at relative humidities greater than 60%; a maximum 20% wood moisture content is preferred to minimize nail corrosion within the wood. The Structural Board Association states that Exposure 1 OSB materials are intended to be used in covered and well ventilated areas; this exposure relates to an OSB moisture content of 8% to 9% or a solid wood moisture content of 12% to 15%. Relative humidities above 80% are not recommended. ASHRAE, ANSI, and EPA concur that the relative humidity of interior ambient air should be between 30% and 60% with temperatures between 20° ° C. (68° F.) and 27° ° C. (81° F.) at all times.

Excess moisture compromises structural components. WUFI Analyses

WUFI, developed with the assistance of Oak Ridge National Laboratory (ORNL), analyzes the hygrothermal properties of building envelopes. WUFI allows realistic calculations of the transient coupled heat and moisture transport in multi-layer building envelopes based upon exterior weather data and normal interior climates. WUFI is an acronym for Wärme Und Feuchte Instationär, which translated means heat and moisture transiency.

Referring now to FIG. **17**, an office building was analyzed using WUFI. The occupant requires 1.6 Air Changes per Hour (ACH) located in Houston, Texas having the indicated dimensions. The office building, generally **1700**, includes walls **1710**. The walls **1710** includes a brick veneer **1712**, an air gap **1714**, a Tyvek® (moisture barrier) layer **1716**, an OSB sheathing **1718** supported by stud framing, batt insulation **1720**, and gypsum board supported by stud framing **1722**.

The office building **1700** also include a roof **1730**. The roof **1730** includes a composite shingle roofing **1732**, an OSB sheathing supported rafters **1734**, a spray-on insulation layer for structure ventilation systems **1736**, attic air **1738**, batt insulation **1740**, and gypsum board ceiling supported joists **1742**.

Attic Assembly

Referring now to FIGS. **18A&B**, plots of temperature and water content versus cross-section of attic air and ceiling insulation of a conventional HVAC system and attic air of a structural ventilation (SV) HVAC system of the office building of FIG. **17**, respectively, are shown. The roof **1730** is at the top, and the ceiling **1742** is at the bottom.

Looking at FIG. **18A**, the plot of the conventional HVAC system temperature profile versus cross-section, generally, **1800**, and a plot of a conventional HVAC system moisture content and humidity versus cross-section, generally **1820**.

The plot **1800** displays a temperature line **1802** across the roof **1730**, attic air **1738** in an attic interior **1739**, the ceiling insulation **1740**, and the gypsum board **1742**.

The plot **1820** displays a moisture content line **1822**, an area representing the relative humidity of attic air **1824**, an average attic air humidity line **1826**, an area representing a moisture content of the attic air **1828**, the interior attic air **1804**, and the ceiling insulation **1806** across the roof **1730**, attic air **1738** in an attic interior **1739**, the ceiling insulation **1740**, and the gypsum board **1742**.

Over a one year period, at the conventional system, the surface relative humidity exceeds 80%. The this humidity level mold growth would be expected, nail corrosion would be expected, and OSB moisture contents are elevated.

Looking at FIG. **18B**, the plot of a SV HVAC system temperature profile versus cross-section, generally, **1850**, and a plot of a SV HVAC system moisture content and humidity versus cross-section, generally **1870**.

The plot **1850** displays an attic air temperature line **1852** measured across the roof **1730**, the attic air **1738** in an attic interior **1739**, the ceiling insulation **1740**, and the gypsum board **1742**.

The plot **1870** displays a moisture content line **1872**, an area **1874** representing the relative humidity of attic air **1738**, an average attic air humidity line **1876**, an area **1878** representing a moisture content of the attic air **1738** in an attic interior **1739**, and the ceiling insulation **1740** measured across the roof **1730**, attic air **1738** in an attic interior **1739**, the ceiling insulation **1740**, and the gypsum board **1742**.

At the SV system, these conditions do not exist except at the base of the roof insulation.

Referring now to FIGS. **19A&B**, a plot of temperature values and relative humidity values over a year at the top

surface of the ceiling insulation (0.18407 inch fiber glass insulation) in the conventional HVAC system and a plot of temperature values and relative humidity values over a year at the top surface of the ceiling insulation (0.204576 fiber glass insulation) in the SV HVAC system of the office building of FIG. 17, respectively, are shown. With the SV system, relative humidity is reduced 35% in the summer months and the temperatures increase 15° C. (25° F.) in the winter months and decrease 12° C. (10° F.) in the summer months. The RSI-6 (R-35) assembly becomes RSI_{HT}-38 (R_{HT}-215). R_{HT} is the Hygrothermal Resistance as defined herein.

Wall Assembly

Referring now to FIGS. 20A&B, plots of temperature and water content versus cross-section of the wall assembly of a conventional HVAC system and a structural ventilation (SV) HVAC system of the office building of FIG. 17, respectively, are shown measured across the brick veneer 1712 is on the left, the gypsum board 1724 is on the right, and the moisture barrier 1716, the air gap 1714, air 1715 in the air gap, the sheathing 1718, and the insulation 1722 are interposed therebetween.

Looking at FIG. 20A, a plot of a conventional HVAC system temperature profile versus cross-section, generally, 2000, and a plot of a conventional HVAC system moisture content and humidity versus cross-section, generally 2020.

The plot 2000 displays a temperature line 2002 measured across the brick veneer 1712, the moisture barrier 1716, the air gap 1714, the air 1715 in the air gap, the sheathing 1718, the insulation 1722, and the gypsum board 1724.

The plot 2020 displays a wall assembly moisture content line 2022, an area 2024 representing a moisture content measured across the brick veneer 1712, the moisture barrier 1716, the air gap 1714, the gap air 1715, the sheathing 1718, the insulation 1722, and the gypsum board 1724, an average wall assembly relative humidity line 2026 measured across the brick veneer 1712, the moisture barrier 1716, the air gap 1714, the gap air 1715, the sheathing 1718, the insulation 1722, and the gypsum board 1724, and an area 2028 representing a relative humidity measured across the brick veneer 1712, the moisture barrier 1716, the air gap 1714, the gap air 1715, the sheathing 1718, the insulation 1722, and the gypsum board 1724.

Over a one year period, at the conventional system, exclusive of components outside the air cavity, the surface relative humidity exceeds 80%. Mold growth is expected. Nail corrosion is expected. OSB moisture contents are elevated.

Looking at FIG. 20B, a plot of a SV HVAC system temperature profile versus cross-section, generally, 2050, and a plot of a SV HVAC system moisture content and humidity versus cross-section, generally 2070.

The plot 2050 displays a wall assembly temperature line 2052 measured across the brick veneer 1712, the moisture barrier 1716, the air gap 1714, the gap air 1715, the sheathing 1718, the insulation 1722, and the gypsum board 1724.

The plot 2070 displays a wall assembly moisture content line 2072, an area 2074 representing a moisture content measured across the brick veneer 1712, the moisture barrier 1716, the air gap 1714, the gap air 1715, the sheathing 1718, the insulation 1722, and the gypsum board 1724, an average relative humidity line 2076 measured across the brick veneer 1712, the moisture barrier 1716, the air gap 1714, the gap air 1715, the sheathing 1718, the insulation 1722, and the gypsum board 1724, and an area 2078 representing a relative humidity measured across the brick veneer 1712, the mois-

ture barrier 1716, the air gap 1714, the gap air 1715, the sheathing 1718, the insulation 1722, and the gypsum board 1724.

At the SV system, these conditions do not exist.

Referring now to FIGS. 21A&B, plots of temperature values and relative humidity values over a year at the exterior surface of the oriented strand board (OSB) sheathing in the conventional HVAC system (0.007039 inches) and the SV HVAC system (0.007039 inches) of the office building of FIG. 17, respectively, are shown, the temperatures and relative humidities of the conventional system and SV system, respectively, are shown. With the SV system, relative humidity is reduced 15% in the summer months; the temperatures increase 8° C. (15° F.) in the winter months and decrease 12° C. (10° F.) in the summer months. The RSI-3.5 (R-20) assembly becomes RSI_{HT}-7 (R_{HT}-40).

Window Assembly

Referring now to FIGS. 22A&B, plots of temperature and water content in the conventional system and the SV system versus cross-section of the window assembly of the office building of FIG. 17, respectively, are shown measured across the air gap 1714 from the left window to the right window.

Looking at FIG. 22A, a plot of a conventional HVAC system temperature profile versus cross-section, generally, 2200, and a plot of a conventional HVAC system moisture content and humidity versus cross-section, generally 2220.

The plot 2200 displays a temperature line 2202 measured across measured across the air gap 1714.

The plot 2220 displays a window assembly moisture content line 2222, an area 2224 representing a moisture content measured across the air gap 1714, an average window assembly relative humidity line 2226 measured across the air gap 1714, and an area 2228 representing a relative humidity measured across the air gap 1714.

Looking at FIG. 22B, a plot of a SV HVAC system temperature profile versus cross-section, generally, 2250, and a plot of a SV HVAC system moisture content and humidity versus cross-section, generally 2270.

The plot 2250 displays a wall assembly temperature line 2252 measured across the air gap 1714.

The plot 2270 displays a wall assembly moisture content line 2272, an area 2274 representing a moisture content measured across the air gap 1714, an average relative humidity line 2276 measured across the air gap 1714, and an area 2278 representing a relative humidity measured across the air gap 1714.

The interior glass surface humidity is lowered 50%. In this instance, it would be preferable to use insulated glass at the exterior pane to minimize condensation.

Referring now to FIGS. 23A&B, the temperatures and relative humidities at the interior surface of the window of the conventional HVAC system and SV HVAC system, respectively, are shown. With the SV system, relative humidity is reduced 50% in the winter months; the temperatures increase 15° C. (25° F.) in the winter months and decreases 8° C. (15° F.) in the summer months. The RSI-0.2 (R-1 assembly becomes RSI_{HT}-0.9 (R_{HT}-5).

If insulated glass was used at both panes, the thermal resistances would be additive. So, if each insulated pane is RSI-0.35 (R-2), ΣR=RSI_{HT}-1.6 (R_{HT}-9) of the window assembly. SV controls the building envelope temperatures and humidities.

Additional WUFI Analyses

The School Building, defined in FIG. 24, was also analyzed using WUFI. The building requires 3.1 ACH. Location: Houston, Texas ***

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Thermal Resistance (RSI and R) and Hygrothermal Resistance (RSI_{HT} and R_{HT})

The designation RSI (R) is the static thermal resistance of a component, or assembly, due to the physical characteristics of the materials.

Hygrothermal Resistance, RSI_{HT} (R_{HT}) combines the static thermal resistance of the material with the dynamic coupled heat and moisture transport through the materials generated by the conditioned airflow through the building envelope.

Hygrothermal Resistance increases with exhaust air cavity height/width and exhaust airflow rate.

Attics

Referring now to FIGS. 25A&B, the R_{HT} of the attic of the Office Building of FIG. 17 and the R_{HT} of the attic of the School of FIG. 24, respectively, are shown. In each, static thermal resistance is approximately RSI-9 (R-50). With SV, the hygrothermal resistance increases significantly with greater attic height and greater air flow rates. Doubling the attic height increases R_{HT} by a factor of 1.7. Doubling the air flow rate increases R_{HT} by a factor of 1.7. At the Office Building, the Attic ACH=3 resulting in an RSI_{HT} -35 (R_{HT} -200); an increase by a factor of 4. At the School Building, the Attic ACH=4.7 resulting in an RSI_{HT} -50 (R_{HT} -275); an increase by a factor of 5.5.

Walls

Referring now to FIGS. 25C&D, the R_{HT} of the walls of the Office Building FIG. 17 and the R_{HT} of the walls of the School Building FIG. 24, respectively, are shown. In each, static thermal resistance is approximately RSI-3.5 (R-20). With SV, the hygrothermal resistance increases significantly with greater air flow rates. At the Office Building, southern exposure of the brick veneer wall results in little change. Doubling the air flow rate increases R_{HT} by a factor of 1.6. The Cavity ACH=180 resulting in an RSI_{HT} -7 (R_{HT} -40); an increase by a factor of 2. At the School Building, southern exposure of the brick veneer wall results in an increase of R_{HT} . Doubling the air flow rate increases R_{HT} by a factor of 1.5. The Cavity ACH=585 resulting in an RSI_{HT} -19 (R_{HT} -110); an increase by a factor of 5.5.

Windows

Referring now to FIG. 26, the R_{HT} of the windows of both buildings are shown. In each, static thermal resistance of a single pane clear window is approximately RSI-0.2 (R-1). Southern exposure and dark tint of the window results in an increase of R_{HT} by a factor of 1.5.

With SV, the hygrothermal resistance increases significantly with greater air flow rates. Doubling the air flow rate increases R_{HT} by a factor of 1.3. At the Office Building, the Window ACH=180 resulting in RSI_{HT} -0.9 (R_{HT} -5). At the School Building, the Window ACH=585 resulting in RSI_{HT} -1.4 (R_{HT} -8).

As stated earlier, R and R_{HT} are additive. If we install thermal windows each side of the air stream that are each RSI-0.35 (R-2), the Office Building windows would attain RSI_{HT} -1.6 (R_{HT} -9). The School Building windows would attain RSI_{HT} -2.1 (R_{HT} -12). Each increase is significant.

SV significantly increases the hygrothermal resistance of the building envelope.

HVAC Sizing

The Office Building was analyzed with Manual J based upon the lowest RHT of the winter months at various locations with-in the United States.

Referring now to FIG. 27, the heating and cooling requirements of the building in various cities are shown. At each city, the conventional heating and cooling requirements are shown as the tall orange and blue bars. The SV heating

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and cooling requirements are shown as the short orange and blue bars. The grey bars are the percent SV unit sizing is of conventional unit sizing. The percentages range from 69% to 75%.

SV results in smaller equipment sizes regardless of location.

Energy Usage for Heating

The Office Building was analyzed with the WUFI Post Processor based upon the RHT of winter monthly averages that are based upon hourly calculations. Typically, the RHT is greater before and after the coldest month. Referring now to FIG. 28, the annual heating energy usage of the building at several cities are shown. The conventional heating and SV heating requirements are presented as separate groups. The SV group shows percentages in the bars that indicates the percent SV energy usage is of conventional usage. The percentages range from 49% to 71%.

SV results in significant energy savings regardless of location.

The School Building was also analyzed with the WUFI Post Processor. Referring now to FIG. 29, the annual heating energy usage of the school building at several cities are shown. The conventional heating and SV heating requirements are presented as separate columns. The SV columns show percentages in the bars that indicates the percent SV energy usage is of conventional usage. The percentages range from 66% to 75%.

SV results in significant energy savings regardless of location.

Moisture Issues

Moisture infiltration is a common problem with building envelopes. Moisture, typically from the exterior environment, passes through defective joints in the exterior finishes and roofing. The joints typically occur at material terminations such as window and door perimeters, roofing and wall penetrations, roof to wall interfaces, and the like.

As noted earlier, moisture can also develop due to naturally occurring moisture due to hygrothermal activity within the building envelope. This is most prevalent when low permeability materials are installed.

The following building envelope moisture content readings were compared to WUFI analyses. Red is well above expected values, yellow is slightly above expected values, green similar to expected values, and blue is below expected values.

Expected moisture content readings of interior components were determined by air temperature and humidity readings.

Referring now to FIGS. 30A-C, moisture content readings at the walls of a one-story, one-year old, daycare are shown. Gypsum sheathing was measured at the exterior faces of the wall studs and gypsum board was measured at the interior of the structure. The interior gypsum board finish moisture contents are elevated at the perimeter of the building. Window leaks are evident. The gypsum sheathing behind the brick veneer is very wet.

In this instance, the weep holes at the base of the wall were plugged with mortar, rainwater was seeping behind the veneer, and the windows leaked.

Referring now to FIGS. 31A-C, moisture content readings at the third floor walls of a three-story, three-year old, house are shown. Gypsum board, baseboards, and windowsills were measured at the interior of the structure.

The gypsum board finish moisture contents are moderate at the perimeter walls and the interior walls. The baseboards are typically moist beneath windows and at doors. Window-sill reading are elevated.

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Window and door leakage are the prime contributors. Moisture is collecting within the building envelope.

Referring now to FIGS. 32A-D, moisture content readings at the third floor walls and ceiling of a three-story, four-year old, house are shown. Gypsum board, baseboards, window-sills, and the ceiling were measured at the interior of the structure. The interior gypsum board moisture contents are elevated at the central area of the house and some exterior walls. The baseboards are more moist at the interior of the building as opposed to the perimeter. Windowsill readings are elevated. The ceiling is typically moist with greater moisture at the right side of the drawing.

Moisture is collecting within the attic due to the lack of adequate ventilation. Moisture from this source is infiltrating the interior and perimeter walls of the upper floor. Window and door leakage are also issues.

Embodiments of the Disclosure

Embodiment 1. A structure comprising:

a floor;

a roof;

one or more exterior walls, each exterior wall including:

an exterior wall air path way including:
one or more exterior wall air pathway inlets, and one or more exterior wall air pathway outlets;

a plurality of interior walls;

one or more of exterior window assemblies, each of the

one or more of exterior window assemblies including:
an exterior window,

an interior window, and

a window air pathway interposed between the exterior window and the interior window and including:

a window air flow pathway inlet, and

a window air flow pathway outlet,

a plurality of interior spaces defined by the exterior and interior walls;

an HVAC unit including:

one or more HVAC inlet conduits, each of the one or more HVAC inlet conduits including:

an HVAC inlet including:

an HVAC inlet control valve configured to control a supply of unconditioned air or a mixture of unconditioned air and exhaust conditioned air to the HVAC unit;

one or more HVAC exhaust conduits, each of the one or more HVAC exhaust conduits including:

an HVAC exhaust inlet, and

an HVAC exhaust outlet configured to support an air flow of the exhaust conditioned air out of the structure, to the HVAC inlet, or any combination thereof;

one or more HVAC interior space outlet conduits, each of the one or more HVAC interior space outlet conduits including:

an HVAC interior space outlet including:

an HVAC interior space outlet control valve configured to control a supply of conditioned air to each of the interior spaces;

one or more HVAC exterior wall conduits, each of the one or more HVAC exterior wall conduits including:

an HVAC exterior wall outlet including:

an HVAC exterior wall control valve configured to support an exterior wall pathway air flow of conditioned air to each of the exterior wall pathways,

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wherein wall pathway air flows reduce, remove, and/or eliminate moisture in the exterior walls and increase heating and cooling efficiencies, and

an HVAC window conduit for each of the one or more of exterior window assemblies, each HVAC window conduit including:

an HVAC window conduit inlet, and

an HVAC window conduit inlet,

wherein window pathway air flows reduce, remove, and/or eliminate moisture in the exterior windows and increase heating and cooling efficiencies.

Embodiment 2. The structure of Embodiment 1, wherein: one, some, or all of the exterior wall air flow pathway outlets include:

a passive one-way or unidirectional valve or an active control valve to stop air or water flow into each of the exterior wall air flow pathway outlets, and/or

one, some, or all of the exterior window air flow pathway outlets include:

a passive one-way or unidirectional valve or an active control valve to stop air or water flow into each of the exterior wall air flow pathway outlets.

Embodiment 3. The structure of Embodiment 1, wherein each of the plurality of interior walls includes:

one or more interior wall air flow pathways, each of the one or more interior wall air flow pathways including:

a interior wall air pathway inlet, and

a interior wall air pathway outlet; and

the HVAC unit further including:

one or more HVAC interior wall conduits, each of the one or more HVAC interior wall conduits including:

an HVAC interior wall outlet including:

an HVAC interior wall control valve configured to support an interior wall pathway air flow of conditioned air to each of the interior wall pathways,

wherein the interior air flows reduce, remove, and/or eliminate moisture in the interior walls, and

one, some, or all of the interior wall air flow pathway outlets include:

a passive one-way or unidirectional valve or an active control valve to stop air or water flow into each of the exterior wall air flow pathway outlets.

Embodiment 4. The structure of Embodiment 1, wherein each of the one or more exterior walls includes:

an outer surface including:

an insulating layer;

a weather-resistant layer disposed on an outer surface of the insulating layer,

a sheathing,

a liquid barrier, or

any combination thereof, and/or

an inner surface including:

an insulating layer,

a weather-resistant layer disposed on an outer surface of the insulating layer,

a sheathing,

a liquid barrier, or

any combination thereof.

Embodiment 5. The structure of Embodiment 1, further comprising:

an air flow control system including:

a processing unit; and

plurality of sensors;

wherein one, some, or all of the control valves are active control valves,

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wherein the processing unit is in communication with the active control valves, and
 wherein the processing unit is configured to:
 receive input from the sensors, and
 control the HVAC unit, and
 control the active control valves so that air flow rates are controlled into the interior spaces and into the pathways, and
 during a 24 hour period of time, the processing unit is further configured to:
 adjust the air flow into each of the interior spaces and into each pathway so that one or more interior space properties and/or one or more pathway properties are maintained at desired values.

Embodiment 6. The structure of Embodiment 5, wherein the sensors comprise temperature sensors, water/moisture sensors, air flow sensors, chemical sensors, other physical sensors, and combinations thereof.

Embodiment 7. The structure of Embodiment 5, wherein one or more of the sensors are disposed in the air flow pathways, in the interior spaces, or in both.

Embodiment 8. The structure of Embodiment 5, wherein: all of the control valves are active control valves, and
 during a 24 hour period of time, the processing unit is further configured to:
 adjust the air flow into each of the interior spaces and into each pathway so that one or more interior space properties and/or one or more pathway properties are maintained at desired values.

Embodiment 9. The structure of Embodiment 8, wherein the one or more properties include moisture concentrations in the interior spaces, moisture concentrations in the pathways, temperatures in the interior spaces, temperatures in the pathways, pressures in the interior spaces, pressures in the pathways, air flow rates in the interior spaces, air flow rates in the pathways, sensed chemical concentrations in the interior spaces, sensed chemical concentrations in the pathways, microbial concentrations in the interior spaces, microbial concentrations in the pathways or mixtures and combinations thereof.

Embodiment 10. The structure of Embodiment 1, further comprising:

a dispensing assembly associated with one, some, or all of the pathways and including:
 a reservoir containing a treating composition;
 a dispensing supply conduit having:
 a dispensing assembly control valve,
 a dispensing head at its distal end,

wherein:
 the dispensing assembly control valve is configured to control an amount of treating solution or gas introduced into one, some, or all of the pathways, the treating composition comprising a mold control composition, a pesticide, an insecticide, a fungicide, a bactericide, a rodent poison, other control compositions, a gas, a fire retardant compositions, a foam, or mixtures or combinations thereof; the gas comprising nitrogen, nitrogen enriched air, argon, haloalkanes or mixtures thereof;

the haloalkanes comprising iodomethane; bromomethane, bromochloromethane, carbon tetrachloride, tribromofluoromethane, dichlorofluoromethane, bromodifluoromethane, dibromodifluoromethane, bromochlorodifluoromethane, dichlorodifluoromethane, bromotrifluoromethane, tetrafluorometh-

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ane, 1,2-dichlorotetrafluoroethane, dibromotetrafluoroethane, hexafluoroethane, or mixtures thereof.

Embodiment 11. A multistory structure comprising:

a plurality of stories, each of the stories including:
 a floor,
 a ceiling, and
 a dead space disposed above the ceiling;
 one or more exterior walls, each of the one or more exterior walls including:
 an exterior wall air flow pathway; and
 at least one exterior wall exhaust conduit;
 a plurality of exterior window assemblies, each of the plurality of exterior window assemblies including:
 an exterior window,
 an interior window, and
 a window air pathway interposed between the exterior window and the interior window and including:
 a window air flow pathway inlet, and
 a window air flow pathway outlet,
 a plurality of interior walls, each of the interior walls including:
 an interior wall air flow pathway; and
 at least one interior wall outlet conduit;
 a plurality of interior spaces defined by the one or more exterior walls and interior walls;
 an air handling system comprising:
 an HVAC unit;
 an HVAC inlet conduit including:
 an HVAC surroundings inlet conduit having:
 a surrounding inlet including:
 a surrounding inlet active control valve configured to supply unconditioned air to the HVAC unit;
 a return conditioned air conduit including:
 a return conduit active control valve configured to control an amount of exhausted conditioned air to the HVAC unit;
 a plurality of floor return conduits, each of the floor return conduits including:
 an inlet disposed in the each of the floor dead spaces including:
 a floor return conduit active control valve configured to control an amount of exhaust conditioned air supplied to the return conduit from each of the dead spaces;
 a main HVAC outlet conduit including:
 a plurality of floor outlet conduits, each of the floor outlet conduits including:
 a floor outlet conduit active control valve configured to control an amount of conditioned air supplied to each of the floors;
 a plurality of interior space outlet conduits, each of the interior space outlet conduits having:
 a plurality of interior space outlets, each of the interior space outlets having an interior space outlet conduit active control valve configured to control an amount of conditioned air supplied to each of the interior space outlets; and
 exterior window outlet conduit active control valves configured to control an amount of conditioned air supplied to each of the plurality of exterior windows;
 one or more HVAC exhaust conduits, each of the one or more HVAC exhaust conduits including:

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a HVAC exhaust conduit active control valve configured to control an amount of conditioned air exhausted to the surroundings; and
an air flow control system including:

a processing unit coupled to or associated with a plurality of sensors, the active control valves and the air handling system;

one or more of the sensors is disposed in each of the interior spaces;

one or more of the sensors is disposed in the pathways; wherein the processing unit is configured to:

receive input from the sensors; and

control the HVAC unit and the active control valves based on the sensor inputs to control unconditioned air flow rates into the HVAC, interior space air flow rates from the HVAC, return air flow rates into the HVAC, exhaust air flow rate to the surroundings, and pathway air flow rates through the pathways and into the surroundings.

Embodiment 12. The structure of Embodiment 11, wherein each of the floor return conduits further including: at least one return air active control valve configured to control an amount of the return conditioned air to the return conditioned air conduit.

Embodiment 13. The structure of Embodiment 11, wherein the sensors comprise temperature sensors, water/moisture sensors, air flow sensors, chemical sensors, other physical sensors, and combinations thereof.

Embodiment 14. The structure of Embodiment 13, wherein, during a 24 hour period of time, the processing unit is further configured to:

adjust the air flow into each of the interior spaces and into each pathway based the received sensor input so that one or more interior space properties and/or one or more pathway properties are maintained at desired values.

Embodiment 15. The structure of Embodiment 14, wherein the one or more properties include moisture concentrations in the interior spaces, moisture concentrations in the pathways, temperatures in the interior spaces, temperatures in the pathways, pressures in the interior spaces, pressures in the pathways, air flow rates in the interior spaces, air flow rates in the pathways, sensed chemical concentrations in the interior spaces, sensed chemical concentrations in the pathways, microbial concentrations in the interior spaces, microbial concentrations in the pathways or mixtures and combinations thereof.

Embodiment 16. The structure of Embodiment 11, further comprising:

a dispensing assembly associated with one, some, or all of the pathways and including:

a reservoir containing a treating composition;

a dispensing supply conduit including:

a dispensing assembly control valve,

a dispensing head at its distal end,

wherein:

the dispensing assembly control valve is configured to control an amount of treating solution or gas introduced into one, some, or all of the pathways, the treating composition comprising a mold control composition, a pesticide, an insecticide, a fungicide, a bactericide, a rodent poison, other control compositions, a gas, a fire retardant compositions, a foam, or mixtures or combinations thereof; the gas comprising nitrogen, nitrogen enriched air, argon, haloalkanes, or mixtures thereof;

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the haloalkanes comprising iodomethane; bromomethane, bromochloromethane, carbon tetrachloride, tribromofluoromethane, dichlorofluoromethane, bromodifluoromethane, dibromodifluoromethane, bromochlorodifluoromethane, dichlorodifluoromethane, bromotrifluoromethane, tetrafluoromethane, 1,2-dichlorotetrafluoroethane, dibromotetrafluoroethane, hexafluoroethane, or mixtures thereof.

CLOSING PARAGRAPH OF THE DISCLOSURE

All references cited herein are incorporated by reference. Although the disclosure has been disclosed with reference to its preferred embodiments, from reading this description those of skill in the art may appreciate changes and modification that may be made which do not depart from the scope and spirit of the disclosure as described above and claimed hereafter.

I claim:

1. A structure comprising:

a floor;

a roof;

one or more exterior walls, each exterior wall including:

an exterior wall air path way including:

one or more exterior wall air pathway inlets, and

one or more exterior wall air pathway outlets;

a plurality of interior walls;

one or more of exterior window assemblies, each of the

one or more of exterior window assemblies including:

an exterior window,

an interior window, and

a window air pathway interposed between the exterior window and the interior window and including:

a window air flow pathway inlet, and

a window air flow pathway outlet,

a plurality of interior spaces defined by the exterior and interior walls;

an heating, ventilation, and air conditioning (HVAC) unit including:

one or more HVAC inlet conduits, each of the one or more HVAC inlet conduits including:

an HVAC inlet including:

an HVAC inlet control valve configured to control a supply of unconditioned air or a mixture of unconditioned air and exhaust conditioned air to the HVAC unit;

one or more HVAC exhaust conduits, each of the one or more HVAC exhaust conduits including:

an HVAC exhaust inlet, and

an HVAC exhaust outlet configured to support an air flow of the exhaust conditioned air out of the structure, to the HVAC inlet, or any combination thereof;

one or more HVAC interior space outlet conduits, each of the one or more HVAC interior space outlet conduits including:

an HVAC interior space outlet including:

an HVAC interior space outlet control valve configured to control a supply of conditioned air to each of the interior spaces;

one or more HVAC exterior wall conduits, each of the one or more HVAC exterior wall conduits including:

an HVAC exterior wall outlet including:

an HVAC exterior wall control valve configured to support an exterior wall pathway air flow of conditioned air to each of the exterior wall pathways, wherein wall pathway air flows reduce,

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remove, and/or eliminate moisture in the exterior walls and increase heating and cooling efficiencies, and

an HVAC window conduit for each of the one or more of exterior window assemblies, each HVAC window conduit including:

an HVAC window conduit inlet, and

an HVAC window conduit inlet,

wherein window pathway air flows reduce, remove, and/or eliminate moisture in the exterior windows and increase heating and cooling efficiencies.

2. The structure of claim 1, wherein:

one, some, or all of the exterior wall air flow pathway outlets include:

a passive one-way or unidirectional valve or an active control valve to stop air or water flow into each of the exterior wall air flow pathway outlets, and/or one, some, or all of the exterior window air flow pathway outlets include:

a passive one-way or unidirectional valve or an active control valve to stop air or water flow into each of the exterior wall air flow pathway outlets.

3. The structure of claim 1, wherein each of the plurality of interior walls includes:

one or more interior wall air flow pathways, each of the one or more interior wall air flow pathways including:

a interior wall air pathway inlet, and

a interior wall air pathway outlet; and

the HVAC unit further including:

one or more HVAC interior wall conduits, each of the one or more HVAC interior wall conduits including:

an HVAC interior wall outlet including:

an HVAC interior wall control valve configured to support an interior wall pathway air flow of conditioned air to each of the interior wall pathways,

wherein the interior air flows reduce, remove, and/or eliminate moisture in the interior walls, and

one, some, or all of the interior wall air flow pathway outlets include:

a passive one-way or unidirectional valve or an active control valve to stop air or water flow into each of the exterior wall air flow pathway outlets.

4. The structure of claim 1, wherein each of the one or more exterior walls includes:

an outer surface including:

an insulating layer;

a weather-resistant layer disposed on an outer surface of the insulating layer,

a sheathing,

a liquid barrier, or

any combination thereof, and/or

an inner surface including:

an insulating layer,

a weather-resistant layer disposed on an outer surface of the insulating layer,

a sheathing,

a liquid barrier, or

any combination thereof.

5. The structure of claim 1, further comprising:

an air flow control system including:

a processing unit; and

plurality of sensors;

wherein one, some, or all of the control valves are active control valves,

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wherein the processing unit is in communication with the active control valves, and

wherein the processing unit is configured to:

receive input from the sensors, and

control the HVAC unit, and

control the active control valves so that air flow rates are controlled into the interior spaces and into the pathways, and

during a 24 hour period of time, the processing unit is further configured to:

adjust the air flow into each of the interior spaces and into each pathway so that one or more interior space properties and/or one or more pathway properties are maintained at desired values.

6. The structure of claim 5, wherein the sensors comprise temperature sensors, water/moisture sensors, air flow sensors, chemical sensors, other physical sensors, and combinations thereof.

7. The structure of claim 5, wherein one or more of the sensors are disposed in the air flow pathways, in the interior spaces, or in both.

8. The structure of claim 5, wherein:

all of the control valves are active control valves, and

during a 24 hour period of time, the processing unit is further configured to:

adjust the air flow into each of the interior spaces and into each pathway so that one or more interior space properties and/or one or more pathway properties are maintained at desired values.

9. The structure of claim 8, wherein the one or more properties include moisture concentrations in the interior spaces, moisture concentrations in the pathways, temperatures in the interior spaces, temperatures in the pathways, pressures in the interior spaces, pressures in the pathways, air flow rates in the interior spaces, air flow rates in the pathways, sensed chemical concentrations in the interior spaces, sensed chemical concentrations in the pathways, microbial concentrations in the interior spaces, microbial concentrations in the pathways or mixtures and combinations thereof.

10. The structure of claim 1, further comprising:

a dispensing assembly associated with one, some, or all of the pathways and including:

a reservoir containing a treating composition;

a dispensing supply conduit having:

a dispensing assembly control valve,

a dispensing head at its distal end,

wherein:

the dispensing assembly control valve is configured to control an amount of treating solution or gas introduced into one, some, or all of the pathways, the treating composition comprising a mold control composition, a pesticide, an insecticide, a fungicide, a bactericide, a rodent poison, other control compositions, a gas, a fire retardant compositions, a foam, or mixtures or combinations thereof;

the gas comprising nitrogen, nitrogen enriched air, argon, haloalkanes or mixtures thereof;

the haloalkanes comprising iodomethane; bromomethane, bromochloromethane, carbon tetrachloride, tribromofluoromethane, dichlorofluoromethane, bromodifluoromethane, dibromodifluoromethane, bromochlorodifluoromethane, dichlorodifluoromethane, bromotrifluoromethane, tetrafluoromethane, 1,2-dichlorotetrafluoroethane, dibromotetrafluoroethane, hexafluoroethane, or mixtures thereof.

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11. A multistory structure comprising:
 a plurality of stories, each of the stories including:
 a floor,
 a ceiling, and
 a dead space disposed above the ceiling;
 one or more exterior walls, each of the one or more exterior walls including:
 an exterior wall air flow pathway; and
 at least one exterior wall exhaust conduit;
 a plurality of exterior window assemblies, each of the plurality of exterior window assemblies including:
 an exterior window,
 an interior window, and
 a window air pathway interposed between the exterior window and the interior window and including:
 a window air flow pathway inlet, and
 a window air flow pathway outlet;
 a plurality of interior walls, each of the interior walls including:
 an interior wall air flow pathway; and
 at least one interior wall outlet conduit;
 a plurality of interior spaces defined by the one or more exterior walls and interior walls;
 an air handling system comprising:
 an heating, ventilation, and air conditioning (HVAC) unit;
 an HVAC inlet conduit including:
 an HVAC surroundings inlet conduit having:
 a surrounding inlet including:
 a surrounding inlet active control valve configured to supply unconditioned air to the HVAC unit;
 a return conditioned air conduit including:
 a return conduit active control valve configured to control an amount of exhausted conditioned air to the HVAC unit;
 a plurality of floor return conduits, each of the floor return conduits including:
 an inlet disposed in the each of the floor dead spaces including:
 a floor return conduit active control valve configured to control an amount of exhaust conditioned air supplied to the return conduit from each of the dead spaces;
 a main HVAC outlet conduit including:
 a plurality of floor outlet conduits, each of the floor outlet conduits including:
 a floor outlet conduit active control valve configured to control an amount of conditioned air supplied to each of the floors;
 a plurality of interior space outlet conduits, each of the interior space outlet conduits having:
 a plurality of interior space outlets, each of the interior space outlets having an interior space outlet conduit active control valve configured to control an amount of conditioned air supplied to each of the interior space outlets; and
 exterior window outlet conduit active control valves configured to control an amount of conditioned air supplied to each of the plurality of exterior windows;
 one or more HVAC exhaust conduits, each of the one or more HVAC exhaust conduits including:
 a HVAC exhaust conduit active control valve configured to control an amount of conditioned air exhausted to the surroundings; and

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an air flow control system including:
 a processing unit coupled to or associated with a plurality of sensors, the active control valves and the air handling system;
 one or more of the sensors is disposed in each of the interior spaces;
 one or more of the sensors is disposed in the pathways; wherein the processing unit is configured to:
 receive input from the sensors; and
 control the HVAC unit and the active control valves based on the sensor inputs to control unconditioned air flow rates into the HVAC, interior space air flow rates from the HVAC, return air flow rates into the HVAC, exhaust air flow rate to the surroundings, and pathway air flow rates through the pathways and into the surroundings.

12. The structure of claim 11, wherein each of the floor return conduits further including:

at least one return air active control valve configured to control an amount of the return conditioned air to the return conditioned air conduit.

13. The structure of claim 11, wherein the sensors comprise temperature sensors, water/moisture sensors, air flow sensors, chemical sensors, other physical sensors, and combinations thereof.

14. The structure of claim 13, wherein, during a 24 hour period of time, the processing unit is further configured to:
 adjust the air flow into each of the interior spaces and into each pathway based the received sensor input so that one or more interior space properties and/or one or more pathway properties are maintained at desired values.

15. The structure of claim 14, wherein the one or more properties include moisture concentrations in the interior spaces, moisture concentrations in the pathways, temperatures in the interior spaces, temperatures in the pathways, pressures in the interior spaces, pressures in the pathways, air flow rates in the interior spaces, air flow rates in the pathways, sensed chemical concentrations in the interior spaces, sensed chemical concentrations in the pathways, microbial concentrations in the interior spaces, microbial concentrations in the pathways or mixtures and combinations thereof.

16. The structure of claim 11, further comprising:

a dispensing assembly associated with one, some, or all of the pathways and including:
 a reservoir containing a treating composition;
 a dispensing supply conduit including:
 a dispensing assembly control valve,
 a dispensing head at its distal end,
 wherein:

the dispensing assembly control valve is configured to control an amount of treating solution or gas introduced into one, some, or all of the pathways, the treating composition comprising a mold control composition, a pesticide, an insecticide, a fungicide, a bactericide, a rodent poison, other control compositions, a gas, a fire retardant compositions, a foam, or mixtures or combinations thereof;

the gas comprising nitrogen, nitrogen enriched air, argon, haloalkanes, or mixtures thereof;

the haloalkanes comprising iodomethane; bromomethane, bromochloromethane, carbon tetrachloride, tribromofluoromethane, dichlorofluoromethane, bromodifluoromethane, dibromodifluoromethane, bromochlorodifluoromethane, dichlorodifluoromethane, bromotrifluoromethane, tetrafluoromethane,

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ane, 1,2-dichlorotetrafluoroethane, dibromotetrafluoroethane, hexafluoroethane, or mixtures thereof.

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